

# Mathematics for Civil Engineers

**Introduction**  
*Read Me First!*

**Further  
Material**

**CORE  
MATHS**

**Refresher  
Topics**

**See Your  
Scores!**



**UNIVERSITY OF LEEDS**

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### About the Authors

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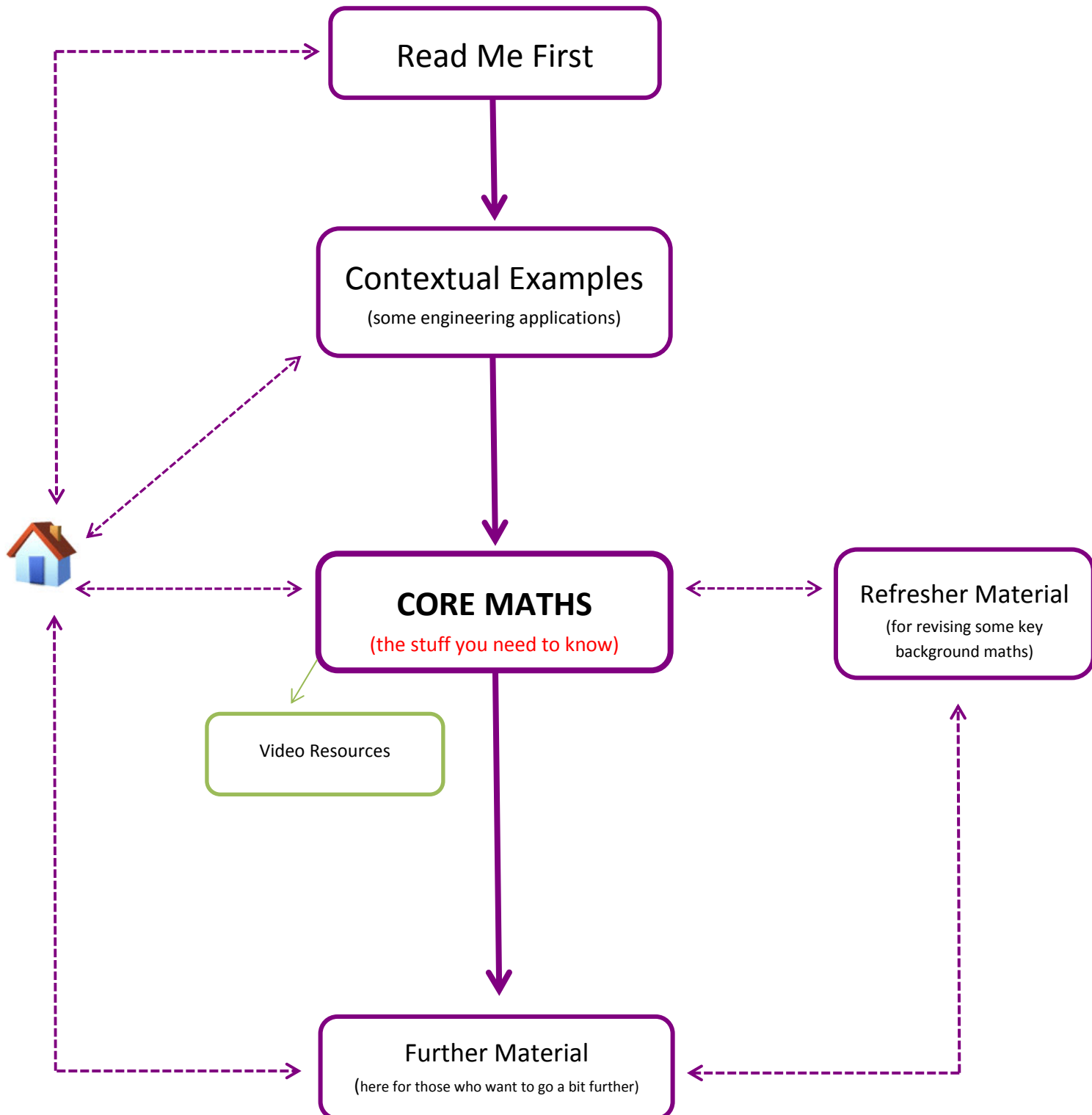


**It is important that you complete this mathematics resource in the order outlined below.**

1. You should start by looking at the **flow chart** on the next page. This helps to explain how you should progress through this resource and shows you what and where the key areas of the document are.
2. Next it is important you read through **the perspectives from past students and the programme leader**, this should provide you with some context as to why this resource is useful. You should take the time here to **watch a first year lecture** to give you an idea of the Level One Engineering Mathematics content at Leeds.
3. There are also some **examples of the applications** of complex mathematics in **famous civil engineering structures**. This section is designed to give you an idea of the importance and power of maths in civil engineering.
4. You should then begin working through the **Core Mathematics** section. This is the **priority area** of the resource and contains the most important content. It is intended to get you up to speed with the maths you will encounter at Level One so you should take some time to work through this. There are numerous examples for you to go through as well as 'Test Yourself Sections'. These are self-test only and can be done as many times as you wish - they are designed simply to check your understanding as you progress through the document. If you are ever uncertain about any of the content in this section you can refer to the **Refresher Material** which are small revision topics covering some of the maths you will be more familiar with.
5. There is also a **Further Material** section at the end of the document for those who want to go a bit further and cover some extra topics. This is a low priority section and should be done only if you have completed the Core Maths section.
6. If at any time you want to go back and **revisit** any of the content anywhere in the document you should click the  button and click on the relevant bubbles – this will navigate you quickly to the part of the document you want to find.



This **flow chart** below provides a guide as to how you should **use, progress through** and **navigate around** the resource. Please note, it is the **Core Maths** section that is the **priority area** for you to get up to speed with, the other sections will give some related examples and also extension material.



Why will this resource help?

**Read This First!**

Why is it important I use and complete this maths resource?

### *A personal context*

"My name is Adam and I'm just about to go into my second year of Civil Engineering at Leeds. Before starting my degree I completed an **Access Course** in engineering at Leeds City College, I thought this would put me in good stead for my endeavours into civil engineering. The course I'd done gave me a good overview into the maths that I would see here at University but having only done this course for a year - and having been out of education for a long-time meant that I was not as quick and confident in my ability to do maths as some of the students that had come from an A-Level background.

I found that my colleagues who had come via a **BTEC route** also had a similar experience. They found the level of maths they had done during their BTEC course meant they still had a lot of catching up to do (on applying mathematical concepts) compared with students who had completed A-level mathematics. There are **extra lectures** provided at Leeds to support this, but we virtually **all wished** we had been able to do more before arriving so that we were **far more prepared** from **day one** of the **degree course**.

Going into the first few maths lectures was rather intimidating since they were delivered at quite a pace. The entirety of the content that I had learnt from my access course had been covered in the first four lectures or so. This left ten more lectures of material that I had not seen before - just for calculus! The reason that **I like the idea of this resource** is that it would have given me a **real grasp** of the **fundamental** aspects of the **mathematical** content required **for an engineering degree**. Just spending a small amount of time each day before arriving to build on what I already knew, I feel, would have **prepared me better** for my first year.



The areas that I found particularly difficult have been addressed in this resource: some things may seem rather basic while other topics may seem a little more daunting. You may not need to go through everything in its entirety, but I know that if I had access to this before starting at Leeds I would have felt happier and more confident in the maths that I have now learnt this year. The **areas that I would say would have helped me the most** would have been: the practice and use of different **differentiation and integration** rules, and the manipulation of algebraic expressions including **re-arranging equations**."

Click here: to watch reflections of recent students who've studied the course

### *Programme Leader perspective*

"If you are a **BTEC student or someone** coming to study Civil Engineering at Leeds via an **Access Course** and have not completed A-level maths (or equivalent) you are likely to find the **pace of lectures** quite fast and **potentially** intimidating when you arrive and start your course. Spending time working through and building experience and skills, solving the key mathematical topics **in this resource**, will give you a big head-start and a much smoother transition into your degree course. **This is why we ask all our students who have not completed A-level Maths** (or equivalent) **to complete it**.

Please ensure you commit the time to this resource before arriving- *it is developed by past students for you!*"



## Maths for Engineers

Maths is the language of engineering and is used by engineers extensively every day. Without maths, engineers would be unable provide solutions to typical problems ranging from the design of superstructures to modelling the flow of fluids. This resource aims to familiarise you with fundamental and key concepts of mathematics, but also to provide, some context as to how and where they can be applied to real engineering situations.

When you are accessing this resource, you may not have a large amount of experience (or confidence) in some of the key mathematical concepts (such as differentiation and integration). Differentiation and integration, or calculus as it is collectively known, is an essential tool in engineering. As a prospective engineer you should be capable of performing calculus for quite challenging problems.

**Many pre-university mathematics courses, however, do not stress enough the importance of calculus and how it is an essential tool for solving real & applied engineering problems.**

### So, where do we use calculus?

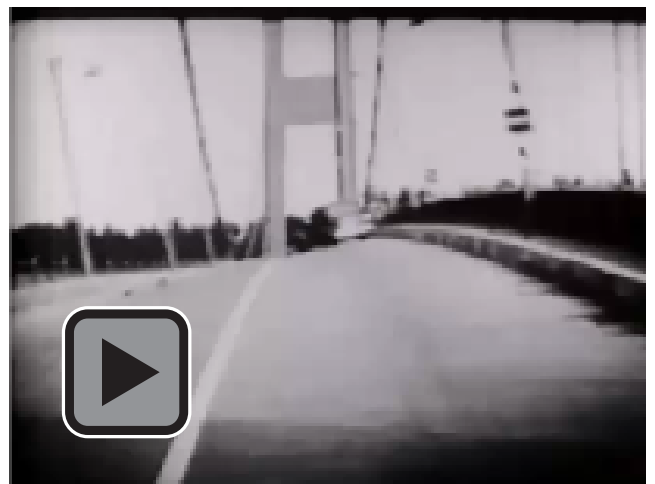
#### Bridges

As engineers we sometimes need to form and solve differential equations to help us explain certain phenomena. For example, the forced **vertical oscillations** caused by the wind, which led (in part) to the **collapse of the Tacoma Narrows Bridge**, can be described by a **differential equation**:

This **differential equation** allows us to see how an object is displaced ( **$x$** ) over time( **$t$** ) when an oscillating external forced of amplitude ( **$F$** ) is applied (e.g due to the wind) in terms of the bridge mass ( **$m$** ), spring damping constant ( **$b$** ) and spring constant ( **$k$** )

By understanding differential equations such as the one above, engineers can improve upon their designs for a safer and more effective structure. **In short – differentiation can save lives!**

$$m \frac{d^2x}{dt^2} + b \frac{dx}{dt} + kx = F e^{i\omega t}$$



[Tacoma-Narrows Bridge Collapse](#), No Copyright, [CC BY-PD 1.0](#)



## Dams

To find the centre of mass for a complex 3D shape (like a Dam) we can use **integration**.

For example if the object can be described as a revolution of a curve we can find the centroid

$$\bar{x} = \frac{\pi}{V} \int_a^b x(f(x))^2 dx$$

Where V is volume given by the **integral**.

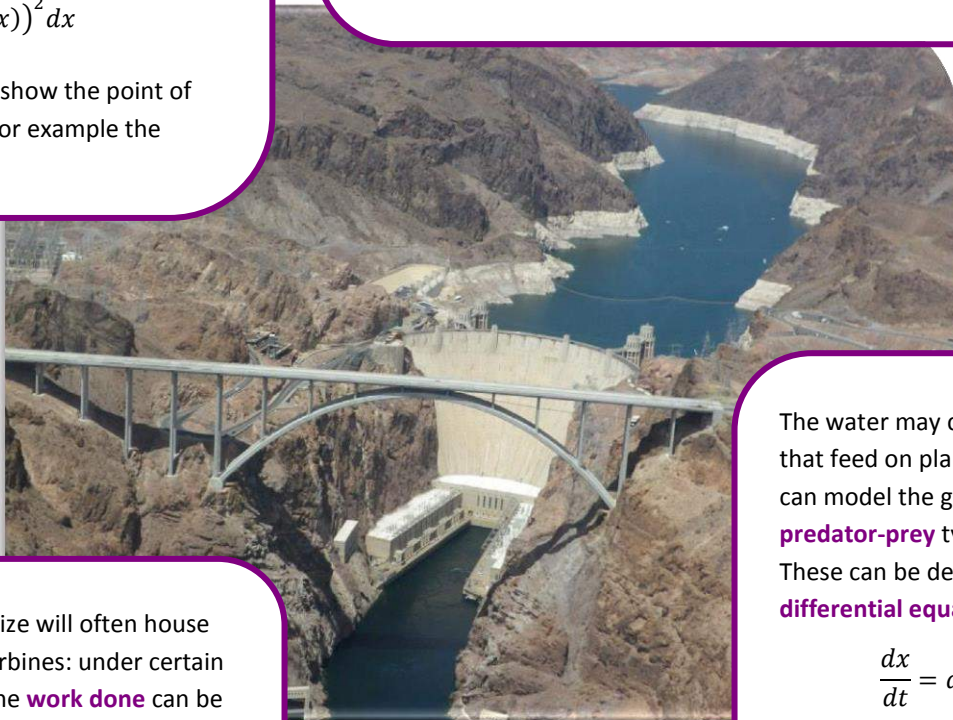
$$V = \pi \int_a^b (f(x))^2 dx$$

This can be useful as it can show the point of action of different forces (for example the **centre of mass**).

We can calculate the pressure behind the dam with **integration**. For example we can consider how the weight of water acts over the height and length of a dam.

$$F = \int_0^h \frac{dF}{dy} dy = \rho g l \int_0^h y dy = \frac{\rho g l h^2}{2}$$

The integral describes a relationship between the lateral Force (**F**) on the dam and the density of water (**ρ**) the length (**l**) and height of the dam. (**h**)



A dam of this size will often house hydroelectric turbines: under certain assumptions - the **work done** can be calculated by the **integral**:

$$W = \int_a^b F \cdot ds$$

Where **F** is the force needed to move an object from *a* and *b* along a curve *s*.

The water may contain organisms that feed on plant species. We can model the growth of such **predator-prey** type **relationships**. These can be described by a **set of differential equations**.

$$\frac{dx}{dt} = ax - axy$$

$$\frac{dy}{dt} = -cy + \gamma xy$$

Where **x** is the prey (plant) species **y** is the predator species and **a, c, γ** are all constants specific to the case.



## Super Structures

If we look at the cables that strengthen the steel exoskeleton of the structure the **differential equation**

$$\frac{dy}{dx} = \sinh\left(+\frac{w_s}{H}x + C_1\right)$$

can show how the cables will deform under load  $w_s$ .

The deflection of each member in the structure can be modelled using the **second order differential equation:**

$$EI \frac{d^2y}{dx^2} = \frac{w_1[x]^2}{2} + w_2[x - L]$$

The solution will vary depending on the load on each beam ( $w_1, w_2$ ), the beams flexural rigidity  $EI$  with respect to the distance  $x$  along the beam.

We need to use integration to calculate the deflections.

Engineers need to be able to interpret **calculus** effectively in order to design such impressive structures.

[Buri Al-Arab Dubai](#) © Joi Ito [CC BY 2.0](#).





# How to Use the Resource

This **resource** has been designed to be interactive, so you should find navigating around the document simple. The **homepage** is designed so that you can **move to** different parts of the document: each '**bubble**' directs you to a different topic within the resource. At the bottom of each and every page you will see the following symbol:



The house symbol will direct you back to the **homepage** should you wish to go there.

As you go through the booklet you will see black boxes like this:

'These contain important **equations** or **rules**'

You will also see various boxes which have been coloured coded according to their purpose:



## Example:

Blue boxes are **Example Boxes**. These are general mathematical examples to help reinforce your learning as you progress through the booklet

## Engineering Example:

Green boxes are **Engineering Examples**. These are real examples of where you may see the mathematics learnt applied directly in Civil Engineering.



## Test Yourself – Questions

When you see a Yellow Box you are invited to answer some questions to **Test Your Knowledge**. Some questions will be easy, some hard.

Once you have had a go at the questions you can access the solutions **by clicking on the link in the box** or you can go to the back of the booklet.

External **Video Links** are also available throughout the booklet. You can access these really useful resources by clicking on the hyperlink. Video links look like this



# 1 Algebra

## 1.1 Rearranging Equations

Rearranging equations is changing the arrangement of terms in an equation. Consider the pressure equation  $P = \rho gh$ . We could rearrange the equation so that we have  $h$  in terms of the other variables, so in this case  $h = \frac{P}{\rho g}$ . This rearranging is called making  $h$  the subject of the formula. When rearranging we perform operations to both sides of the equation so we may:

- Add or subtract the same quantity to or from both sides.
- Multiply or divide both sides by the same quantity.

### Order of Operations

BODMAS is used to decide which operations we should undo to make our quantity the subject. It is used in reverse, so **the operations are undone** in the following order:

1. Subtraction
2. Addition
3. Multiplication
4. Division
5. Orders
6. Brackets

**Note:** As mentioned in **Section 7.1**, addition and subtraction have the same priority so can be done in either order. The same is true for multiplication and division.

### Rearranging with Powers and Roots

The inverse operations of powers are roots, so to reverse a power we take a root and vice versa. We complete rearrangement in reverse order of BODMAS therefore we rearrange powers and roots in the orders section.



**Example:** Einstein's equation for mass energy equivalence is often seen in the form:

$$E = mc^2$$

Express it with  $c$  being the subject.

Solution: To start with we would divide each side of the equation by  $m$ .

$$\frac{E}{m} = \frac{mc^2}{m} \Rightarrow \frac{E}{m} = c^2$$

Now  $c$  is raised to the power of 2. To undo this operation we take the square root of both sides.

$$\frac{E}{m} = c^2 \Rightarrow \sqrt{\frac{E}{m}} = \sqrt{c^2} \Rightarrow \sqrt{\frac{E}{m}} = c$$



**Note:** If we were to make  $a$  the subject in the equation  $a^2 = b + c$  we take the square root of the whole of both sides of the equation to get  $a = \sqrt{b + c}$  and not  $a = \sqrt{b} + \sqrt{c}$ .





### Engineering Example:

The equation below calculates the flow rate over a rectangular weir. If we already know the flow rate we require then we can rearrange the equation to find the height (H) needed to achieve this.

$$Q = C_d \frac{2}{3} B \sqrt{2g} H^{\frac{3}{2}}$$

Rearrange this equation to make H the subject.

Solution: We identify the operations in use are  $\times, \div$  and a fractional power. Then we undo (or invert) each of the operations. Remember that we can do anything as long as we do it on both sides of the  $=$  sign

We start by undoing the multiplication as we go in the order of BODMAS backwards. On the right hand side we have  $H^{\frac{3}{2}}$  being multiplied to the terms involving  $C_d \frac{2}{3} B \sqrt{2g}$ . We then invert the operation by doing the opposite and dividing  $C_d \frac{2}{3} B \sqrt{2g}$  from both sides of the equation. This is possible because all these factors are multiplied by the  $H^{\frac{3}{2}}$  value. So

$$\frac{Q}{C_d \frac{2}{3} B \sqrt{2g}} = \frac{C_d \frac{2}{3} B \sqrt{2g} H^{\frac{3}{2}}}{C_d \frac{2}{3} B \sqrt{2g}}$$

$$\frac{Q}{C_d \frac{2}{3} B \sqrt{2g}} = H^{\frac{3}{2}}$$

We can now see that the only other operation acting on our  $H$  is the power  $\frac{3}{2}$  which we undo by raising both sides to the 2 and taking the 3<sup>rd</sup> root to make  $H$  the subject of the equation:

$$\sqrt[3]{\frac{Q}{C_d \frac{2}{3} B \sqrt{2g}}^2} = H^{\frac{3}{2} \times \frac{2}{3}}$$

$$H = \sqrt[3]{\frac{Q}{C_d \frac{2}{3} B \sqrt{2g}}^2}$$



[River Calder Weir](#) © Aontrotar [CC BY-SA 3.0](#)





### Test Yourself – Re-arranging equations

Re-arrange the following equations:

1. For the equation  $PV = nRT$  make T the subject of the equation (1 mark)
2. For the equation  $V = \pi r^2 \frac{h}{3}$  make h the subject of the equation. (1 mark)
3. For the equation  $K = \frac{mv^2}{2}$  make m the subject of the equation (1 mark)
4. For the equation  $v^2 = u^2 + 2as$  make u the subject of the equation (2 marks)
5. For the equation  $\frac{P_1}{\rho g} + \frac{u_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{u_2^2}{2g} + z_2$  make  $P_2$  the subject of the equation (2 marks)

For the solutions click [here](#)

Enter Score /7



## 1.2 Simultaneous Equations

The simplest simultaneous equations are two equations with two variables. They are called simultaneous because they must both be solved at the same time to find a pair of numbers that satisfy both equations. The two equations below form a pair simultaneous of equations:

$$2y = x + 1 \quad \textcircled{1}$$

$$x + y = 5 \quad \textcircled{2}$$

In order to solve these we use a method called substitution shown in the example below. We start by making one of the variables the subject of one of the equations. We then substitute this new expression into the other equation and solve.

**Note:** Similar methods can be applied to 3 equations with 3 unknowns and more generally for  $n$  equations with  $n$  unknowns.



**Example:** Solve the simultaneous equations:  $y + x = 5$   $\textcircled{1}$  and  $2y - x = 1$   $\textcircled{2}$ .

Solution: To solve these we will make one of the variables the subject. Take equation  $\textcircled{2}$  and make  $x$  the subject:

$$\begin{aligned} 2y &= x + 1 \\ \Rightarrow 2y - 1 &= x \end{aligned}$$

**Note:** We could have found  $y$  as the subject or used equation  $\textcircled{1}$  it does not matter!

As  $x = 2y - 1$  we can **substitute** this expression into  $\textcircled{1}$  to get that: (We must always substitute our expression into the other equation that we did not rearrange in the step before).

$$\begin{aligned} y + x &= 5 \\ \Rightarrow y + (2y - 1) &= 5 \\ \Rightarrow 3y - 1 &= 5 \\ \Rightarrow 3y &= 6 \\ \Rightarrow y &= 2 \end{aligned}$$

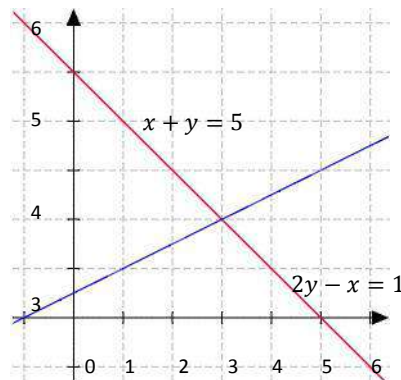
We have the value of  $y = 2$ . To find out what  $x$  is we need to substitute  $y = 2$  into either one of the equations (both should give you the same answer). So if we substitute this into  $\textcircled{1}$  we get:

$$\begin{aligned} y + x &= 5 \\ \Rightarrow 2 + x &= 5 \\ \Rightarrow x &= 3 \end{aligned}$$

**We have the solution of  $x = 3$  and  $y = 2$ .**



Solving these simultaneous equations can also be done graphically. Below are both equations plotted on the same graph. Notice the intersection point of the lines is (3,2) which is what we calculated.



**Example:** Stacy's mother was 30 years old when she was born. At what age is she half her mother's age?

Solution: While you may be able to spot the answer to this question directly, you can solve this using simultaneous equations. To solve this problem we first need to form equations that describe the information given to us. Let the variables  $s$  and  $m$  denote the age of Stacy and her mother respectively then we can write:

①  $m = s + 30$       Meaning that Stacy's mother is 30 years older than her.

②  $s = \frac{1}{2} m$       Meaning that Stacy's age is half her mother's.

We want to find Stacy's age  $s$  so to do that we shall substitute ① into ② resulting in:

$$s = \frac{1}{2}(s + 30) \quad \text{Multiply each side by 2}$$

$$2s = s + 30 \quad \text{Subtracts from both sides}$$

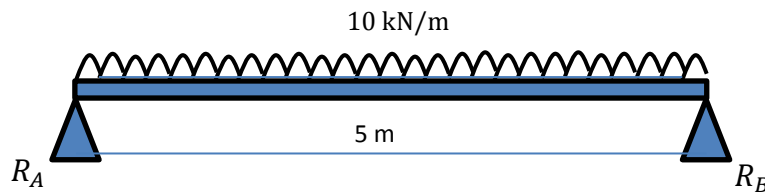
$$s = 30$$

**Stacy will be 30 when she's half her mother's age.**



### Engineering Example:

We often see simultaneous equations in engineering. An example of this is when we are finding the forces pushing back from the supports of a beam (reactions).



Take the beam above: to work out the reactions at each end we have to set up a few equations. We know that the total force down must equal the total force up. The load of 10 kN/m over 5m is 50kN (acting downwards.) We equate this to the upward forces due to the reactions at the supports,  $R_A$  and  $R_B$

$$\sum_y = R_{Ay} + R_{By} - 50 = 0 \quad (\text{the sum of all the forces in the vertical direction is equal to zero})$$

$$R_{Ay} + R_{By} = 50$$

This is the first of our simultaneous equations

Now we work out a moment about the one of the supports, this is the turning effect due to the forces about this point. We know that the moment created by the reaction on the right hand side of the beam,  $R_B$  must equal the moment that is created by the load on the beam.

$$\sum_{mA} = 2.5 \times (10 \times 5) - 5 \times R_{By} = 0 \quad (\text{the sum of moments about A is equal to zero})$$

$$125 - 5 \times R_{By} = 0$$

$$125 = 5 \times R_{By}$$

$$R_{By} = 25$$

This becomes our second equation. We can then substitute this back into our first equation to find the reaction for  $R_{Ay}$ .

$$R_{Ay} + 25 = 50$$

$$R_{Ay} = 25$$

$$R_{Ay} = 25 \text{ kN}$$

$$R_{By} = 25 \text{ kN}$$

**Note:** Don't worry if you are unfamiliar with taking moments as you will see this in your first year



## 1.3 Quadratics

A quadratic (in the variable  $x$ ) is any expression of the form:

$$ax^2 + bx + c$$

Where  $a, b$  and  $c$  are constants. It is called a quadratic because it contains terms in  $x$  up to and including  $x^2$ .

A quadratic function has the form:

$$y = ax^2 + bx + c$$

Where  $a, b$  and  $c$  are constants. There are a number of examples of such functions illustrated in the pages that follow.

### Expanding Brackets to Produce a Quadratic

Quadratic expressions can also be factorised and written in the form:

$$(x + A)(x + B)$$

where  $A$  and  $B$  are constants. We have a method called **FOIL** of expanding quadratic equations in the above form so that they appear in the form  $ax^2 + bx + c$ .

1. The **F**irst terms will be the  $x$ 's since they are both first in their brackets.
2. The **O**uter terms will be  $x$  and  $B$  because they are on the outer side (nearest to the edges).
3. The **I**nnner terms are  $A$  and the other  $x$  for the same reason.
4. The **L**ast terms are  $A$  and  $B$  because they are last in their brackets.

We then expand our quadratic  $(x + A)(x + B)$  as shown below:

$$(x + A)(x + B) = \overbrace{x \times x}^{\text{F}} + \overbrace{x \times B}^{\text{O}} + \overbrace{A \times x}^{\text{I}} + \overbrace{A \times B}^{\text{L}} = x^2 + (A + B)x + AB$$

**Note:** This is in the form  $ax^2 + bx + c$  where  $a = 1$ ,  $b = A + B$  and  $c = AB$ .



**Example:** Expand the brackets of the following:

1.  $(x + 5)(x + 7)$
2.  $(x - 3)(x + 2)$

Solution:

$$1. \text{ We use } \mathbf{FOIL} \text{ to get } (x + 5)(x + 7) = \overbrace{x \times x}^{\text{F}} + \overbrace{x \times 7}^{\text{O}} + \overbrace{5 \times x}^{\text{I}} + \overbrace{5 \times 7}^{\text{L}}$$

$$\text{Multiplying these out gives } x^2 + 7x + 5x + 35 = x^2 + 12x + 35$$

2. There are some negative numbers here so be careful when using **FOIL**.

$$(x - 3)(x + 2) = \overbrace{x \times x}^{\text{F}} + \overbrace{x \times 2}^{\text{O}} + \overbrace{-3 \times x}^{\text{I}} + \overbrace{-3 \times 2}^{\text{L}} = x^2 + 2x - 3x - 6 = x^2 - x - 6$$



## Solving Quadratic Equations

Solving a quadratic equation involves finding all possible values of  $x$  that satisfy  $ax^2 + bx + c = 0$ . The values of  $x$  that satisfy this equation are called the roots.

Note that when  $a = 0$ , the quadratic equation reduces to a simple linear equation  $bx + c$  which is easy to solve. Also, when  $a \neq 0$  we can divide throughout by  $a$  to get an equivalent equation of the form:  
 $x^2 + bx + c = 0$

There are three methods that can be used to solve a quadratic:

1. Completing the square.
2. Inspection.
3. The quadratic formula.

## Completing the Square

When a quadratic is in the form  $x^2 + 2bx + b^2$  it is known as a perfect square because it can be factorised into a squared linear term  $(x + b)^2$ . However when a quadratic is not a perfect square we have the relationship below:

$$x^2 + 2bx + c = (x + b)^2 + (c - b^2)$$

(To see this simply expand out the right hand side.)

Hence if we are trying to solve  $x^2 + 2bx + c = 0$  then we can rewrite this as  $(x + b)^2 + (c - b^2) = 0$  and then rearrange to find  $x$ .



**Example:** Find the values of  $x$  that make  $x^2 + 4x - 1 = 0$

Solution: we can see that  $2b = 4$  and  $c = -1$ . From the equation mentioned above we can change the quadratic into the following form:

$$\begin{aligned}x^2 + 4x - 1 &= 0 \\ \Rightarrow (x + 2)^2 + (-1 - (2^2)) &= 0 \\ \Rightarrow (x + 2)^2 - 5 &= 0\end{aligned}$$

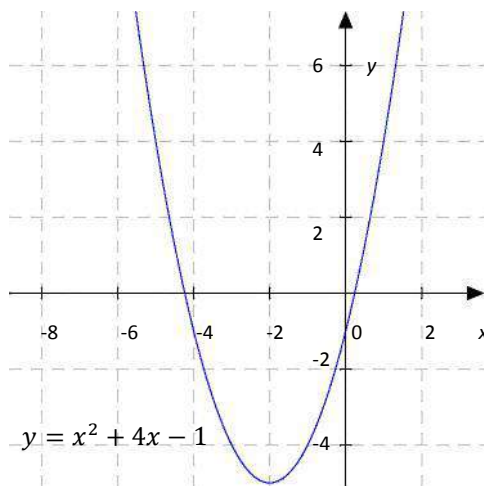
We can now re-arrange and take the square root of both sides:

$$\begin{aligned}\Rightarrow (x + 2)^2 &= 5 \\ \Rightarrow (x + 2)^2 &= \pm\sqrt{5} \\ \Rightarrow x &= -2 \pm \sqrt{5}\end{aligned}$$



**Note:** When we take the square root of both sides we need to remember that both  $(+\sqrt{5})^2$  and  $(-\sqrt{5})^2$  are equal to 5; the expression  $\pm\sqrt{5}$  indicates that both signs are included. The two different roots for this equation are shown on the graph on the next page.





The  $x$ -axis is where  $y = 0$  so the points where the curve crosses the  $x$ -axis are the roots for  $x^2 + bx + c = 0$ . The graph goes through a point just to the left of  $x = -4$  and just to the right of  $x = 0$ . Those correspond to  $x = -2 - \sqrt{5}$  and  $x = -2 + \sqrt{5}$  respectively.

**Note:** When solving a quadratic equation that has come from a scientific situation, there may be non-mathematical reasons for disregarding, say, a negative root as it might not make physical sense. A good example would be if the roots represented a time since a having negative time doesn't make sense.

### Solving by Inspection

If we have a quadratic with integer coefficients it is sometimes possible to solve it by inspection. So if the quadratic  $x^2 + bx + c$  can be factorised to give  $(x + A)(x + B)$  then the roots of the equation are  $-A$  and  $-B$ . We have the following rules for solving by inspection:

- $c$  will be the product of  $A$  and  $B$
- $b$  will be the sum of  $A$  and  $B$ .

In other words if we are trying to solve  $x^2 + bx + c = 0$  then we need to find two numbers  $A$  and  $B$  such that their sum is  $b$  and product is  $c$ . We can then write the equation as  $(x + A)(x + B) = 0$  and our roots will be  $x = -A$  and  $x = -B$ .



**Example:** Find  $x$  by inspection in the following  $x^2 + 5x + 6 = 0$ .

Solution: Since we want to factorise this we need to find two numbers that add to 5 and multiply to 6. The best way to do this is to think about what the factors of 6 are. We have 1 and 6 as well as 2 and 3. We can instantly see that  $2 + 3 = 5$  so these are the numbers required. We can rewrite the original equation as:

$$x^2 + 5x + 6 = (x + 2)(x + 3)$$

$$\Rightarrow (x + 2)(x + 3) = 0$$

Now that we have the quadratic factorised we can now find out what  $x$  is. We notice if either of the terms in the brackets is 0 then the whole thing will be 0. This gives the two situations

$$x + 2 = 0$$

OR

$$x + 3 = 0$$

When we find out what  $x$  is for each of those we see that the roots are  $-2$  and  $-3$ . These are both solutions to this quadratic equation.





## Inspection with Negative Coefficients

Inspection is easy when all the coefficients are positive but when they are negative you need to consider a few things. First if  $c$  is negative then one of  $A$  or  $B$  will be negative. This is because when a negative number multiplies a positive number the result will be negative. Secondly you would look at  $b$ , if it is positive then the larger number of  $A$  and  $B$  is positive and vice versa.

Take the example  $x^2 - 4x - 5$ . We first look at what factors 5 has: here the only factors are 1 and 5. Then we notice that since  $c$  is negative (here it is -5), one of  $A$  and  $B$  is negative and one is positive. Finally, since  $b = -4$  is negative the sum  $A + B$  is negative and this forces us to choose  $A = -5$  and  $B = 1$  giving us,  $x^2 - 4x - 5 = (x - 5)(x + 1)$



**Example:** Factorise the following:

1.  $x^2 + 4x - 12$

2.  $x^2 - 6x - 40$

Solution:

1. From the signs we can deduce that one of  $A$  or  $B$  is negative and, since  $b = 4$ ,  $A + B = 4$ , i.e. is positive. The factors of 12 are 1 and 12, 2 and 6 and 3 and 4. The pair of numbers that we need are therefore  $A = 6$  and  $B = -2$  giving us  $x^2 + 4x - 12 = (x + 6)(x - 2)$ .
2. The signs tell us that one of  $A$  or  $B$  is negative and, since  $b = -6x - 40$ , i.e. is negative. Looking at the factors of 40 we can see that -10 and 4 would produce the required result ( $-10 + 4 = -6$ ) giving us,  $x^2 - 6x - 40 = (x - 10)(x + 4)$

Finally, if  $c$  is positive but  $b$  negative then both  $A$  and  $B$  have to be negative, since they would give a positive product but a negative sum.



**Example:** Factorise  $x^2 - 8x + 15$

Solution: Since  $c = 15$  is positive we know that both  $A$  and  $B$  have the same sign and, since  $b = -8$ ,  $A + B = -8$  we see that they must both be negative. Looking at the factors of 15 we can see that -5 and -3 would produce the required result ( $-5 - 3 = -8$ ) giving us,  $x^2 - 8x + 15 = (x - 5)(x - 3)$ .

## Quadratic where $a \neq 1$

So far we have looked at quadratics with only an ' $a$ ' term of 1. However we will often find quadratics with an  $a$  term that is  $< 1$  or  $> -1$ . We can factorise these in a similar way to as we have seen before.

We do this by first multiplying the  $a$  term by the  $c$  term. We then find two numbers that add together to give the  $b$  term but also multiply together to give the ' $ac$ ' term. This is sometimes known as the '**a-c method**'. From here we take out common factors to allow us to fully factorize the equation. This is best explained through the example on the next page.





**Example:** Find the roots of the equations  $2x^2 + 13x + 15$  and  $5x^2 - 14x - 3$

Solution:

$$2x^2 + 13x + 15$$

We can solve this in a similar way to solving by inspection with a few small differences. We take the first term and multiply it by the last term to find 'ac'.

This gives us  $2 \times 15 = 30$ .

Now we want to find two numbers that multiply together to make 30 and add together to get 13.

$$? \times ? = 30$$

$$? + ? = 13$$

In this case this we get 10 and 3. We then put these numbers into the equation to replace our  $x$  term

$$2x^2 + 10x + 3x + 15$$

We can then look at the first half of the equation and take out a common factor:

$$2x(x + 5)$$

We know we have done this right if we can take a common factor out the second half of the equation to leave us with the same bracketed term for the second half of the equation.

$$2x(x + 5) + 3(x + 5)$$

We can now take common factors out of the equation.

$$(x + 5)(2x + 3)$$

From here we can find values of  $x$  that will give use the roots we are looking for.

$$x + 5 = 0 \quad \text{and} \quad 2x + 3 = 0$$

$$\Rightarrow x = -5 \text{ and } x = -\frac{3}{2}$$

$$5x^2 - 14x - 3$$

We need to find two numbers that multiply together to give  $-15$  and add together to give  $-14$

$$? \times ? = -15$$

$$? + ? = -14$$

These are  $-15$  and  $+1$

$$5x^2 - 15x + 1x - 3$$

Now we take a common factor out of the 'first half' and 'second half' of the equation:

$$5x(x - 3) + 1(x - 3)$$

Taking  $(x - 3)$  as a common factor for the whole equation this time:

$$(x - 3)(5x + 1)$$

From here we can find values of  $x$  that will give use the roots we are looking for.

$$x - 3 = 0 \quad \text{and} \quad 5x + 1 = 0$$

$$\Rightarrow x = +3 \text{ and } x = -\frac{1}{5}$$



## The Quadratic Formula

If all else fails we can always rely on the quadratic formula. If we have a quadratic equation of the form  $ax^2 + bx + c = 0$  then the quadratic formula for the roots is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

**Note:** The  $\pm$  symbol means plus or minus. So we have two solutions one for when you add the square root and one where you subtract.

**Note:** If the total of the sum inside the square root is negative this will give you an error in your calculator as you cannot square root a negative number.



**Example:** Use the quadratic formula to solve the equation  $3x^2 - 5x + 2$

Solution: As we are using the formula all we need to do is put in the values for  $a, b$  and  $c$ .

So  $a = 3, b = -5$  and  $c = 2$  putting these values into the equation yields:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(3 \times 2)}}{2 \times 3}$$

(Take care to bracket negative functions  $(-5)^2 \neq -5^2$ )

$$\Rightarrow x = \frac{5 \pm \sqrt{25 - 24}}{6}$$

$$\Rightarrow x = \frac{5 \pm \sqrt{1}}{6}$$

$$\Rightarrow x = \frac{5+1}{6} = 1 \quad \text{corresponding to the positive square root } +\sqrt{1}$$

$$\Rightarrow x = \frac{5-1}{6} = \frac{4}{6} = \frac{2}{3} \quad \text{corresponding to the positive square root } -\sqrt{1}$$

So  $x = 1$  and  $x = \frac{2}{3}$  are the two roots





### Test Yourself – Quadratic Equations

Find the roots of the following equation:

1.  $x^2 - 6x + 9$  (1 mark)
2.  $x^2 - 2x - 15$  (1 mark)
3.  $x^2 - 10x + 21$  (1 mark)
4.  $2x^2 - 7x + 6$  (2 marks)
5.  $3x^2 + 5x - 2$  (2 marks)
6.  $x^2 + 4x - 3$  [Hint: This quadratic does not solve easily by factorizing] (2 marks)
7.  $x^3 + 6x^2 + 7x$  [Hint: Find a quadratic in this equation] (3 marks)

Enter Score / 12

For the solutions click [here](#)

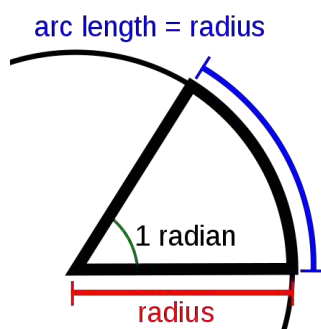


## 2 Trigonometry

### 2.1 Radians

We often use degrees to measure angles however we also have another measurement of angles called radians. Instead of splitting a full circle into  $360^\circ$ , it is split into  $2\pi$  radians. We must use radians in calculus (differentiation and integration), complex numbers and polar coordinates.

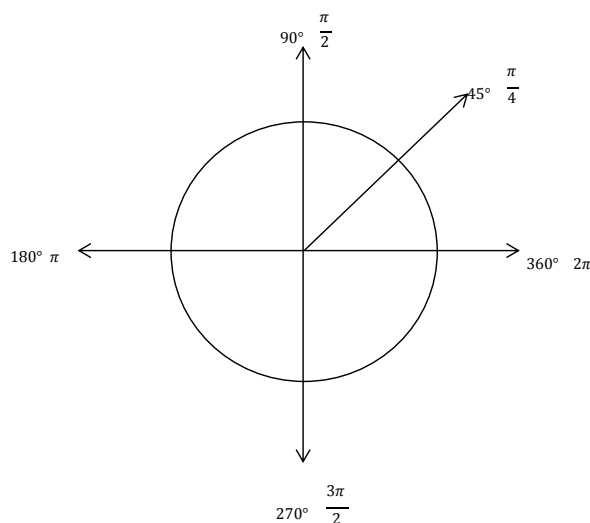
One radian is equal to the angle made at the centre of a circle where the arc describing that angle is equal to the radius. The graph below describes this more clearly



[Definition of a radian](#) © Antonsusi [CC BY 2.5](#)

This may seem new and complicated at first however it is no different to thinking that we could measure length in miles or metres.

Below we have a full circle showing some angles in degrees and what they are in radians.



We may wish to convert between degree and radians. In order to turn  $x^\circ$  degrees into radians we use the formula:

$$\frac{x^\circ}{360^\circ} \times 2\pi$$

In order to convert  $x$  radians into degrees we use the formula:

$$\frac{x}{2\pi} \times 360^\circ$$





**Example:** Convert the following:

1.  $60^\circ$  into radians.
2.  $\frac{3\pi}{8}$  radians into degrees.

Solutions:

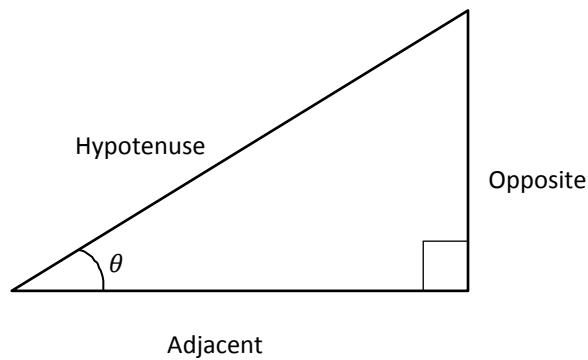
1. We need to use the formula  $\frac{x^\circ}{360^\circ} \times 2\pi$

$$\frac{x^\circ}{360^\circ} \times 2\pi = \frac{60^\circ}{360^\circ} \times 2\pi = \frac{\pi}{3} \text{ radians.}$$

2. We need to use the formula  $\frac{x}{2\pi} \times 360^\circ$

$$\frac{x}{2\pi} \times 360^\circ = \frac{\left(\frac{3\pi}{8}\right)}{2\pi} \times 360^\circ = \frac{3}{16} \times 360^\circ = 67.5^\circ$$

## 2.2 Right-Angled Triangles



As shown in the diagram above when given a right-angled triangle with one of the non-right angles labelled  $\theta$  then we label the sides as follows:

- The longest side is called the **hypotenuse** this is always the side opposite the right angle.
- The side that is opposite the angle  $\theta$  is called the **opposite**.
- The side that is next to the angle  $\theta$  is called the **adjacent**.

### Pythagoras' Theorem

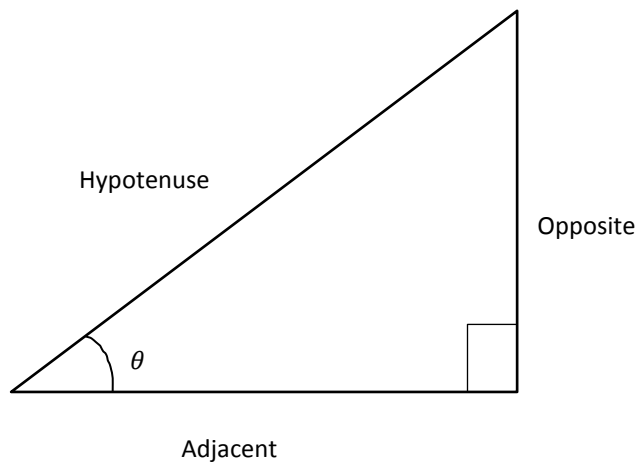
Pythagoras' theorem relates only to right-angled triangles. Given a right-angled triangle such as the one below it states:

$$c^2 = a^2 + b^2$$

where  $c$  is the length of the hypotenuse and  $a$  and  $b$  are the lengths of the other two sides of the triangle. This can be re-arranged to find the each side of the triangle.



## 2.3 SOHCAHTOA



Given our right-angled triangle we start by defining sine, cosine and tangent before reviewing what their graphs look like.

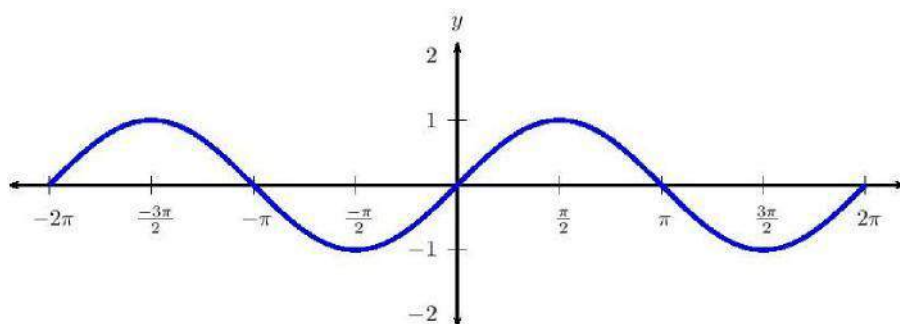
**Note:** The angle  $\theta$  is measured in radians in the following graphs on the  $x$ -axis.

### Sine Function

We define sine as the ratio of the opposite to the hypotenuse.

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

Part of the graph of  $y = \sin(\theta)$  is shown below.

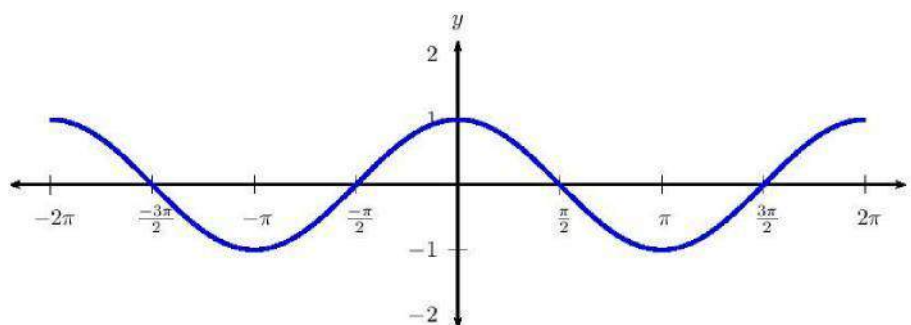


## Cosine Function

We define cosine as the ratio of the adjacent to the hypotenuse.

$$\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

Part of the graph of  $y = \cos(\theta)$  is shown below.



## Tangent Function

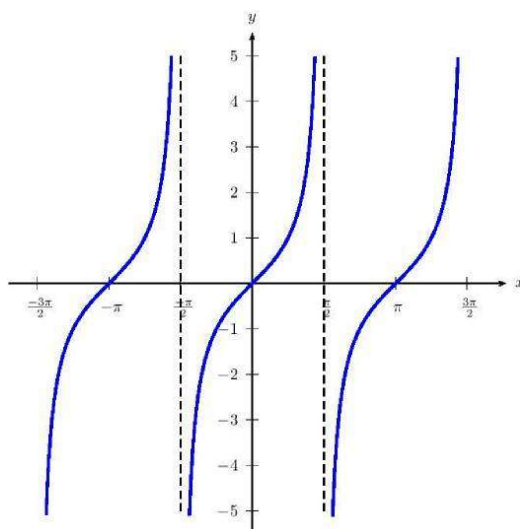
We define tangent as the ratio of the opposite to the adjacent.

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

This leads to the relationship  $\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$  shown below:

$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{\text{Opposite}}{\text{Adjacent}} \times \frac{\text{Hypotenuse}}{\text{Hypotenuse}} = \frac{\left(\frac{\text{Opposite}}{\text{Hypotenuse}}\right)}{\left(\frac{\text{Adjacent}}{\text{Hypotenuse}}\right)} = \frac{\sin(\theta)}{\cos(\theta)}$$

The graph of  $y = \tan(\theta)$  is shown below.





The definitions of sine, cosine and tangent are often more easily remembered by using SOHCAHTOA. We can use the formulas in SOHCAHTOA to calculate the size of angles and the length of sides in **right-angled triangles**.

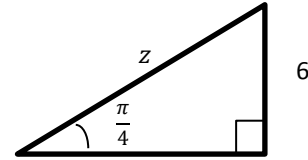
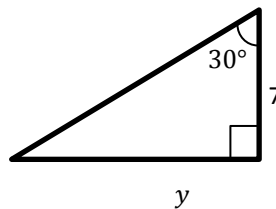
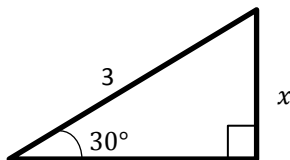
$$\sin(\theta) = \frac{\overbrace{SOH}^{\text{Opposite}}}{\text{Hypotenuse}}$$

$$\cos(\theta) = \frac{\overbrace{CAH}^{\text{Adjacent}}}{\text{Hypotenuse}}$$

$$\tan(\theta) = \frac{\overbrace{TOA}^{\text{Opposite}}}{\text{Adjacent}}$$



**Example:** Using SOHCAHTOA find the length of the angles  $x$ ,  $y$  and  $z$  in the right-angled triangles below.



**Solution:** For shorthand we have used H to denote the hypotenuse, O the opposite and A the adjacent.

1. We know the length of the hypotenuse and wish to find the opposite hence we need to use

$$\sin(\theta) = \frac{O}{H} \Rightarrow \sin(30^\circ) = \frac{O}{3} \Rightarrow O = 3 \sin(30^\circ) = \frac{3}{2}$$

**Note:** This question is in degrees.

2. We know the length of the adjacent and wish to find the opposite hence we need to use

$$\tan(\theta) = \frac{O}{A} \Rightarrow \tan(30^\circ) = \frac{O}{7} \Rightarrow O = 7 \tan(30^\circ) = \frac{7\sqrt{3}}{3}$$

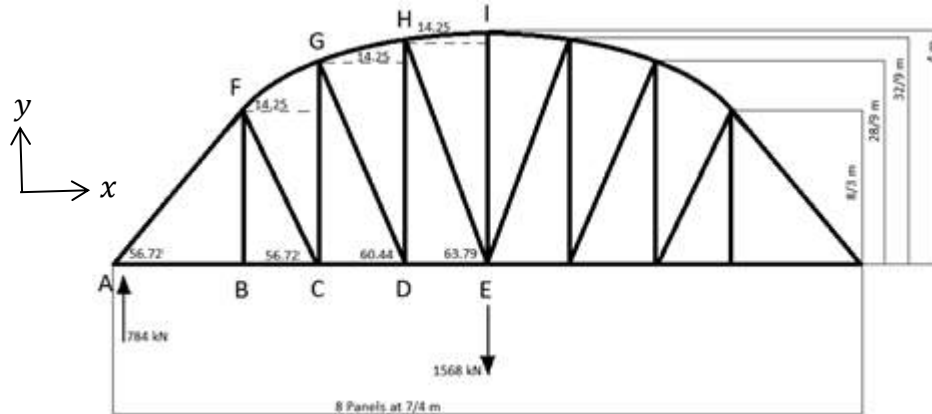
**Note:** This question is in degrees.

3. We know the length of the adjacent and wish to find the hypotenuse hence we need to use

$$\cos(\theta) = \frac{A}{H} \Rightarrow \cos\left(\frac{\pi}{4}\right) = \frac{6}{H} \Rightarrow H = \frac{6}{\cos\left(\frac{\pi}{4}\right)}$$



We use trigonometric functions in engineering all the time as very rarely do we have forces acting parallel or at  $90^\circ$  to one another. One example of this is the forces acting on a truss we use sin a cos functions to find the forces on the members and joints of the truss.



Joint A

**Sum of the forces in the horizontal direction= 0**

$$\Sigma_{FX} = 0$$

$$F_{AB} + F_{AF} \cos(56.72)$$

**Sum of the forces in the vertical direction = 0**

$$\Sigma_{Fy} = 0$$

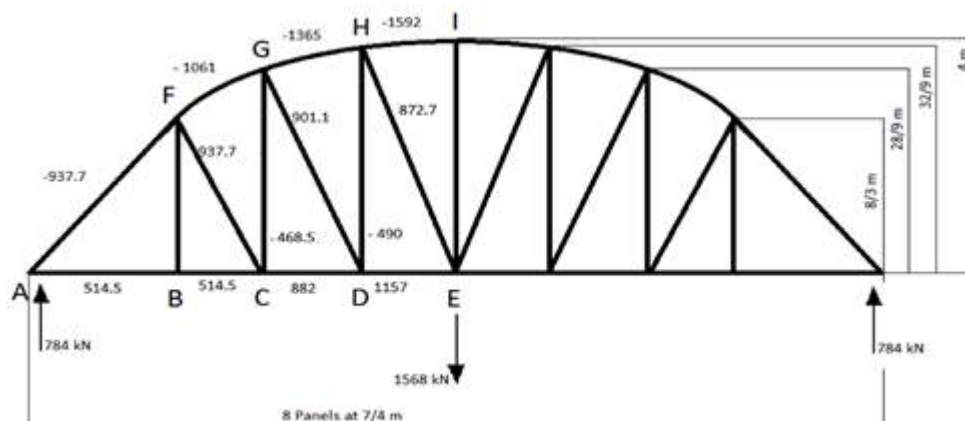
$$784 + F_{AF} \sin (56.75)$$

$$F_{AF} = -937.7 \text{ kN (c)}$$

$$F_{AB} = 514.5 \text{ kN (t)}$$

These equations show how the reaction will push up through the members AF and AB

We can do this for the entire truss to calculate the forces in each member and to forces that will be applied to each joint. This can give us something that looks like this.



The negative numbers show compressive forces in the member and the positive number are tensile forces.

## 2.4 Inverse Function and Rearranging Trigonometric Functions

We may be given the lengths of two sides of a right-angled triangle and be asked to find the angle  $\theta$  between them or be asked to rearrange an equation with a trigonometric function. In order to do these we need to introduce the inverse trigonometric functions:

- The inverse function of *sine*  $\sin(\dots)$  is  $\sin^{-1}(\dots)$  also can be denoted as  $\arcsin(\dots)$
- The inverse function of *cosine*  $\cos(\dots)$  is  $\cos^{-1}(\dots)$  also can be denoted as  $\arccos(\dots)$ .
- The inverse function of *tangent*  $\tan(\dots)$  is  $\tan^{-1}(\dots)$  also can be denoted as  $\arctan(\dots)$ .

We use the inverse trigonometric functions with SOHCAHTOA to find the values of angles in right-angled triangles in which two of the lengths of the sides have been given. We do this as follows, suppose we know that:

$$\sin(\theta) = \frac{1}{2}$$

To find  $\theta$  we can apply the inverse sine function to both sides of the equation.

$$\sin^{-1}(\sin(\theta)) = \sin^{-1}\left(\frac{1}{2}\right)$$

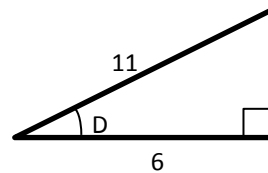
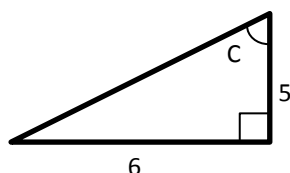
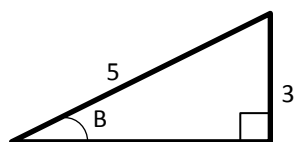
$$\Rightarrow \theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\Rightarrow \theta = 30^\circ$$

We can use a calculator to find  $\sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$  or  $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$  radians.



**Example:** Using SOHCAHTOA find the size of the angles  $B$ ,  $C$  and  $D$  in the right-angled triangles below.



**Solution:** For shorthand we have used H to denote the hypotenuse, O the opposite and A the adjacent.

1. We know the length of the hypotenuse and the opposite.

$$\sin(\theta) = \frac{O}{H} \Rightarrow \sin(B) = \frac{3}{5} \Rightarrow B = \sin^{-1}\left(\frac{3}{5}\right) = 36.9^\circ \text{ or } 0.64 \text{ radians.}$$

2. We know the length of the adjacent and the opposite.

$$\tan(\theta) = \frac{O}{A} \Rightarrow \tan(C) = \frac{6}{5} \Rightarrow C = \tan^{-1}\left(\frac{6}{5}\right) = 50.2^\circ \text{ or } 0.88 \text{ radians.}$$

3. We know the length of the hypotenuse and the adjacent.

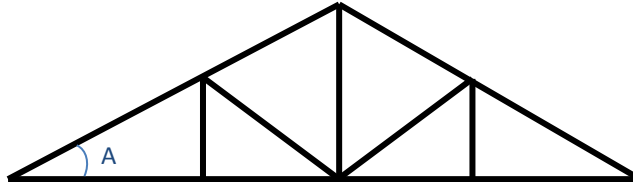
$$\cos(\theta) = \frac{A}{H} \Rightarrow \cos(D) = \frac{6}{11} \Rightarrow D = \cos^{-1}\left(\frac{6}{11}\right) = 56.9^\circ \text{ or } 0.99 \text{ radians.}$$



**Note:** When squaring trigonometric functions such as  $(\sin(x))^2$ ,  $(\cos(x))^2$  and  $(\tan(x))^2$  we use the notation  $\sin^2(x)$  for  $(\sin(x))^2$  the same applies for  $\cos^2(x)$  and  $\tan^2(x)$ . This is also true more generally for example  $(\sin(x))^n$  is denoted by  $\sin^n(x)$ .

### Engineering Example:

We often want to find the angle between two forces, as denoted by 'A' in the truss below.



If the truss is **8m wide** and **3m** tall we can calculate the angle at A we can do this with the *tan* function. Looking at the problem we can pick out the triangle that makes up the truss to find the angle.

This triangle will have an opposite of 3m and an adjacent of 4m (as 4m is the distance from A to the the mid-point - so half of 8m)

$$\tan(\theta) = \frac{3}{4}$$

$$\theta = \tan^{-1} \frac{3}{4}$$

$$36.9^\circ$$

## 2.5 Trigonometric Identities

Trigonometric Identities relate different trigonometric functions to one another. These can be useful when we are re-arranging or solving equations containing multiple trig functions. We may need these to more easily differentiate or integrate particular functions. Some of these can be seen below.

**Note:** these are useful to know about and to have an understanding of but you do not need to memorize all of these.

$$\tan(\theta) = \frac{\sin(\theta)}{\cos(\theta)}$$

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

$$\sin(2\theta) = 2 \sin(\theta) \cos(\theta) = \frac{2 \tan(\theta)}{1 + \tan^2(\theta)}$$

$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = 2 \cos^2(\theta) - 1 = 1 - 2 \sin^2(\theta) = \frac{1 - \tan^2(\theta)}{1 + \tan^2(\theta)}$$

There are many more of these identities these are just a few

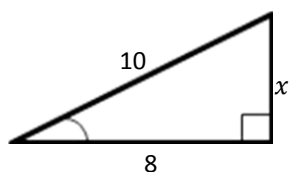




### Test Yourself – Trigonometry

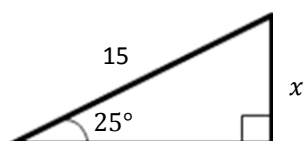
1. Find the length of the side  $x$  :

(1 mark)



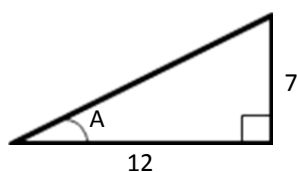
2. Find the length of the side  $x$ :

(1 mark)



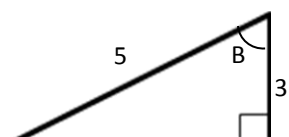
3. Find the angle A in radians and degrees:

(2 marks)



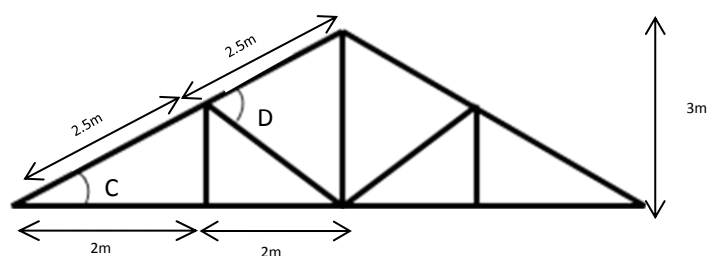
4. Find the angle B in radians and degrees:

(2 marks)



5. A truss has the following dimensions find angle C and D in degrees

(4 marks)



For the solutions click [here](#)

Enter your score /10



## 3 Differentiation

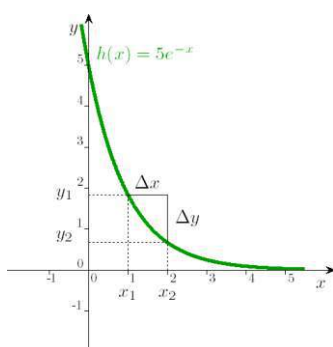
### 3.1 Introduction to Differentiation

Recall that the equation of a straight line is of the form  $y = mx + c$  where  $m$  is the gradient and  $c$  is the intercept. We can find the gradient  $m$  of a straight line using the triangle method with the formula  $m = \frac{\Delta y}{\Delta x}$ .

**Note:** Remember that the gradient is a measure of how steep a graph is at a certain point.

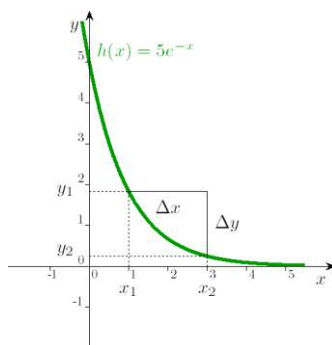
As the gradient of a *straight line* is the same at every point on the line it is easy to find. With a *curve* this is not the case. Consider the graph of  $y = 5e^{-x}$  as shown below. We will use the triangle method with two different triangles and show we get two different values for the gradient, this implies the gradient is not constant.

Using the triangle method with the triangle in the graph below gives:



$$\text{gradient} = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0.68 - 1.85}{3 - 1} = -0.17$$

Now using the triangle method again but this time with a larger triangle as in the graph below:



$$\text{gradient} = \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0.25 - 1.85}{3 - 1} = -0.8$$

According to the triangle method we have two different values for the gradient of the curve. This is because the value of the gradient is not the same at all points on the curve, unlike a straight line. This should be obvious as we can see that the steepness of the graph changes along the curve. This means it is quite hard to find the gradient of non straight line using the triangle method and so we need a new method.



**Differentiation** allows us to find the gradient of a graph. When we differentiate we are calculating how one variable changes due to a change in another (in other words finding the rate of change). This is incredibly useful in Civil Engineering, for example we may want to find out how the compressive strength of a concrete element changes with respect to time.

## Notation

When we use the triangle method we are finding  $\frac{\Delta y}{\Delta x}$  where  $\Delta$  means 'the difference'. Similarly when we differentiate we are effectively drawing an infinitely small triangle to calculate the rate of change (or gradient). This might seem quite an abstract concept and a full explanation lies beyond the scope of this resource where we present the main properties of differentiation. If you would like to see the concept of differentiation from first principles [click here for an excellent 'Maths Tutor' resource](#).

When differentiating we use the notation:

$$\frac{dy}{dx}$$

where the  $d$  stands for an infinitely small difference. (We don't use  $\Delta$  anymore as this represents a larger, not infinitely small, difference).



### Example

The derivative of  $x^2$ , as you will soon learn, is  $2x$ . Using the notation above we would write this as:

$$y = x^2 \Rightarrow \frac{dy}{dx} = 2x$$

Recall how we can write  $y$  or  $f(x)$  this leads to other notation for differentiation displayed in the table below:

| Function | Derivative           | Pronunciation                   |
|----------|----------------------|---------------------------------|
| $y$      | $\frac{dy}{dx}$      | "d-y by d-x" or "d-y d-x"       |
| $f(x)$   | $f'(x)$              | "f prime of x" or "f dash of x" |
| $f(x)$   | $\frac{d}{dx}(f(x))$ | "d by d-x of f of x"            |

**Note:** We may sometimes refer to  $\frac{d}{dx}$  as the differential operator.



### Example

So if  $f(x) = x^2$  then we can write the derivative in all the following ways:

$$f'(x) = 2x \qquad \frac{dy}{dx} = 2x$$

$$\frac{d}{dx}(x^2) = 2x$$



**IMPORTANT:** When we find  $\frac{dy}{dx}$  we say we are 'differentiating  $y$  with respect to  $x$ '. Suppose instead we now we have the equation  $\theta = t^2$  then:

$$\frac{d\theta}{dt} = 2t$$

This works in the same way as we have for  $x$  and  $y$ . In this case we would say we are 'differentiating  $\theta$  with respect to  $t$ '



## 3.2 Differentiating Polynomials

A polynomial is an equation in the form

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

where

- $a_n, \dots, a_2, a_1, a_0$  are constant coefficients
- $x^n, x^{n-1}, \dots, x^2, x$  are powers of a variable  $x$ .

A rather simple example of a polynomial is the quadratic  $x^2 + 3x + 1$ .

As differentiation finds the gradient of a curve at any given point, suppose we want to find the gradient of a function that is a polynomial such as  $y = x^3 + 1$  how do we find this differential?

As differentiation may be new to many readers we will begin slowly and first explain how to differentiate the individual terms of a polynomial, for example  $x^2$ , before moving on in the section 'Differentiating a function with Multiple Terms.'

The general rule is as follows

$$y = x^n \Rightarrow \frac{dy}{dx} = nx^{n-1}$$

Sometimes people remember this rule as 'times by the power then drop the power by one.'

The main points to note are that:

- If  $y = 1$  then  $\frac{dy}{dx} = 0$  (since  $x^0 = 1$ , so  $n = 0$  here). In fact the same can be applied to any constant not just one so:

$$\text{If } y = c \text{ where } c \text{ is any constant} \Rightarrow \frac{dy}{dx} = 0$$

- If we have  $y = x$  then this is the same as  $y = x^1$  so we have  $\frac{dy}{dx} = 1 \times x^{1-1} = 1$
- If we have  $y = x^2$  then we have  $\frac{dy}{dx} = 2 \times x^{2-1} = 2x$ .

We have only considered examples of polynomials up to the power of 2. The next example shows that we can apply this rule to any power.



**Example:** Differentiate  $y = x^5$

**Solution:** From above we know that if  $y = x^n$  then  $\frac{dy}{dx} = nx^{n-1}$ . So in this example we have that  $n = 5$ . This gives the following answer:

$$\frac{dy}{dx} = 5x^{5-1} = 5x^4$$

We can also differentiate using this rule when  $x$  is raised to a negative power as shown in the next example.



**Example:** Differentiate  $y = x^{-1}$

**Solution:** From above we know that if  $y = x^n$  then  $\frac{dy}{dx} = nx^{n-1}$ . So in this example we have that  $n = -1$ . This gives the following answer:

$$\frac{dy}{dx} = -1 \times x^{-1-1} = -x^{-2} \text{ or } = -\frac{1}{x^2}$$







**Example:** Differentiate  $y = x^{\frac{1}{2}} = \sqrt{x}$

*Solution:* From above we know that if  $y = x^n$  then  $\frac{dy}{dx} = nx^{n-1}$ . So in this example we have that  $n = \frac{1}{2}$ . This gives the following answer:

$$\frac{dy}{dx} = \frac{1}{2} \times x^{\frac{1}{2}-1} = \frac{x^{-\frac{1}{2}}}{2} = \frac{1}{2\sqrt{x}}$$

So far we have only considered polynomials which have a coefficient of 1. But how do we differentiate  $y = 4x^2$  where the coefficient is equal to 4? When they have a coefficient, denoted  $a$  for example the rule is:

$$y = ax^n \Rightarrow \frac{dy}{dx} = a \times nx^{n-1}$$

**Note:** The first rule is just a special case of the above where  $a = 1$ .

**IMPORTANT:** When differentiating any function  $f(x)$  multiplied by some constant  $a$ , finding  $\frac{d}{dx}[af(x)]$ , we can take the constant out of the derivative so:

$$\frac{d}{dx}[af(x)] = a \frac{d}{dx}[f(x)]$$

For example  $\frac{d}{dx}(2x^3) = 2 \times \frac{d}{dx}(x^3) = 2 \times 3x^2 = 6x^2$

**Example:** Differentiate  $y = 4x^{-\frac{1}{2}}$

*Solution:* This combines everything we have looked at thus far. So using the rule if  $y = ax^n$  then  $\frac{dy}{dx} = a \times nx^{n-1}$ . So in this example we have  $a = 4$ ;  $n = -\frac{1}{2}$ . This gives the following answer:

$$\frac{dy}{dx} = 4 \times -\frac{1}{2} \times x^{-\frac{1}{2}-1} = -2x^{-\frac{3}{2}}$$



**Engineering Example:** A fluid is moving along a pipe so that the distance travelled along the pipe from the origin is given by the expression  $s = 0.25t^2$ . An engineer believes that the velocity 100m down the pipe is too high. Find that velocity.

*Hint:* Velocity, by its definition is the rate of change of distance with respect to time:

$$v = \frac{ds}{dt}$$

*Solution:* Here we are given an expression relating  $s$ , the distance, and  $t$ , the time, for the fluid moving along the pipe. As stated in the question we need to differentiate our expression for distance to give us another expression, but this time for the velocity.

$$s = 0.25t^2 \text{ so } v = \frac{ds}{dt} = 2 \times 0.25t^{2-1} = 0.5t^1$$

Now we have a relationship between  $v$ , the velocity and  $t$  the time.

We want to find the velocity 100m down the pipe, first we need to find the time when this occurs. Using our initial expression:

$$s = 100 = 0.25t^2 \therefore t^2 = \frac{100}{0.25} = 400 \therefore t = 20 \text{ secs}$$

Now we can substitute our time into our derivative to find the velocity:

$$v = \frac{ds}{dt} = 0.5t = 0.5(20) = \mathbf{10m/s}$$



[Click here for a video tutorial](#)



[Water Pipes, Ladybower Reservoir](#), © Mike Garratt [CC BY-SA](#)



### 3.3 Differentiating a Function with Multiple Terms

If we have multiple terms in the form of a sum for example  $y = x^3 + 8x + 1$  then in order to find  $\frac{dy}{dx}$  we apply the differential operator on each of the terms in turn. So in other words we differentiate each part of the sum one by one.

Consider,  $y = x^3 + 8x + 1$  If we want to find  $\frac{dy}{dx}$  then we would apply the operator to both sides of the equation giving us:

$$\frac{dy}{dx} = \frac{d}{dx}(x^3 + 8x + 1) \quad \text{Now we apply the } \frac{d}{dx} \text{ operator to each term.}$$

$$\frac{dy}{dx} = \frac{d}{dx}(x^3) + \frac{d}{dx}(8x) + \frac{d}{dx}(1) \quad \text{Now we differentiate each of terms individually.}$$

$$\frac{dy}{dx} = 3x^2 + 8x + 0$$

**Note:** These rules apply to all types of operators, whether they are exponentials logarithms or trigonometric functions (which we will see shortly), if the expression is a sum you can always split it up into and differentiate each constituent in term.



**Example:** Differentiate the following:

1.  $y = 4x^6 - 6x^3 + 3$
2.  $y = 10x^{-3} + 4x^{-\frac{1}{3}} + 2x^2 + 3x$

**Solution**

- 1 To differentiate  $y$  we simply differentiate all the terms individually, you should take special care of whether the term is positive or negative:

$$\frac{d}{dx}(4x^6 - 6x^3 + 3) = \frac{dy}{dx} = 24x^5 - 18x^2$$

- 2 The same rule applies here again just be careful with the negative powers and fractional powers:

$$\frac{d}{dx}\left(10x^{-3} + 4x^{-\frac{1}{3}} + 2x^2 + 3x\right) = \frac{dy}{dx} = -30x^{-4} + 4\left(-\frac{1}{3}\right)x^{-\frac{4}{3}} + 4x + 3$$

Simplifying gives:

$$\frac{dy}{dx} = 4x + 3 - \frac{4(x)^{-\frac{4}{3}}}{3} - 30x^{-4}$$



**Engineering Example:** The moment along a portion of a 10m steel beam is given by the equation  $M(x) = 100x - \frac{20(x)^2}{2}$  where  $x$  is the distance, in metres, along the beam and  $M$  is the moment in kNm. Given that the shear force  $V$ , is equal to  $\frac{dM}{dx}$  and that the maximum moment occurs at  $V = 0$  find where the maximum moment occurs and its magnitude.

*Solution:*

First we need to find the derivative of our expression for the moment. (This is equal to the shear force) :

$$V = \frac{dM}{dx} = 100 - \frac{40(x)^1}{2} = 100 - 20x$$

The moment is max when:

$$V = 0 \text{ so } \frac{dM}{dx} = 0 \therefore 100 - 20x = 0 \Rightarrow 20x = 100 \Rightarrow x = \mathbf{5m}$$

So the maximum moment occurs at 5m. Putting this back into our expression for the moment:

$$M(5) = 100(5) - \frac{20 \times 5^2}{2} = \mathbf{250kNm}$$



### 3.4 Differentiating Trigonometric Functions

Below we have the rules for differentiating trigonometric functions:

$$y = \sin(x) \Rightarrow \frac{dy}{dx} = \cos(x)$$

$$y = \cos(x) \Rightarrow \frac{dy}{dx} = -\sin(x)$$

When we have constants multiplying the trigonometric functions and coefficients multiplying our  $x$  we can use the following rules:

$$y = a \sin(bx) \Rightarrow \frac{dy}{dx} = a \times b \times \cos(bx)$$

$$y = a \cos(bx) \Rightarrow \frac{dy}{dx} = -a \times b \times \sin(bx)$$

**Note:** These rules only work when  $x$  is being measured in radians.



**Example:** Differentiate the following:

1.  $y = \sin(9x)$
2.  $y = 4 \cos\left(\frac{x}{2}\right)$
3.  $y = -2 \cos(-7x)$

**Solution:**

1 Using the rule:

$$\text{If } y = a \sin(bx) \text{ then } \frac{dy}{dx} = a \times b \times \cos(bx)$$

$$a = 1 \text{ and } b = 9. \text{ So } \frac{dy}{dx} = 9\cos(9x)$$

2 Using the rule:

$$\text{If } y = a \cos(bx) \text{ then } \frac{dy}{dx} = -a \times b \times \sin(bx)$$

$$a = 4 \text{ and } b = \frac{1}{2}. \text{ So } \frac{dy}{dx} = -4 \times \frac{1}{2} \times \sin\left(\frac{x}{2}\right) = -2 \sin\left(\frac{x}{2}\right)$$

3 Using the rule:

$$\text{If } y = a \cos(bx) \text{ then } \frac{dy}{dx} = -a \times b \times \sin(bx)$$

$$a = -2 \text{ and } b = -7. \text{ So } \frac{dy}{dx} = -(-2) \times (-7 \sin(-7x)) = -14 \sin(-7x)$$



## 3.5 Differentiating Exponential and Logarithmic Functions

### Differentiating Exponentials

The exponential function,  $e^x$ , is special in that when differentiated it gives itself. This is shown in the rule below:

$$y = e^x \Rightarrow \frac{dy}{dx} = e^x$$

**Note:**  $e^x$  may also be written as  $\exp(x)$

We can also differentiate when the ' $x$ ' has a coefficient e.g.  $y = e^{4x}$  and also when the exponential is multiplied by a constant e.g.  $y = 3e^x$ . We can combine these rules into the rules below:

$$y = ae^{bx} \Rightarrow \frac{dy}{dx} = a \times b \times e^{bx}$$



**Example:** Differentiate the following:

1.  $y = e^{2x}$
2.  $y = 8 \exp\left(-\frac{x}{3}\right)$
3.  $y = 3e^{-3x} - \cos(2x)$

**Solution:**

1 Using the rule:

$$\text{If } y = ae^{bx} \text{ then } \frac{dy}{dx} = a \times b \times e^{bx}$$

$$a = 1 \quad b = 2 \text{ hence } \frac{dy}{dx} = 2e^{2x}$$

2 Using the rule:

$$\text{If } y = ae^{bx} \text{ then } \frac{dy}{dx} = a \times b \times e^{bx}$$

$$a = 8 \quad b = -\frac{1}{3} \text{ hence } \frac{dy}{dx} = 8 \times \left(-\frac{1}{3}\right) \times \exp\left(-\frac{x}{3}\right) = -\frac{8}{3} \exp\left(-\frac{x}{3}\right) = -\frac{8}{3} e^{-x/3}$$

3 Using the rules:

$$\text{If } y = ae^{bx} \text{ then } \frac{dy}{dx} = a \times b \times e^{bx} \quad \text{and} \quad \text{If } y = a \cos(bx) \text{ then } \frac{dy}{dx} = -a \times b \times \sin(bx)$$

$$\begin{aligned} \frac{dy}{dx} &= 3 \times (-3) \times e^{-3x} - (-2) \times \sin(2x) \\ &= -9e^{-3x} + 4\sin(2x) \end{aligned}$$



## Differentiating Logarithms

We differentiate logarithmic functions using the rule shown below:

$$\text{If } y = \ln(x) \text{ then } \frac{dy}{dx} = \frac{1}{x}$$

We can differentiate when the 'x' has a coefficient for example  $y = \ln(4x)$  and also when the logarithmic function has a coefficient for example  $y = 3 \ln(x)$ . We combine these into the rule below:

$$y = a \times \ln(bx) \Rightarrow \frac{dy}{dx} = \frac{a}{x}$$

**Note:** In the above rule the 'b' disappears and it forms a nice exercise in the practice of using the laws of logarithms hence we have included the algebraic proof in the next example. This rule does not apply if the equation is not of the form above e.g  $y = \ln(3x^2 + 2x + 1)$  you will be shown how to differentiate this later on.



**Example:** For  $y = a \times \ln(bx)$  find  $\frac{dy}{dx}$

**Solution:** Using laws of logarithms we can rewrite  $y = a \ln(bx)$  as:  
 $y = a(\ln(x) + \ln(b))$

Now differentiating this as a sum gives:

$$\frac{dy}{dx} = \frac{d}{dx}(a(\ln(x) + \ln(b))) = a \frac{d}{dx}[\ln(x)] + a \frac{d}{dx}[\ln(b)]$$

Now as  $\frac{d}{dx}[\ln(x)] = \frac{1}{x}$  and  $\frac{d}{dx}[\ln(b)] = 0$  since  $\ln(b)$  is a constant we get:

$$\frac{dy}{dx} = a \times \frac{1}{x} = \frac{a}{x}$$



**Example:** Differentiate:

1.  $y = \ln(3x)$
2.  $y = -\frac{4}{5} \ln\left(\frac{x}{2}\right)$

**Solution:**

1 Using the rule:

$$\text{If } y = a \times \ln(bx) \text{ then } \frac{dy}{dx} = \frac{a}{x}$$

$$a = 1 \text{ and } b = 3 \text{ so } \frac{dy}{dx} = \frac{1}{x}$$

2 Using the rule:

$$\text{If } y = a \times \ln(bx) \text{ then } \frac{dy}{dx} = \frac{a}{x}$$

$$a = -\frac{4}{5}; b = \frac{1}{2} \quad \therefore \frac{dy}{dx} = -\frac{4}{5} \times \frac{1}{x} = -\frac{4}{5x}$$





**Example:** Differentiate the following expressions:

3.  $y = 3 \ln\left(\frac{x}{2}\right) + e^{3x}$

4.  $P = 2 \ln(2\theta) - \frac{1}{2} \cos(2\theta) + e^{2\theta} + 3(\theta)^2 + 6$

*Solution:*

3. Using the rules above:

$$\frac{dy}{dx} = 3 \frac{d}{dx} \left[ \ln\left(\frac{x}{2}\right) \right] + \frac{d}{dx} [e^{3x}]$$

$$\frac{dy}{dx} = \frac{3}{x} + 3e^{3x}$$

4 Using all the techniques we have learnt so far:

$$\frac{dP}{d\theta} = \frac{d}{d\theta} (2 \ln(2\theta) - \frac{1}{2} \cos(2\theta) + e^{2\theta} + 3(\theta)^2 + 6)$$

Differentiating each term individually we get:

$$\begin{aligned} \frac{dP}{d\theta} &= \frac{2}{\theta} - \frac{1}{2} \times 2 \times (-\sin 2\theta) + 2e^{2\theta} + 6\theta + 0 \\ \frac{dP}{d\theta} &= \frac{2}{\theta} + \sin 2\theta + 2e^{2\theta} + 6\theta \end{aligned}$$



**Test Yourself – Differentiation (1):** Differentiate the following functions:

1.  $y = 2x^2 + 3x - 10$  (2 marks)

2.  $y = 12x^3 - \frac{3}{x^2} + 6x$  (3 marks)

3.  $y = \frac{9}{x^4}$  (2 marks)

4.  $T = 2 \ln(12k)$  (2 marks)

5.  $p(r) = \frac{4}{3} \pi r^3 - 2e^{2r}$  (3 marks)

6.  $y = 12x^5 + 6x^{-3} - \sin\left(-\frac{1}{3}x\right) + 8 \ln(4x) - \frac{3}{e^{3x}}$  (4 marks)

For the solutions click [here](#)

Enter mark / 16





### 3.6 Chain Rule

When we have one function inside another function (this is called a composite function) we differentiate using the chain rule. An example of a composite function is  $y = \sin(x^2 + 1)$ . There are two functions:  $\sin(\ )$  and  $x^2 + 1$ . We are applying the  $\sin$  function to the ' $x^2 + 1$ ', thus making it a function inside another function or a composite function.

The general form of a composite function is:

$$y = f(g(x))$$

where  $f$  and  $g$  are both functions. In the above example  $y = \sin(x^2 + 1)$  we would have  $f(g(x)) = \sin(g(x))$  and where  $g(x) = x^2 + 1$

$$y = \sin(x^2 + 1)$$

Outer function  $f(\cdot)$       Inner function  $g(x)$

To use the chain rule we use the following steps:

1. Introduce a new variable  $u$  to be equal to  $u = g(x)$
2. Substitute  $u = g(x)$  into the expression  $y = f(g(x))$  so that  $y = f(u)$
3. Find  $\frac{dy}{du}$  and  $\frac{du}{dx}$
4. The derivative  $\frac{dy}{dx}$  can be found by the equation:  $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

**Note:** Check to see if the  $du$  would cancel back to  $\frac{dy}{dx}$  to make sure you have the correct chain-rule equation e.g  $\frac{dy}{dx} = \frac{dy}{\cancel{du}} \times \frac{\cancel{du}}{dx} = \frac{dy}{dx}$

5. Then substitute  $u = g(x)$  back in to get your function in terms of what it was written in.

The chain rule is summarised in the box below:

$$y = f(g(x)) \quad \text{let } u = g(x) \quad \Rightarrow \quad y = f(g(x)) = f(u)$$
$$\therefore \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Below we follow the steps of the chain rule to differentiate  $y = \sin(x^2 + 1)$ :

1. Introduce a new variable, let's call it  $u$ , to be equal to  $g(x)$  (the inside function, in this case  $u = x^2 + 1$ )
2. Substitute  $u = g(x)$  into the expression  $y = f(g(x))$  so now we have  $y = f(u)$  therefore  $y = \sin(u)$



3.  $y = \sin(u)$  and  $u = x^2 + 1$  are now both expressions we should be familiar with differentiating so we should be able to find  $\frac{dy}{du}$  and  $\frac{du}{dx}$ :

$$y = \sin(u) \Rightarrow \frac{dy}{du} = \cos(u)$$

and

$$u = x^2 + 1 \Rightarrow \frac{du}{dx} = 2x$$

4. The derivative we want to find  $\frac{dy}{dx}$  can now be found by the equation  $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$$\frac{dy}{dx} = \cos(u) \times 2x$$

5. We earlier said let  $u = x^2 + 1$  so we can now substitute back in for  $u = g(x)$

$$\frac{dy}{dx} = 2x \cos(x^2 + 1)$$



**Example:** Differentiate  $y = (x^2 + 2)^3$  using the chain rule.

*Solution:*

Take  $u = x^2 + 2$  as this is the 'inside function'. (The outside function is the cubing)

$$\therefore y = f(g(x)) = f(u) = u^3$$

Then we find the derivative of each:

$$\frac{du}{dx} = 2x \quad \frac{dy}{du} = 3u^2$$

From the chain rule:  $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

So substitute our values for  $\frac{dy}{du}$  and  $\frac{du}{dx}$

$$\frac{dy}{dx} = 3u^2 \times 2x$$

Now substitute  $u$  back in

$$\frac{dy}{dx} = 3(x^2 + 2)^2 \times 2x$$

$$\frac{dy}{dx} = 6x(x^2 + 2)^2$$

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**Example:** Differentiate  $y = \cos(e^{2x})$

*Solution:*

Take  $u = e^{2x}$  and therefore  $y = \cos(u)$ . Differentiate to find:

$$\frac{du}{dx} = 2e^{2x} \quad \frac{dy}{du} = -\sin(u)$$

From the chain rule:  $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$$\frac{dy}{dx} = 2e^{2x} \times (-\sin(u))$$

Substitute  $u$  back in

$$\frac{dy}{dx} = -2e^{2x} \sin(e^{2x})$$





**Example:** Differentiate  $y = \ln(x^3 + x)$

{2ftzúú2y}

Take  $u = x^3 + x$  as this is the 'inside function'. (The outside function is the  $\ln$ )

$$\therefore y = f(g(x)) = f(u) = \ln(u)$$

Then we find the derivative of each:

$$\frac{du}{dx} = 3x^2 + 1 \quad \frac{dy}{du} = \frac{1}{u}$$

From the chain rule:  $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

So substitute our values for  $\frac{dy}{du}$  and  $\frac{du}{dx}$

$$\frac{dy}{dx} = \frac{1}{u} \times (3x^2 + 1)$$

Now substitute  $u$  back in

$$\frac{dy}{dx} = \frac{1}{x^3 + x} \times (3x^2 + 1)$$

$$\frac{dy}{dx} = \frac{3x^2 + 1}{x^3 + x}$$

**Engineering Example:** A large water tank is emptying with water. The volume of water leaving the tank is modelled by the equation:  $\frac{dV}{dt} = 3e^{-(2.5 \times 10^{-3})t} \text{ m}^3/\text{s}$ . An engineer wants to know how the height of water changes with time. The radius of the water tank is 8m



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Using the chain rule, find an expression in  $t$  for the rate at which the water height changes.

{2ftzúú2y}

While this question appears challenging it is really just a different way of applying the chain rule. We are asked to find the  $\frac{dh}{dt}$  at which the height of the water is changing with time i.e we want to find  $\frac{dh}{dt}$ .

We also know that the volume of the reservoir is  $V = \pi r^2 h$  (it's a cylinder) which means we can find  $\frac{dV}{dh}$

Using the chain rule:  $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$

Therefore we need to find  $\frac{dV}{dh}$  and  $\frac{dV}{dt}$

$$\frac{dV}{dh} = \pi r^2 = \pi \times 8^2 = 64\pi$$

(Note:  $\pi$  and  $r$  are just constants)

$$\frac{dV}{dt} = 3e^{-(2.5 \times 10^{-3})t}$$

$$\therefore \frac{dh}{dt} = \frac{dV}{dt} \div \frac{dV}{dh} = \frac{3e^{-(2.5 \times 10^{-3})t}}{64\pi} = 0.0149e^{-(2.5 \times 10^{-3})t}$$

Find the rate at which water height is changing 10 seconds after emptying:

{2ftzúú2y}

To find out the rate at which the volume is changing after 8 seconds we simply need to substitute  $t=10$  into our equation for  $\frac{dh}{dt}$

$$\frac{dh}{dt} = 0.0149e^{-(2.5 \times 10^{-3})10} = 0.0145 \text{ ms}^{-1} = 1.45 \text{ cms}^{-1}$$



## 3.7 Product Rule

Suppose we want to differentiate  $y = x^2 \sin(x)$  then we encounter the problem: how do we differentiate a function that is the product of two other functions, in this case  $x^2$  and  $\sin(x)$

To do this we need the product rule:

**If the function  $y = f(x)$  is written as the product of two functions say  $u(x)$  and  $v(x)$  so:**

$$y = f(x) = u(x)v(x) \text{ then } \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Returning to the example we have  $f(x) = x^2 \sin(x)$  with  $u = x^2$  and  $v = \sin(x)$

First we find that  $\frac{du}{dx} = 2x$  and  $\frac{dv}{dx} = \cos(x)$  then we substitute into  $\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

$$\Rightarrow \frac{dy}{dx} = x^2 \cos(x) + \sin(x) 2x$$

$$\Rightarrow \frac{dy}{dx} = x(x \cos(x) + 2 \sin(x))$$

Sometimes, the rule is better remembered by saying 'differentiate the first function and leave the second function, then differentiate the second function and leave the first function then add them together'

**Note:** The product rule still applies even when the function is the product of more than two functions – this will be seen later in an example.



**Example:** Differentiate using the product rule:

1.  $y = x \ln(x)$



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2.  $y = 6e^x \cos(x)$



[Click here for a video tutorial](#)

*{2ftzú12y}*

1. Take  $u = x$  and  $v = \ln(x)$  we can now differentiate these to find that:

$$\frac{du}{dx} = 1 \text{ and } \frac{dv}{dx} = \frac{1}{x}$$

Using the product rule:

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{dy}{dx} = x \left( \frac{1}{x} \right) + \ln(x) (1) \quad \therefore \quad \frac{dy}{dx} = 1 + \ln(x)$$

2. For this take  $u = 6e^x$  and  $v = \cos(x)$  again we can differentiate to find that:

$$\frac{du}{dx} = 6e^x \text{ and } \frac{dv}{dx} = -\sin(x)$$

Using the product rule:

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{dy}{dx} = 6e^x (-\sin(x)) + \cos(x) (6e^x)$$

$$\frac{dy}{dx} = 6e^x (\cos(x) - \sin(x))$$





**Example:** This next example requires you to use the product rule three times.

1. Differentiate  $y = \frac{1}{3}(x^2 + 5x) \sin(x) (e^{2x})$



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*Solution*

Set  $u = \frac{1}{3}(x^2 + 5x)$        $v = \sin(x)$        $w = e^{2x}$

We can differentiate each individual function to find:

$$\frac{du}{dx} = \frac{1}{3}(2x + 5) \quad \frac{dv}{dx} = \cos(x) \quad \text{and} \quad \frac{dw}{dx} = 2e^{2x}$$

Using the product rule, but for three functions, we get:

$$\frac{dy}{dx} = uv \frac{dw}{dx} + uw \frac{dv}{dx} + vw \frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{1}{3}(x^2 + 5) \times \sin(x) \times 2e^{2x} + \frac{1}{3}(x^2 + 5x) \times (e^{2x}) \times \cos(x) + \sin(x) \times 2e^{2x} \times \left(\frac{1}{3}(2x + 5)\right)$$

Taking out common factors, we get:

$$\frac{dy}{dx} = \frac{1}{3}e^{2x}(2(x^2 + 5) \sin(x) + (x^2 + 5x) \cos(x) + 2(2x + 5) \sin(x))$$



### 3.8 Quotient Rule

A quotient looks like a fraction with one function being divided by another function for example:

$$y = \frac{x^3}{e^{3x}}$$

When we have a function in the form of a quotient we can differentiate it using the quotient rule:

If the function  $y = f(x)$  is written as a quotient so:

$$y = \frac{u(x)}{v(x)} \Rightarrow \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Returning to  $y = \frac{x^3}{e^{3x}}$  we have  $u = x^3$  and  $v = e^{3x}$

Hence:  $\frac{du}{dx} = 3x^2$  and  $\frac{dv}{dx} = 3e^{3x}$  and  $v^2 = e^{6x}$ . Then using the quotient rule  $\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

$$\Rightarrow \frac{dy}{dx} = \frac{e^{3x} 3x^2 - x^3 3e^{3x}}{e^{6x}}$$

$$\frac{dy}{dx} = 3x^2 e^{-3x} (1 - x)$$



**Example:** Differentiate  $y = \frac{\sin(x)}{3 \ln(x)}$

**Solution:** Just like the previous example we have  $u$  to be the numerator and  $v$  to be the denominator of the fraction so  $u = \sin(x)$  and  $v = 3 \ln(x)$ . Differentiating these we get:

$$\frac{du}{dx} = \cos(x) \text{ and } \frac{dv}{dx} = \frac{3}{x}$$

Using the quotient rule we get:

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \Rightarrow \frac{3 \ln(x) \times \cos(x) - \sin(x) \times \frac{3}{x}}{(3 \ln(x))^2} \\ &= \frac{3 \ln(x) \times \cos(x) - \frac{3 \sin(x)}{x}}{9 \ln(x)^2} \\ &= \frac{\ln(x) \cos(x) - \frac{\sin(x)}{x}}{3 \ln(x)^2} \end{aligned}$$

Simplifying gives

Taking out a factor of 3



**Note:** For all quotients you can also find the derivative by using the product rule, since

$y = \frac{u(x)}{v(x)}$  is the same as  $y = u(x) \times (v(x))^{-1}$ . For example the quotient  $y = \frac{e^{2x}}{6(x)}$  is identical to the product  $y = \frac{1}{6} (e^{2x} \times x^{-1})$ . Which method you decide to implement is entirely down to preference. The next example will be differentiated by using both methods to prove how we achieve the same answer regardless of the method used.





**Example:** Differentiate  $y = \frac{x}{e^{2x}}$

### Using Quotient Rule

*Solution:* Just like the previous example we have  $u$  to be the numerator and  $v$  to be the denominator of the fraction so  $u = x$  and  $v = e^{2x}$ . Differentiating these we get:

$$\frac{du}{dx} = 1 \text{ and } \frac{dv}{dx} = 2e^{2x}$$

Using the quotient rule we get:

$$\begin{aligned} \frac{dy}{dx} &= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} \Rightarrow \frac{e^{2x} \times 1 - x \times 2e^{2x}}{(e^{2x})^2} && \text{Simplifying by taking a factor of } e^{2x} \text{ out} \\ &= \frac{e^{2x}(1-2x)}{(e^{2x})^2} && \text{The } e^{2x} \text{ on top cancels with one on bottom} \\ &= e^{-2x}(1-2x) \end{aligned}$$

### Using Product Rule

*Solution:*

Rearrange the equation so it looks like a product so  $y = x \times e^{-2x}$

Take  $u = x$  and  $v = e^{-2x}$  we can now differentiate these to find that:

$$\frac{du}{dx} = 1 \text{ and } \frac{dv}{dx} = -2e^{-2x}$$

Using the product rule:

$$\begin{aligned} \frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} \Rightarrow x(-2e^{-2x}) + e^{-2x}(1) && \text{Take } e^{-2x} \text{ out as a factor} \\ \frac{dy}{dx} &= e^{-2x}(1-2x) \end{aligned}$$

Which is the same answer achieved using the Quotient Rule



**Test Yourself – Differentiation(2):** Differentiate the following functions: Hint: You may need to use a combination of more than one rule

1.  $y = \cos(3x^3 + 2x^2)$  (2)

2.  $y = \ln(x) \times 4x^2$  (3)

3.  $f(x) = \frac{4e^{4x} + 4x^3}{7x^2}$  (4)

4.  $P(t) = \ln(t^3 + 3t + t) \times \sin(t)$  (3)

5.  $y = \sin(x)(x^3 + 6)e^{-x}$  (5)

(Bonus) :  $y = 7e^{-(3x^2+2x+1)} \sin(x^2 + 6)x^3$

For the solutions click [here](#)

Enter Mark /17



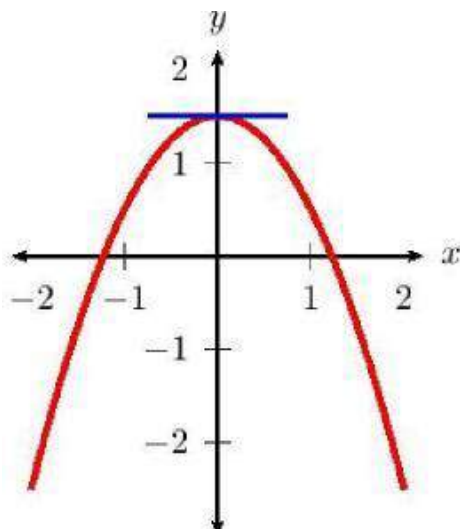
### 3.9 Stationary Points

Stationary points occur when we have  $\frac{dy}{dx} = 0$ ; this represents when the gradient of a curve is horizontal, i.e. the gradient is zero.

**Note:** remember the derivative of a function is also the gradient function so it can be used to find where stationary points exist.

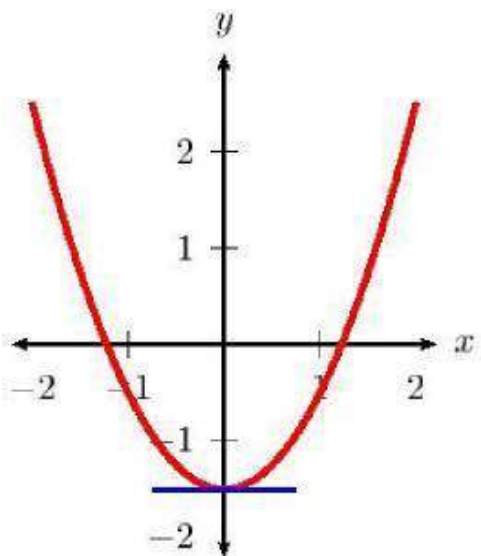
There are three types of stationary points:

1. Local maximum



We can see that when we pass through a maximum at the gradient goes from being **positive** ( $f'(x) > 0$  or  $\frac{dy}{dx} > 0$ ) to being **negative** ( $f'(x) < 0$  or  $\frac{dy}{dx} < 0$ )

2. Local minimum

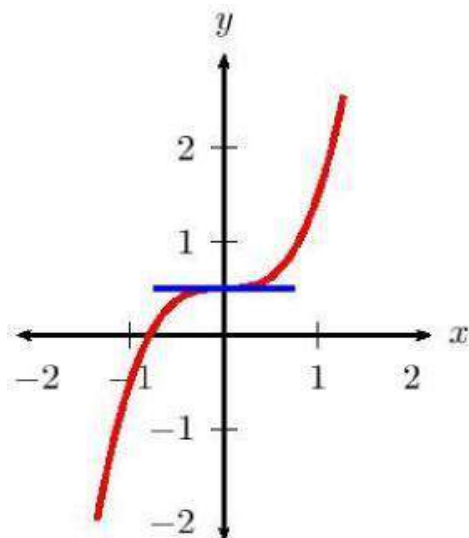


We can see that when we pass through a minimum the gradient goes from being **negative** i.e.  $f'(x) < 0$  or  $\frac{dy}{dx} < 0$  to being **positive** ( $f'(x) > 0$  or  $\frac{dy}{dx} > 0$ )





### 3. Point of inflection

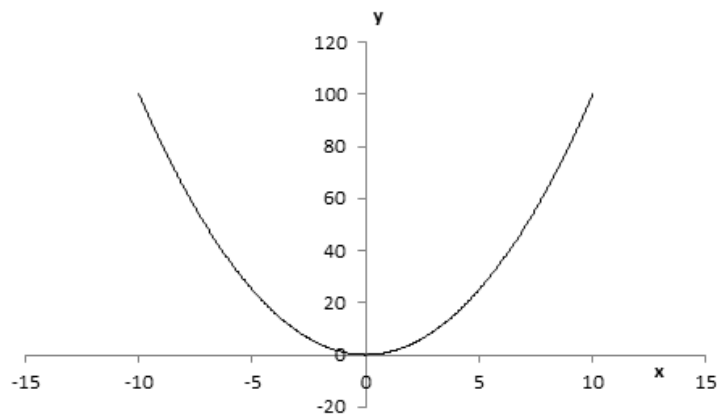


Finally when we pass through a point of inflection we should note that the gradient remains of the same sign (so either positive both sides or negative both sides).



**Example:** find the stationary point of the graph  $y = x^2$

*Solution:* From the graph of  $y = x^2$  we can see that there is a minimum at  $x = 0$ , however we need to find this algebraically.



First we need to find when  $\frac{dy}{dx} = 0$ . Now we should be able to differentiate  $x^2$  with relative ease now so:

$$\frac{dy}{dx} = 2x \therefore 2x = 0$$

$$x = 0$$

So we know we have a stationary point at  $x = 0$  in the next section we will discuss how to decide whether this is a maximum, minimum or point of inflection.



## Classifying Stationary Points

We now need to introduce the second derivative denoted:

$$\frac{d^2y}{dx^2} \text{ or } f''(x)$$

This means that we differentiate a function twice so for example if we have  $y = x^3$  we would first find:

$$\frac{dy}{dx} = 3x^2$$

which we would then differentiate again to find:

$$\frac{d^2y}{dx^2} = 6x$$

### First Test

The second derivative is used to decide whether a stationary point is a maximum or a minimum. If we have the value of  $x$  for a stationary point then we substitute this value into the second derivative, if:

- $\frac{d^2y}{dx^2} > 0$  (so positive) then we have a **minimum**
- $\frac{d^2y}{dx^2} < 0$  (so negative) then we have a **maximum**
- $\frac{d^2y}{dx^2} = 0$  - Here the **test fails**. In this cases we could have a maximum, minimum, or a point of inflection and we must carry out the following further test in order to decide

### Second Test

First find two  $x$  values just to the left and right of the stationary point and calculate  $f'(x)$  for each. Let us denote the  $x$  value just to the left as  $x_L$  and the one to the right as  $x_R$  then if:

- $f'(x_L) > 0$  and  $f'(x_R) < 0$  then we have a **maximum**
- $f'(x_L) < 0$  and  $f'(x_R) > 0$  then we have a **minimum**
- If  $f'(x_L)$  and  $f'(x_R)$  have the same sign ( so both are either negative or positive) then we have a point of inflection.





**Example:** Find and classify the stationary points of  $y = \frac{x^4}{4} - \frac{x^2}{2}$ .



[Click here for a video tutorial](#)

*Solution:* First we find when  $\frac{dy}{dx} = 0$  to find where the stationary points are

$$\frac{dy}{dx} = x^3 - x = 0$$

$$\Rightarrow x(x^2 - 1) = 0$$

$$\Rightarrow x(x - 1)(x + 1) = 0$$

Therefore we have stationary points at  $x = 0, x = 1$  and  $x = -1$  to classify them we must first find the second derivative:

$$\frac{d^2y}{dx^2} = 3x^2 - 1$$

Now we substitute our  $x$  - values for the location of the stationary point into  $\frac{d^2y}{dx^2}$ .

- When  $x = 0$ ,  $\frac{d^2y}{dx^2} = -1 \therefore \frac{d^2y}{dx^2} < 0$  (negative) so this is a **maximum**
- When  $x = 1$ ,  $\frac{d^2y}{dx^2} = 2 \therefore \frac{d^2y}{dx^2} > 0$  (positive) so this is a **minimum**
- When  $x = -1$ ,  $\frac{d^2y}{dx^2} = 2 \therefore \frac{d^2y}{dx^2} > 0$  (positive) so this is a **minimum**



**Example:** Find and classify the stationary points of  $f(x) = x^3$ .

*Solution:* First we find when  $\frac{dy}{dx} = 0$  to find where the stationary points are :

$$\frac{dy}{dx} = 3x^2 = 0$$

Therefore we have one stationary point at  $x = 0$ . To classify this point we shall first try the first test, so must find the second derivative:

$$\frac{d^2y}{dx^2} = 6x$$

Now when  $x = 0$  we have  $\frac{d^2y}{dx^2} = 0$  hence the test has failed and we must carry out further tests.

Using the second test we need to pick  $x$  values close to either side of the stationary point at  $x = 0$  so appropriate values may be  $x_L = -0.5$  and  $x_R = 0.5$ .

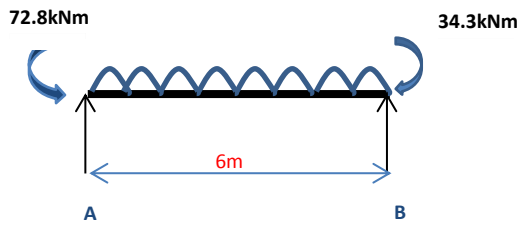
As  $\frac{dy}{dx} = f'(x) = 3x^2$  we have that:

- $f'(x_L) = 3(x_L)^2 = 3 \times (-0.5)^2 = 0.75 > 0$
- $f'(x_R) = 3(x_R)^2 = 3 \times 0.5^2 = 0.75 > 0$

So our gradient before the stationary point (at  $x = -0.5$ ) is positive – at the stationary point it is 0, and then after the stationary point (at  $x = +0.5$ ) it is positive again, this means that the stationary point is a **point of inflection**.



**Engineering Example:** A section of the following beam is found to have the following parameters:



An equation for the bending moment  $x$  metres along the beam from A can be found to be:

$$M(x) = 10x^2 - 66.42x + 72.8$$

Given the loading,  $w$ , is the *second derivative* of moment with respect to  $x$ , find:

1. The position of maximum moment
2. The magnitude of maximum moment
3. The loading on the beam

*Solution:*

1. Since we are looking for the position of the 'maximum magnitude' we are looking for a maximum or minimum – so a turning point. Therefore we need to find the differential:

$$\frac{dM}{dx} = 20x - 66.42 = 0 \text{ (for a stationary point)}$$

$$20x = 66.42 \Rightarrow x = 3.32\text{m (from A)}$$

2. To find what this maximum equals we can put it back into our equation for  $M(x)$

$$\begin{aligned} M(3.32) &= 10(3.32)^2 - 66.42(3.32) + 72.8 \\ &= -37.49\text{kNm} \end{aligned}$$

3. We are told that the loading,  $w$ , is equal to the second derivative of the moment i.e.  $\frac{d^2M}{dx^2}$

$$\begin{aligned} \frac{d^2M}{dx^2} &= \frac{d}{dx} \left( \frac{dM}{dx} \right) = \frac{d}{dx} (20x - 66.42) \\ &= 20 \end{aligned}$$

**So the loading is 20kN/m**



## 4 Integration

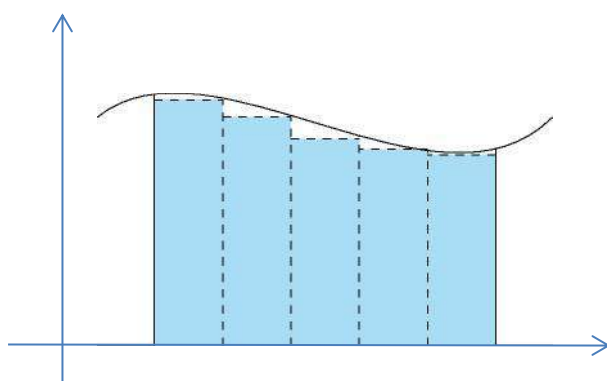
### 4.1 Introduction to Integration

Integration is the opposite of differentiation so principally it helps us find the answer to the question:

'Suppose we have  $\frac{dy}{dx} = 2x$  then what is  $y$ ?'

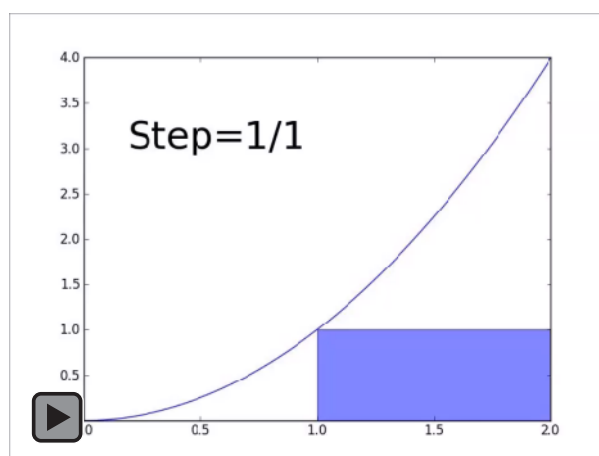
From the previous section we know the answer will be  $x^2 + C$  where  $C$  is a constant that could be anything. So we would say that  $2x$  integrated is:  $x^2 + C$ . This is known as symbolic integration as you are not integrating between limits (you will see this later)

**Note:** We have an infinite number of solutions to the above question as the constant  $C$  could be any number. Remember, any constant on its own differentiated gives 0. Therefore when we integrate we must always account for the possibility that the function has a constant.



Graphically we know differentiation gives us the gradient of a curve. Integration finds the area beneath a curve.

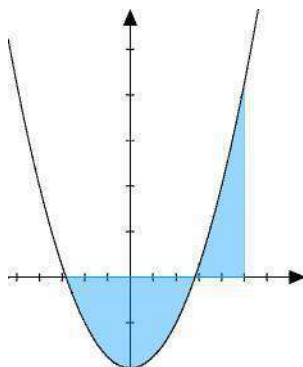
Suppose we wish to find the area under the curve between the  $x$  values  $a$  and  $b$  shaded in the graph above. We could find an approximate answer by creating the dashed rectangles above and summing their area. Now we can see this does not give a perfect answer however the smaller the width of the rectangles the lower the error in our answer would be.



[Riemann Integration and Darboux Lower Sums](#) © IkamusumeFan, [CC BY-SA 3.0](#)



So if we could use rectangles of an infinitely small width we would find an answer with no error. **This is what integration does.** It creates an infinite number of rectangles under the curve all with an infinitely small width and sums their area to find the area under the curve.



So integration could be used to find the area shaded on the graph of  $y = x^2 - 2$  above. Note that when the curve is below the x-axis integration finds the area bounded between the curve and the x-axis.

**Note:** If the area we are finding is below the x-axis then we get a negative answer when we integrate.

## Notation

We use the fact that  $2x$  integrated is  $x^2 + C$  to help introduce the following notation for integration. If we want to integrate  $2x$  we write this as:

$$\int 2x \, dx = x^2 + C$$

- The  $\int$  tells us we need to integrate.
- The  $dx$  tells us to integrate with respect to  $x$ . So we could write  $\int 2z \, dz = z^2 + C$  but in this case we have integrated with respect to  $z$ .
- We call  $\int 2x \, dx = x^2 + C$  the integral.

Another notation *sometimes* used is the following: if we have the function  $f(x)$  then we write this integrated as  $F(x)$ . So if  $f(x) = 2x$  then  $F(x) = x^2 + C$ .

Integration finds the area under the curve. So if we want to find the area between two points (called limits) on the  $x$ -axis, say  $a$  and  $b$ , we use the following notation:

$$\int_a^b f(x) \, dx$$

This tells us we want to find the integral of  $f(x)$  and then use that information to find the area under the curve between  $a$  and  $b$ . We find this area by using the formula below:

$$\int_a^b f(x) \, dx = F(b) - F(a)$$



**Note:** This is explained in detail in the section on definite integrals.

- When we are given limits then we have a definite integral.
- When we are not given limits then we have an indefinite integral.

### Rules for Integrals

1. We can split integrals that have multiple terms in them. For example:

$$\int (x^3 + 2x + 7)dx = \int x^3 dx + \int 2x dx + \int 7 dx$$

2. We can take constants out of the integral. For example:

$$\int 6x^4 dx = 6 \int x^4 dx$$



## 4.2 Integrating Polynomials

When we are **differentiating**  $x^n$  we multiply by  $n$  and then reduce the power on  $x$  by 1. When **integrating** we do the opposite first **we add 1 to the power, then divide by this new power**. In general we have the rule:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

**Note:** This rule works for all values of  $n$  apart from  $n = -1$ . If we were to use the rule with  $n = -1$  we would have  $\frac{x^0}{0} + C$ . This would have us dividing by 0 which is forbidden. The integral of  $x^{-1}$  is discussed on the next page. We also need to account for the possibility of there being a constant ' $C$ ' so we write '+ $C$ ' at the end of our integral. This is discussed in detail later.



**Example:** Find the integrals of the following:

1.  $x^2$
2.  $2x + 6x^2$
3.  $x(x + 3)$

*Solution:*

1. We are first being asked to find  $\int x^2 dx$  so using the rule:  $\int x^n dx = \frac{x^{n+1}}{n+1} + C$

$$\begin{aligned}\int x^2 dx &= \frac{x^{2+1}}{2+1} + C \\ &= \frac{x^3}{3} + C\end{aligned}$$

2. We are being asked to find  $\int (2x + 6x^2) dx$ . When integrating multiple terms we do each term in turn. This means that:

$$\begin{aligned}\int (2x + 6x^2) dx &= \int 2x dx + \int 6x^2 dx && \text{Constants can then be taken out of the integral} \\ &= 2 \int x dx + 6 \int x^2 dx && \text{We can now integrate} \\ &= 2 \frac{x^{1+1}}{1+1} + 6 \frac{x^{2+1}}{2+1} + C \\ &= x^2 + 2x^3 + C\end{aligned}$$

3. First we expand the brackets to produce  $x^2 + 3x$ . Then we would integrate this (term by term) giving:

$$\begin{aligned}\int (x^2 + 3x) dx &= \frac{x^{2+1}}{2+1} + 3 \frac{x^{1+1}}{1+1} + C && \text{We can simplify this down} \\ &= \frac{1}{3}x^3 + \frac{3}{2}x^2 + C\end{aligned}$$





### 4.3 Integrating $x^{-1}$ and exponentials

We recall that if we differentiate  $\ln(x)$  we get  $x^{-1}$ . Since integration is the opposite of differentiation, when we integrate  $x^{-1}$  we get  $\ln(x)$ .

$$\int x^{-1} dx = \ln |x| + c$$

When integrating it is sometimes useful to think: “*what would I need to differentiate to get this answer?*” You can always check your integration by differentiating the result and checking to see if it’s the same as what you started with.

**Note:** The  $|$  signs either side of the  $x$  mean the modulus (or the absolute value) of  $x$ . It makes sure our  $x$  value is positive. For example  $|5| = 5$  and  $|-5| = 5$ . It is important in this integral as we can input a negative value of  $x$  into  $x^{-1}$  however we cannot take the log of a negative  $x$ .



**Example:** Find the integrals of the following:

$$\int \frac{x^2 + 2x^3 - 4x^4}{x^3} dx$$

**Solution:** The fraction can be simplified; by dividing each term on the denominator by  $x^3$ . This reduces the original integral to:

$$\begin{aligned} \int \frac{x^2 + 2x^3 - 4x^4}{x^3} dx &= \int \left( \frac{1}{x} + 2 - 4x \right) dx && \text{Integrate each term individually} \\ &= \ln|x| + 2x + \frac{-4x^2}{2} + C \\ &= \ln|x| + 2x - 2x^2 + C \end{aligned}$$



## Integrating exponentials

We can recall that:

$$y = ae^{bx} \Rightarrow \frac{dy}{dx} = a \times b \times e^{bx}$$

That rule states that when we are differentiating  $e^{bx}$  we multiply the exponential by the coefficient of the power. When integrating we do the opposite to differentiation, therefore we **divide by the coefficient of the power**. In general we have the rule:

$$\int e^{bx} = \frac{e^{bx}}{b} + C$$



**Example:** Find the following:

1.  $\int e^{6x} dx$
2.  $\int (e^x + 4e^{-3x}) dx$

*Solution:*

1. Using the rule  $\int e^{ax} = \frac{e^{bx}}{b} + C$  and dividing by the coefficient of the power we find:

$$\int e^{6x} = \frac{e^{6x}}{6} + C$$

2. First we should break up the integral into its constituent parts:

$$\int (e^x + 4e^{-3x}) dx = \int e^x dx + 4 \int e^{-3x}$$

Now we can apply the above rule to give:

$$\begin{aligned} &= \frac{(1)e^x}{1} + 4 \frac{e^{-3x}}{-3} + C \\ &= e^x - \frac{4}{3}e^{-3x} + C \end{aligned}$$

Again it is sometimes worth checking our answer by differentiating to be extra sure. For example, our solution to (2) was:  $e^x - \frac{4}{3}e^{-3x} + C$  differentiating this we get:

$$\begin{aligned} &e^x - \frac{4(-3)e^{-3x}}{3} + 0 \text{ This simplifies down to:} \\ &e^x + 4e^{-3x} \end{aligned}$$

Which is the same as what we started with, so our integration was correct



## 4.4 Integrating Trigonometric Function

Recall that:

$$y = a \sin(bx) \Rightarrow \frac{dy}{dx} = a \times b \times \cos(bx)$$

$$y = a \cos(bx) \Rightarrow \frac{dy}{dx} = -a \times b \times \sin(bx)$$

As aforementioned, integration is the opposite or the 'undoing' of differentiation, therefore it follows that:

$$\int \cos(ax) dx = \frac{\sin(ax)}{a} + C$$

$$\int \sin(ax) dx = \frac{-\cos(ax)}{a} + C$$



**Example:** Find the following:

1.  $\int \cos(4x) dx$
2.  $\int (5\cos(-x) + \sin(3x)) dx$

*Solution*

1. Using the rule  $\int \cos(ax) dx = \frac{\sin(ax)}{a} + C$  we find:

$$\int \cos(4x) dx = \frac{\sin(4x)}{4} + C$$

2. First we should split up the integral and extract the constants – this will give:

$$\int (5\cos(-x) + \sin(3x)) dx = 5 \int \cos(-x) dx + \int \sin(3x) dx$$

Applying the above rules we get

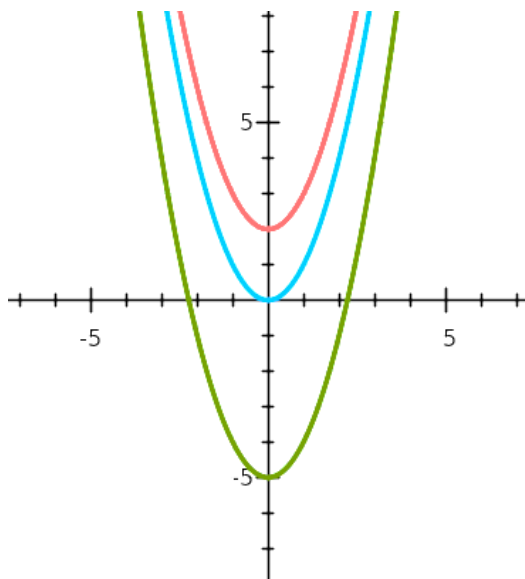
$$\begin{aligned} I \text{ (the integral)} &= 5 \times \frac{\sin(-x)}{-1} + \frac{\cos(3x)}{3} + C \\ &= -5 \sin(-x) - \frac{\cos(3x)}{3} + C \end{aligned}$$



## 4.5 Finding the Constant of Integration

In the integrals so far we have always had a constant of integration denoted  $+C$  at the end. Consider the functions  $y = x^2$ ,  $y = x^2 + 2$  and  $y = x^2 - 5$ , as shown on the graph below. These all differentiate to  $2x$ .

As mentioned earlier when integrating  $2x$  we don't know whether the actual answer is  $y = x^2$ ,  $y = x^2 + 2$  or  $y = x^2 - 5$  and so on, hence why we give the general answer of  $y = x^2 + C$



If we are given the gradient of a curve and a point that the curve goes through, we can find full equation of the curve and therefore a value of  $C$ .



**Example:** A curve of gradient  $4x^5$  passes through the point  $(1,2)$ . What is the full equation of the line?

*Solution:* We know the gradient is  $4x^5$  meaning that  $\frac{dy}{dx} = 4x^5$ . Hence we have:

$$y = \int 4x^5 dx$$

Which we can now integrate

$$y = \frac{4}{6}x^6 + C = \frac{2}{3}x^6 + C$$

We now have an expression for  $y$  in terms of  $x$ . However, there is a ' $C$ ' constant that we want to know. Since we know the curve passes through the point  $(1,2)$  (i.e when  $x = 1$   $y = 2$ ) we can **substitute these values in** to help us find the constant:

$$2 = \frac{2}{3}(1)^6 + C$$

$$C = 2 - \frac{2}{3} = \frac{4}{3}$$

Now we know the value for  $C$  we can provide the exact equation of the curve with gradient  $4x^5$  and through the points  $1,2$  as:

$$y = \frac{2}{3}x^6 + \frac{4}{3}$$



## 4.6 Integrals with Limits

Integrals represent the area under a graph. Various different things across the scientific discipline are represented by areas under graphs, for example the area under a speed vs time graph represents the distance travelled. Moreover, many probabilities can be defined by areas bound by curves and the axes: if you wanted to find the probability between two values we could use integration as a possible method of determining such probability – this would be known as solving a definite integral. Definite integrals are ones evaluated between two values (or limits) and can be summarised as:

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

- The braces, or parentheses,  $[\dots]_a^b$ , with  $a$  and  $b$  signify that we will evaluate the integral between these two limits.
- The numerical answer produced by  $F(b) - F(a)$  represents the area under the curve.

Below is a random function  $f(x)$  integrated between the limits of  $a$  and  $b$  with the shaded area being:  $\int_a^b f(x)$

**Note:** There is no need to find the  $C$  value when integrating between limits since the constant will always cancel out. When evaluating between the two points we will have:

$$(F(b) + C) - (F(a) + C) = F(b) - F(a) + C - C = F(b) - F(a)$$



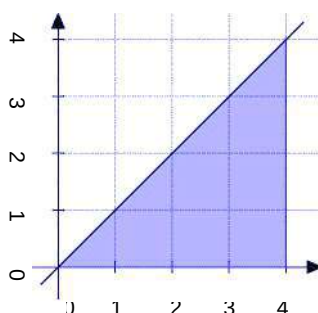
**Example:** Calculate:  $\int_0^4 x dx$

*Solution:* We first integrate with respect to  $x$ .

$$\int_0^4 x dx = \left[ \frac{1}{2} x^2 \right]_0^4$$

$$\left[ \frac{1}{2} x^2 \right]_0^4 = \left[ \frac{1}{2} \times 4^2 \right] - \left[ \frac{1}{2} \times 0^2 \right] = 8$$

We can verify this answer by visualising the area under the graph of  $y = x$  between 0 and 4





**Example:** Find  $\int_0^{\pi} \sin(x) dx$

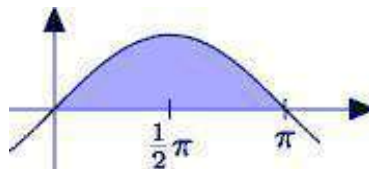
*Solution:* To begin we should integrate the function:

$$\int_0^{\pi} \sin(x) dx = [-\cos(x)]_0^{\pi}$$

Next we should evaluate this by inputting the limits so:

$$\begin{aligned} & [-\cos(\pi)] - [-\cos(0)] \\ & = (-(-1)) - (-(1)) \\ & = 1 + 1 = 2 \end{aligned}$$

The integral is 2. This means that the area between 0 and  $\pi$  is 2 units<sup>2</sup>. This can be seen below.



### Engineering Example:

The work done on a spring can be found by the integral:

$$Work = \int_a^b F(x)$$

Where:  $F = kx$  (Hooke's law and  $k$  is a constant)

:  $a$  and  $b$  are positions from the 'rest' position of the spring.

A force of 2000N compresses a spring from its natural (rest) length of 22cm to 14cm. How much extra work is done if you were to compress it further to 8cm?

First we must find ' $k$ ' the spring constant using the original data.

$$F = k(x)$$

$$2000 = k(8) \Rightarrow k = \frac{2000}{8} = 250$$

So the general equation for the Force on the spring can be given as:

$$F = 250x \quad \text{and the work done as} \quad \int_a^b 250x$$

$$\begin{aligned} \Rightarrow \left[ \frac{250x^2}{2} \right]_8^{14} &= \left[ \frac{250(14^2)}{2} \right] - \left[ \frac{250(8)^2}{2} \right] \\ &= 16500 \text{ N cm} \end{aligned}$$



## 4.7 Separating the Variable

A differential equation is an equation of the form:

$$\frac{dy}{dx} = f(x, y)$$

where  $f(x, y)$  is a function of both  $x$  and  $y$ . The method of separating the variables works by getting our differential equation into the form:

$$f_1(y)dy = f_2(x)dx$$

so we have all the  $y$ 's in the left hand side of the equation and the  $x$ 's on the right hand side then we can integrate both sides as below to solve.

$$\int f_1(y)dy = \int f_2(x)dx$$



**Example:** If  $\frac{dy}{dx} = \frac{4x^3}{6y}$  then use the method of separation of variables to integrate with respect to  $x$ .

*Solution:* So first we need to get the differential equation into the correct form, i.e., get all the  $y$ 's on one side and all the  $x$ 's on the other so:

$$6y dy = 4x^3 dx$$

We can now integrate both sides, the left hand side with respect to  $y$  and the right hand side with respect to  $x$ .

$$\int 6y dy = \int 4x^3$$

$$\frac{6y^2}{2} = \frac{4x^4}{4}$$

$$3y^2 = x^4 + C$$

Usually we have equations of curves in the forms  $y = f(x)$  so simple rearranging of the equation above yields:

$$y^2 = \frac{1}{3}x^4 + \frac{C}{3} = \frac{1}{3}(x^4 + C)$$

$$y = \frac{\sqrt{3}}{3}(\sqrt{x^4 + C})$$

$$y = \frac{\sqrt{3}x^4}{3} + \frac{\sqrt{3}}{3}C = \frac{\sqrt{3}}{3}x^2 + C$$

**Note:** We have swapped  $\frac{\sqrt{3}}{3}C$  for just  $C$  as they both represent constants.  $\frac{\sqrt{3}}{3}C$  is still just a constant so to make things tidier we just call it  $C$  again. When you are efficient at integrating, there is no need to perform the operations on the constant of integration, it is adequate to leave it as just  $+C$  throughout.



## 4.8 Integration by Substitution

**Note:** This section is designed as only an introduction to integration by substitution. There are supplementary materials in this section if you wish to take a more detailed look at integration by substitution; however you visit this in your first year course at University.

Suppose you had the integral  $\int (6 + 2x)^2 dx$  and we wanted to find the integral of that expression. Of course you could expand the brackets to achieve  $\int (4x^2 + 24x + 36) dx$  and then integrate as outlined in the method above.

However, for more complicated functions, this may become very tedious or difficult to do – what if the integral desired was  $\int (6 + 2x)^7 dx$  for example. Multiplying the brackets out would be very arduous, so instead we use a technique known as ‘**integration by substitution.**’

We introduce a new variable  $u$  so that the integral with  $u$ 's becomes something we can integrate a lot easier. In this case  $u = (6 + 2x)$  so therefore the integral becomes:

$$\int u^7 dx$$

Introduced variable  $u$    Tells us to integrate with respect to  $x$  but we want to integrate with respect to  $u$

Now  $u^7$  is a lot easier to integrate than  $(6 + 2x)^7$  however the ' $dx$ ' in the integral is still telling us to integrate with respect to  $x$ , but since we have introduced the new variable  $u$  we actually want to differentiate with respect to  $u$ . To do this we must get  $dx$  in terms of  $du$  so:

$$\begin{aligned} u &= (6 + 2x) \\ \frac{du}{dx} &= 2 \\ dx &= \frac{1}{2} du \end{aligned}$$

We can now replace  $dx$  for  $du$  in order for us to perform the integral so:

$$\begin{aligned} \int u^7 dx &= \int u^7 \times \frac{1}{2} du = \frac{1}{2} \int u^7 du \\ &= \frac{1}{2} \left[ \frac{u^8}{8} + C \right] \\ &= \frac{1}{16} u^8 + C \end{aligned}$$

We can now substitute back in for  $u = (6 + 2x)$  to give:

$$\int (6 + 2x)^7 dx = \frac{1}{16} (6 + 2x)^8 + C$$

Some very basic rules for integrating by substitution are:

- Introduce a variable  $u$  and substitute accordingly
- Find  $\frac{du}{dx}$  and subsequently get  $du$  in terms of  $dx$
- Replace  $dx$  in the original integral for the corresponding  $du$
- Integrate with respect to  $u$
- Replace  $u$  back with the original substitution so that you get the integral in terms of  $x$ .







**Example:** Use the substitution  $u = 1 + x^3$  to find:

$$\int 6x^2 \sqrt{1 + x^3} dx$$

*Solution:* This substitution may not appear to get rid of all the  $x$ 's but you will find that they cancel later. So first put the substitution in:

$$\int 6x^2 \times \sqrt{u} dx$$

Find  $\frac{du}{dx}$  and  $du$ :

$$\frac{du}{dx} = 3x^2$$

$$dx = \frac{du}{3x^2}$$

Substitute back in for  $dx$ :

$$\int 6x^2 \times \sqrt{u} \frac{du}{3x^2}$$

While at first glance this may appear to have complicated things you should notice the  $x^2$  terms cancel:

$$\int 2 \sqrt{u} du$$

Taking the factor out and integrating:

$$= 2 \int \sqrt{u} du = 2 \left[ \frac{2}{3} u^{3/2} \right] + C$$

$$\frac{4}{3} u^{3/2} + C$$

Substituting back in for  $x$  gives:

$$\frac{4}{3} (1 + x^3)^{3/2} + C$$



### Test Yourself – Integration

**A Find the following integrals:**

1.  $\int (6x^2 + 3x) dx$  (2)      2.  $\int (x^7 - \cos(x)) dx$  (2)

3.  $\int_0^4 ((5x^2 + e^{2x}))dx$  (3)      4.  $\int_0^\pi \left(\frac{3}{x^2} + \sin(x)\right) dx$  (3)

5.  $\int \left(\frac{3x^3+4x^2+5x}{x^3}\right) dx$  (3)

**B Use the following substitutions to find the following:**

1.  $\int 2\sin(x) \cos(x)$  (Substitution  $u = \sin(x)$ ) (5)

2.  $\int x^5(1 + 8x^6)^{\frac{1}{3}}$  (Substitution:  $u = 1 + 8x^6$ ) (6)

**C Use the separating the variables method to find an equation in the form  $y = f(x)$  for:**

1.  $\frac{dy}{dx} = \frac{6x^3+3x^2+2x}{y}$  (6)

**D Find a function in the form of  $y = f(x)$  for:**

1.  $\frac{dy}{dx} = 5x^5 + 2$  and passes through the point (3, 4) (4)

2.  $\frac{dy}{dx} = 3e^{3x} + \sin(x)$  and passes through the point (0,1) (4)

For the solutions click [here](#)

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## 5 Vectors

### 5.1 Introduction to Vectors

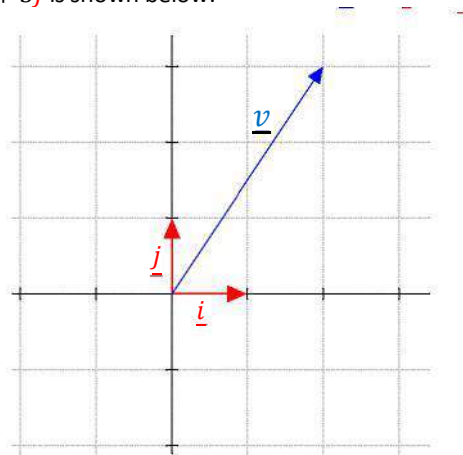
When describing some quantities just a number and units aren't enough. Say we are navigating a ship across the ocean and we are told that the port is 5km away we don't know in which direction we have to sail. 5km due east would give us all the information that we need. This is an example of a **vector**, a quantity that has magnitude and direction.

- Some general engineering related vector quantities are velocity, force, acceleration, linear and angular momentum and moment
- Quantities that have just a magnitude are **scalars**. Examples of these are temperature, mass and speed.

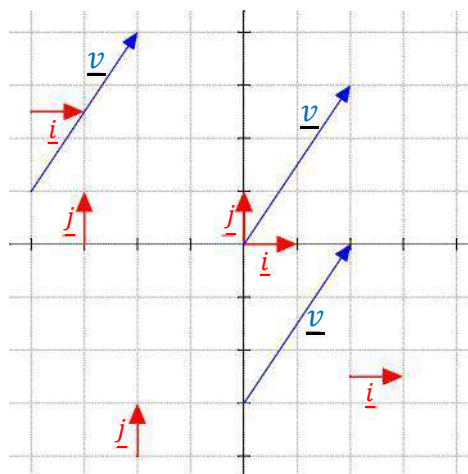
Note: To distinguish vectors from scalars we write them with a line under them, e.g.  $\underline{F}$  is used as the symbol for a force vector. Some books will use bold font for this e.g.  $\mathbf{F}$ .

### Vectors in 2-D Space

A vector in 2-D space can be written as a combination of 2 base vectors,  $\underline{i}$  being a horizontal vector (in the x direction) and  $\underline{j}$  being a vertical vector (in the y direction). They are both unit vectors meaning that their length (magnitude) is 1. For an example the vector  $\underline{v} = 2\underline{i} + 3\underline{j}$  is shown below:



**Note:** The vector  $\underline{v}$  above also describes all vectors that move 2  $\underline{i}$  units and 3  $\underline{j}$  units, they don't have to start at the origin. All the vectors in the graph below are exactly the same:



Writing a general 2-D vector  $\underline{v}$  as a combination of the unit vectors would give:

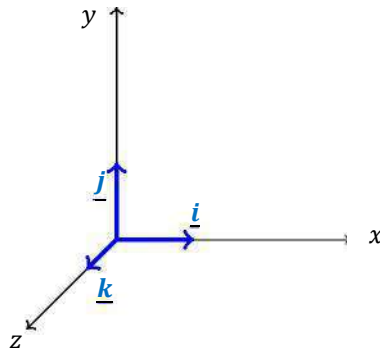
$$\underline{v} = a\underline{i} + b\underline{j}$$

Where  $a$  and  $b$  are real numbers



## Vectors in 3-D Space

Any vector in 3-D space can be written as a combination of 3 base vectors;  $\underline{i}$ ,  $\underline{j}$  and  $\underline{k}$  each being the unit vector in the  $x$ ,  $y$  and  $z$  directions. Below is a graph of the 3 axes with their respective unit vectors:



Writing a general vector  $\underline{v}$  as a combination of the unit vectors would give:

$$\underline{v} = a\underline{i} + b\underline{j} + c\underline{k}$$

Where  $a, b$  and  $c$  are real numbers

## Representation of Vectors

A vector  $\underline{v}$  can be written in the following forms:

### 1. Unit vector notation:

$$\underline{v} = a\underline{i} + b\underline{j} + c\underline{k}$$

The vector is displayed as the clear sum of its base vectors. In 3-D space those are the  $\underline{i}$ ,  $\underline{j}$  and  $\underline{k}$  vectors.

### 2. Ordered set notation:

$$\underline{v} = (a, b, c)$$

The vector is in the same form as unit vector notation ( $a, b$  and  $c$  are the same numbers) but it is more compact.

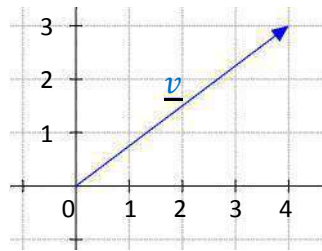
### 3. Column notation:

$$\underline{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

This is identical to ordered set notation apart from it being vertical. This form makes some calculations easier to visualise, such as the dot product.



For example, the 2-D vector  $\underline{v}$  shown below can be written as  $\underline{v} = 4\underline{i} + 3\underline{j}$  or  $\underline{v} = (4,3)$  or  $\underline{v} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$



## Magnitude of Vectors

If we know the lengths of each of the component vectors we can find the length (or magnitude) of the vector using Pythagoras:

For a vector  $\underline{v} = a\underline{i} + b\underline{j} + c\underline{k}$

Magnitude of  $\underline{v} = |\underline{v}| = \sqrt{a^2 + b^2 + c^2}$

For the vector  $\underline{v} = a\underline{i} + b\underline{j} + c\underline{k}$  above we can calculate the magnitude  $|\underline{v}|$  to be  $\sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$



**Example:** Find the magnitude of the following vectors:

1.  $(2,6,3)$
2.  $\underline{i} + 7\underline{j} + 4\underline{k}$
3.  $\begin{pmatrix} \sqrt{2} \\ \sqrt{2} \\ 0 \end{pmatrix}$

Solution:

1.  $|(2,6,3)| = \sqrt{2^2 + 6^2 + 3^2} = \sqrt{4 + 36 + 9} = \sqrt{49} = 7$
2.  $|\underline{i} + 7\underline{j} + 4\underline{k}| = \sqrt{1^2 + 7^2 + 4^2} = \sqrt{1 + 49 + 16} = \sqrt{66}$
3.  $\left| \begin{pmatrix} \sqrt{2} \\ \sqrt{2} \\ 0 \end{pmatrix} \right| = \sqrt{\sqrt{2}^2 + \sqrt{2}^2} = \sqrt{2 + 2} = \sqrt{4} = 2$



## Vectors Representation

| Symbol                                   | Name               | Description                                                                                |
|------------------------------------------|--------------------|--------------------------------------------------------------------------------------------|
| $i, j, k$                                | Axis of the vector | These work in the same manner as the $x, y, z$ axis that we have seen before.              |
| $A \bullet B$                            | Dot Product        | This is multiplication of two vectors that gives you a scalar product.                     |
| $A \times B$                             | Cross product      | This is the multiplication of two vectors that gives you a vector quantity as your answer. |
| $\overrightarrow{AB}$ or $\underline{A}$ | Vector notation    | This shows a vector between two points.                                                    |
| $ A $                                    | Magnitude          | The size or modulus of the vector                                                          |
| $\hat{a}$                                | Unit vector        | This is the vector divided by the magnitude.                                               |



## 5.2 Operations with Vectors

### Scalar Multiplication

Scalar multiplication is when we multiply a vector by a scalar. We do this by multiplying each component of the vector by the scalar. This can be expressed as:

$$\lambda \times (x, y, z) = (\lambda x, \lambda y, \lambda z)$$

Where  $\lambda$  is a real number

**Note:** Scalar multiplication only changes the magnitude of the vector while the direction stays the same.



**Example:** Simplify  $7(-12, 4)$

Solution: we multiply each component of the vector by 7 giving:

$$(7 \times (-12), 7 \times 4) = (-84, 28)$$

### Vector Addition and Subtraction

When we add or subtract two vectors we add or subtract each of the individual components of the vectors. This can be expressed as:

$$(a, b, c) + (x, y, z) = (a + x, b + y, c + z)$$

Where  $(a, b, c)$  and  $(x, y, z)$  are both vectors using i, j and k.

**IMPORTANT:** If the vectors have a different number of base components, for example  $(3,6) + (1,0,7)$  we cannot complete the addition. For all vector on vector operations both vectors need to have the same number of base components.



**Example:** Calculate the following

1.  $(2,2,0) + (5, -1,3)$
2.  $3(a, 4a, 0) - a(7,0, -1)$
3.  $(1,3,8) + (2,1, -5) + (-1,2, -1)$

Solutions:

1.  $(2,2,0) + (5, -1,3) = (2 + 5, 2 - 1, 0 + 3) = (7, 1, 3)$
2.  $3(a, 4a, 0) - a(7,0, -1) = (3a, 12a, 0) - (a7,0, -a) = (-4a, 12a, a)$
3.  $(1,3,8) + (2,1, -5) + (-1,2, -1) = (1 + 2 - 1, 3 + 1 + 2, 8 - 5 - 1) = (2, 6, 2)$



## Vector Multiplication: Dot Product (or Scalar Product)

There are two ways of multiplying two vectors together. The first is the dot (or scalar) product which is defined to be:

$$\underline{A} \cdot \underline{B} = |\underline{A}| |\underline{B}| \cos \theta$$

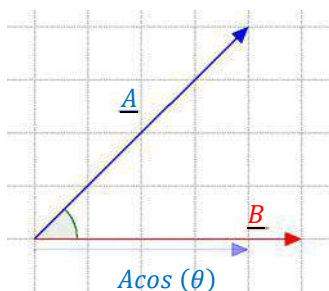
Where  $\theta$  is the angle between vectors  $\underline{A}$  and  $\underline{B}$

**IMPORTANT:** Taking the dot product of two vectors produces a **scalar** quantity.

Given two vectors, the dot product can also be calculated in the following way:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = ax + by + cz$$

**Note:** The dot product can be thought of how much one vector is pointing in the direction of the other.



The part of  $\underline{A}$  that goes in the same direction as  $\underline{B}$  has a length of  $|\underline{A}| \cos(\theta)$ , making the dot product the length of  $\underline{B}$  times the length of  $\underline{A}$  that is in the same direction as  $\underline{B}$



**Example:** Calculate the following

1.  $\begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 10 \end{pmatrix}$

2.  $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} \\ \sqrt{2} \end{pmatrix}$

Solution:

1.  $\begin{pmatrix} 3 \\ 9 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 10 \end{pmatrix} = 3 \times 1 + 9 \times (-2) + 1 \times 10 = 3 - 18 + 10 = -5$

2.  $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} \\ \sqrt{2} \end{pmatrix} = 1 \times \sqrt{2} + 0 \times \sqrt{2} = \sqrt{2}$





## Vector Multiplication: Cross Product

The cross product (or vector product) is defined to be:

$$\underline{A} \times \underline{B} = |\underline{A}| |\underline{B}| \sin(\theta) \underline{\hat{n}}$$

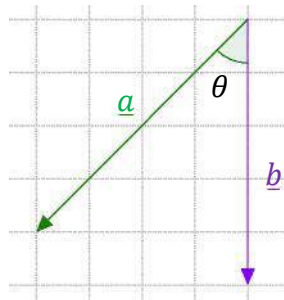
Where  $\theta$  is the angle between the two vectors  
and  $\underline{\hat{n}}$  is the unit vector in the direction of the new vector.

**IMPORTANT:** The cross product takes two vectors and produces a new vector. This new vector is in the direction **perpendicular** to the plane that  $\underline{A}$  and  $\underline{B}$  are in. It is only possible to take the cross product of two 3D vectors (and technically 7-D vectors as well).

The cross product is also dependent on the order the vectors appear in the product. If we change the order of the vectors in a cross product then our answer becomes negative. The vector points in the opposite direction.

$$\underline{a} \times \underline{b} = -\underline{b} \times \underline{a}$$

The right hand rule determines which way the vector produced will point. We use the rule as follows: curl your fingers from the first vector in the cross product to the second vector and the direction your thumb points is the direction of the vector produced by the cross product.



In this case  $\underline{a} \times \underline{b}$  will point out of the page and  $\underline{b} \times \underline{a}$  will point in to the page.

## Calculating the Cross Product

When using the base vector of  $\underline{\hat{i}}$ ,  $\underline{\hat{j}}$  and  $\underline{\hat{k}}$ , we can calculate the cross product by considering the cross product of the base vector with each other:

$$\underline{\hat{i}} \times \underline{\hat{i}} = \underline{\hat{j}} \times \underline{\hat{j}} = \underline{\hat{k}} \times \underline{\hat{k}} = \underline{0}$$

$$\underline{\hat{i}} \times \underline{\hat{j}} = -\underline{\hat{j}} \times \underline{\hat{i}} = \underline{\hat{k}}$$

$$\underline{\hat{j}} \times \underline{\hat{k}} = -\underline{\hat{k}} \times \underline{\hat{j}} = \underline{\hat{i}}$$

$$\underline{\hat{k}} \times \underline{\hat{i}} = -\underline{\hat{i}} \times \underline{\hat{k}} = \underline{\hat{j}}$$



Taking the cross products of two vectors is just like expanding brackets. The cross product two general vectors,  $(a\mathbf{i} + b\mathbf{j} + c\mathbf{k}) \times (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$ , would give  $ax(\mathbf{i} \times \mathbf{i}) + ay(\mathbf{i} \times \mathbf{j}) + az(\mathbf{i} \times \mathbf{k}) + \dots$ . Fully expanded and simplified gives:

$$(a\mathbf{i} + b\mathbf{j} + c\mathbf{k}) \times (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = (bz - cy)\mathbf{i} + (cx - az)\mathbf{j} + (ay - bx)\mathbf{k}$$

Where  $a, b, c, x, y$  and  $z$  are real numbers



**Example:** Calculate the following cross products

$$\begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

Solution: So to find this we use the formula below:

$$(a\mathbf{i} + b\mathbf{j} + c\mathbf{k}) \times (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = (bz - cy)\mathbf{i} + (cx - az)\mathbf{j} + (ay - bx)\mathbf{k}$$

Now we can write the vector in the form of the rule so  $(a\mathbf{i} + b\mathbf{j} + c\mathbf{k}) = (3\mathbf{i} + 1\mathbf{j} - 2\mathbf{k})$

and  $(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}) = (2\mathbf{i} - 1\mathbf{j} + 1\mathbf{k})$

Hence  $a = 3, b = 1, c = -2, x = 2, y = -1$  and  $z = 1$  and therefore:

$$\begin{aligned} & (3\mathbf{i} + 1\mathbf{j} - 2\mathbf{k}) \times (2\mathbf{i} - 1\mathbf{j} + 1\mathbf{k}) \\ &= (1 \times 1 - (-2) \times (-1))\mathbf{i} + ((-2) \times 2 - 3 \times 1)\mathbf{j} + (3 \times (-1) - 1 \times 2)\mathbf{k} \\ &= -1\mathbf{i} - 7\mathbf{j} - 5\mathbf{k} \end{aligned}$$

This could also be written as:

$$\begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -7 \\ -5 \end{pmatrix}$$



## 6 Matrices

### 6.1 What is a Matrix?

A matrix is defined as an array of numbers such as:

$$\begin{pmatrix} 3 & 1 & 8 \\ 0 & -4 & 5 \end{pmatrix}$$

- Matrices come in different sizes. We write the size of the matrix as rows  $\times$  columns. So the matrix above is a  $2 \times 3$  matrix.
- We use capital letters such as  $A$  or  $B$  to denote matrices.
- The numbers contained in a matrix are known as matrix elements. They are denoted  $a_{ij}$  where  $i$  represents the row and  $j$  the column in which the element is in. For example in the above matrix  $a_{11} = 3$  and  $a_{23} = 5$ .
- Matrices with an equal number of rows and columns are called square matrices. For example the  $2 \times 2$  matrix below:

$$\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$



**Example:** For the matrices below find the size of the matrix and the matrix and the value of  $a_{11}$  and  $a_{33}$  if possible.

1.  $A = \begin{pmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \\ 1 & 2 & 3 \end{pmatrix}$

2.  $B = \begin{pmatrix} 1 \\ -1 \\ 40 \end{pmatrix}$

3.  $C = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$

**Solution:**

1. The matrix  $A$  has 4 row and 4 columns, hence is a  $4 \times 3$  matrix. We can find  $a_{21} = 1$  and  $a_{33} = 6$ .
2. The matrix  $B$  has 3 rows and 1 column, so is a  $3 \times 1$  matrix. We can find that  $a_{21} = -1$  but there is no third column so we cannot find  $a_{33}$ .
3. The matrix  $C$  has 3 rows and 3 columns, hence it is a  $3 \times 3$  matrix. We then can find  $a_{21} = d$  and  $a_{33} = i$ .

Matrices are an excellent way of expressing large amounts of information or data in a small space



## 6.2 Matrix Algebra

### Addition and Subtraction

Matrices can only be added or subtracted if they are of the same size. Suppose we have the  $2 \times 2$  matrices below:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \text{ and } B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

Then their sum is found by adding together corresponding matrix elements:

$$A + B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{pmatrix}$$

Subtraction works in exactly the same way:

$$A - B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} - \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11} - b_{11} & a_{12} - b_{12} \\ a_{21} - b_{21} & a_{22} - b_{22} \end{pmatrix}$$

In general if we are adding or subtracting two  $m \times n$  matrices:

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \pm \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{pmatrix} = \begin{pmatrix} a_{11} \pm b_{11} & a_{12} \pm b_{12} & \cdots & a_{1n} \pm b_{1n} \\ a_{21} \pm b_{21} & a_{22} \pm b_{22} & \cdots & a_{2n} \pm b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} \pm b_{m1} & a_{m2} \pm b_{m2} & \cdots & a_{mn} \pm b_{mn} \end{pmatrix}$$

Note: Matrix addition and subtraction is commutative so  $A + B = B + A$ .



**Example:** Calculate the following:

1.  $\begin{pmatrix} 1 & 1 \\ 4 & 0 \end{pmatrix} + \begin{pmatrix} -1 & 4 \\ 2 & -3 \end{pmatrix}$

2.  $\begin{pmatrix} 10 \\ -3 \\ 0 \end{pmatrix} - \begin{pmatrix} -3 \\ 2 \\ 0 \end{pmatrix}$

3.  $\begin{pmatrix} 5 & 1 \\ 3 & 2 \end{pmatrix} + (1 \ 4 \ 2)$

Solution:

1.  $\begin{pmatrix} 1 & 1 \\ 4 & 0 \end{pmatrix} + \begin{pmatrix} -1 & 4 \\ 2 & -3 \end{pmatrix} = \begin{pmatrix} 1 + (-1) & 1 + 4 \\ 4 + 2 & 0 + (-3) \end{pmatrix} = \begin{pmatrix} 0 & 5 \\ 6 & -3 \end{pmatrix}$

2.  $\begin{pmatrix} 10 \\ -3 \\ 0 \end{pmatrix} - \begin{pmatrix} -3 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 - (-3) \\ -3 - 2 \\ 0 - 0 \end{pmatrix} = \begin{pmatrix} 13 \\ -5 \\ 0 \end{pmatrix}$

3. We can not add these matrices as they are different sizes!



## Multiplication by a Constant

The simplest form of multiplication is where we multiply a matrix by a constant or scalar. We show the rule on a  $2 \times 2$  matrix but it follows in the same way for any sized matrix we simply multiply every element by the scalar.

$$\lambda A = \lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix}$$



**Example:** find the following:

1.  $2 \begin{pmatrix} -1 & 2 \\ 0 & 4 \end{pmatrix}$

2.  $\frac{\sqrt{3}}{3} \begin{pmatrix} -9 & 0 & 3 \\ 12 & 12 & 0 \\ 3 & 0 & 3 \end{pmatrix}$

**Solution:**

1.  $2 \begin{pmatrix} -1 & 2 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} -1 \times 2 & 2 \times 2 \\ 0 \times 2 & 4 \times 2 \end{pmatrix} = \begin{pmatrix} -2 & 4 \\ 0 & 8 \end{pmatrix}$

2.  $\frac{\sqrt{3}}{3} \begin{pmatrix} -9 & 0 & 3 \\ 12 & 12 & 0 \\ 3 & 0 & 3 \end{pmatrix} = \begin{pmatrix} -9 \times \frac{\sqrt{3}}{3} & 0 \times \frac{\sqrt{3}}{3} & 3 \times \frac{\sqrt{3}}{3} \\ 12 \times \frac{\sqrt{3}}{3} & 12 \times \frac{\sqrt{3}}{3} & 0 \times \frac{\sqrt{3}}{3} \\ 3 \times \frac{\sqrt{3}}{3} & 0 \times \frac{\sqrt{3}}{3} & 3 \times \frac{\sqrt{3}}{3} \end{pmatrix} = \begin{pmatrix} -3\sqrt{3} & 0 & \sqrt{3} \\ 4\sqrt{3} & 4\sqrt{3} & 0 \\ \sqrt{3} & 0 & \sqrt{3} \end{pmatrix}$

## Matrix Multiplication

Suppose we want to find the product  $AB$  of the matrices  $A$  and  $B$ . Then:

- The number of columns of  $A$  must be equal to the number of rows of  $B$ .
- The product matrix has the same number of rows as  $A$  and the same number of columns as  $B$ .

**Note:** When we multiply matrices the size of the answer is the rows in  $A$  and the columns in  $B$

$$\overbrace{(2 \times 3)}^A \overbrace{(3 \times 1)}^B$$

So we can multiply  $A$  and  $B$  as the two inside numbers are the same and the size will be a  $(2 \times 1)$

Below we have the rule to find the product of two  $2 \times 2$  matrices:

$$AB = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

In general if we want to find the product  $AB$  of a  $m \times l$  matrix  $A$  and  $k \times n$  matrix  $B$ :

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1l} \\ a_{21} & a_{22} & \cdots & a_{2l} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{ml} \end{pmatrix} \text{ and } B = \begin{pmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{k1} & b_{k2} & \cdots & b_{kn} \end{pmatrix} \text{ so that } AB = \begin{pmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{pmatrix}$$

**Important:** we calculate  $c_{ij}$  to be the scalar dot product of row  $i$  and column  $j$ .





**Example:** find the following products:

1.  $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$

2.  $\begin{pmatrix} 3 & 0 & 2 \\ 1 & 5 & 0 \end{pmatrix} \begin{pmatrix} 0 & -6 \\ 2 & -1 \\ 1 & 3 \end{pmatrix}$

3.  $\begin{pmatrix} 10 & 2 & 2 \\ 8 & -1 & 4 \end{pmatrix} \begin{pmatrix} -5 & -3 \\ 2 & 2 \end{pmatrix}$

Solutions:

1. As we are multiplying a  $2 \times 2$  matrix by another  $2 \times 2$  matrix the product matrix will also be a  $2 \times 2$  matrix in the form below:

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

We now calculate the elements of the product matrix above:

To find  $a_{11}$  we take the dot product or row 1 and column 1:

$$a_{11} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 1 \times 3 + 0 \times 2 = 3$$

To find  $a_{12}$  we take the dot product or row 1 and column 2:

$$a_{12} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 \times 1 + 0 \times 1 = 1$$

To find  $a_{21}$  we take the dot product or row 2 and column 1:

$$a_{21} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 2 \times 3 + 1 \times 2 = 8$$

As we get more adept at performing these operations we can just start putting the numbers in the equation to find  $a_{22}$

$$2 \times 1 + 1 \times 1 = 3$$

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 8 & 3 \end{pmatrix}$$

2.

First check we can perform the multiplication ( $2 \times 3$ )( $3 \times 2$ ) we can as the two inside numbers match and the size of our answer will be a ( $2 \times 2$ ).

$$\begin{pmatrix} 3 & 0 & 2 \\ 1 & 5 & 0 \end{pmatrix} \begin{pmatrix} 0 & -6 \\ 2 & -1 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 3 \times 0 + 0 \times 2 + 2 \times 1 & 3 \times -6 + 0 \times -1 + 2 \times 3 \\ 1 \times 0 + 5 \times 2 + 0 \times 1 & 1 \times -6 + 5 \times -1 + 0 \times 3 \end{pmatrix} = \begin{pmatrix} 2 & -12 \\ 10 & -11 \end{pmatrix}$$

3. We cannot multiply these two matrices as the columns of A do not equal the rows of B

$$(2 \times 3)(2 \times 2)$$

**Note:** Matrix multiplication is NOT commutative so  $AB \neq BA$ . Test it yourself with question 1 from the example above



## 6.3 The Identity Matrix, Determinant and Inverse of a Matrix

### The Identity Matrix

The Identity Matrix (or Unit Matrix) is a square matrix which is denoted  $I$  and has '1' along the leading diagonal (from top left to bottom right) with '0' in all other positions.

The  $2 \times 2$  identity matrix is:

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The  $3 \times 3$  identity matrix is:

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

For a  $n \times n$  matrix the identity matrix is:

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

The Identity matrix has the property that when multiplied with another matrix it leaves the other matrix unchanged:

$$AI = A = IA$$



**Example:**

$$\begin{pmatrix} 3 & 4 \\ 8 & 9 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 4 \\ 8 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 4 \\ 8 & 9 \end{pmatrix}$$

### The Transpose of a Matrix

The transpose of a matrix is denoted  $A^T$  and is obtained by interchanging the rows and columns of the matrix. The rule for a  $2 \times 2$  matrix  $A$  is:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$
$$A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

**Note:** A  $3 \times 3$  matrix is more complicated but the same principle applies, we swap the rows and the columns.



**Example:**

$$A = \begin{pmatrix} 3 & 0 & 4 \\ 5 & 1 & 9 \\ 5 & 2 & 7 \end{pmatrix} \quad A^T = \begin{pmatrix} 3 & 5 & 5 \\ 0 & 1 & 9 \\ 4 & 2 & 7 \end{pmatrix}$$



## The Determinant of a Matrix

The determinant of a matrix is denoted  $|A|$ . For finding the determinant of a  $2 \times 2$  matrix  $A$  we have the following formula:

$$\text{For } A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \text{ the determinant is } |A| = ad - bc$$

**Note:** You can find the determinate for any square matrix but it become increasingly complex as you will see in your first year

- A matrix whose determinant is zero so  $|A| = 0$  is said to be singular.
- A matrix whose determinant is non zero so  $|A| \neq 0$  is said to be non-singular.



**Example:** Find the determinate of the matrix  $A$ :

$$A = \begin{pmatrix} 1 & -4 \\ 2 & 5 \end{pmatrix}$$

Solution:

$$|A| = \begin{vmatrix} 1 & -4 \\ 2 & 5 \end{vmatrix} = (1 \times 5) - (-4 \times 2) = 13$$

**Note:** The matrix is non-singular

## Inverse of a Matrix

Only non-singular matrices have an inverse matrix. The inverse of a matrix  $A$  is denoted  $A^{-1}$  and has the following property:

$$AA^{-1} = A^{-1}A = I$$

To find the inverse of a  $2 \times 2$  matrix  $A$  we have the following formula:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

**Where  $|A| = ad - bc \neq 0$**

**Note:** There exists a more complex formula for finding the inverse of larger matrices.





## 7 Refresher topics

### 7.1 Addition, Subtraction, Multiplication and Division

We will all be familiar with the following operations: addition (+), subtraction (−), multiplication (×) and division (÷) but for the sake of completeness we will review some simple rules and conventions.

| Operation      | Representation                           | Ordering                       |
|----------------|------------------------------------------|--------------------------------|
| Addition       | $A + B$                                  | $A + B = B + A$                |
| Subtraction    | $A - B$                                  | $A - B \neq B - A$             |
| Multiplication | $A \times B$ or $A.B$ or $A * B$ or $AB$ | $A \times B = B \times A$      |
| Division       | $A \div B$ or $\frac{A}{B}$              | $\frac{A}{B} \neq \frac{B}{A}$ |

#### Addition and Multiplication are Distributive

For numbers  $x$ ,  $y$  and  $a$  the distributive rule for addition and multiplication is:

$$a \times (x + y) = (a \times x) + (a \times y)$$



**Example:** For Example suppose we have  $x = 3$ ,  $y = 5$  and  $a = 6$  then:

1.  $a \times (x + y) = 6 \times (3 + 5) = 6 \times 8 = 48$

2.  $(a \times x) + (a \times y) = (6 \times 3) + (6 \times 5) = 18 + 30 = 48$

So  $a \times (x + y) = (a \times x) + (a \times y)$  and the distributive rule holds

#### Factorisation

Factorisation is used to tidy up equations in order to make them easier to read and understand.



**Example:**

We can factorise the equation  $y = 3x^3 + 6x$  if we note that both  $3x^3$  and  $6x$  are divisible by  $3x$  this gives us  $y = 3x(x^2 + 2)$ .



## Multiplying Out Terms

Multiplying out terms (or expanding out of brackets) makes use of the distributive law to remove brackets from an equation and is the opposite of factorisation.



### Example:

We can multiply out the equation  $y = 3z(2x^4 + 7)$  to give:

$$y = (3z \times 2x^4) + (3z \times 7)$$

$$\Rightarrow y = 6zx^4 + 21z$$



### Example:

We can multiply out the equation  $3a[(a + 3b) - 5(2b - a)]$  to give:

$$3a[(a + 3b) - (5 \times 2b - 5 \times a)] = 3a[(a + 3b) - (10b - 5a)]$$

$$= 3a(6a - 7b)$$

$$= 18a^2 - 21ab$$

## Adding and Subtracting Negative Numbers

1. Adding a negative number is like subtracting a positive number.
2. Subtracting a negative number is like adding a positive number.



### Example:

Suppose we have  $x = 4$  and  $y = -3$

$$x + y = 4 + (-3) = 4 - 3 = 1$$

$$x - y = 4 - (-3) = 4 + 3 = 7$$



## Order of Operations: BODMAS

Consider the calculation below:

$$6 + (7 \times 3^2 + 1)$$

In what order should we calculate our operations? BODMAS tells us that we should carry out the operations in the order listed below:

1. First Brackets.
2. Then Orders. (This is the powers and roots).
3. Then Division.
4. Then Multiplication.
5. Then Addition.
6. Then Subtraction.

Note: Addition and subtraction have the same priority so can actually be done in either order. The same is true for multiplication and division.



### Example:

So returning back to  $6 + (7 \times 3^2 + 1)$

$$6 + (7 \times 3^2 + 1) \quad \text{Start inside the Brackets and do the Orders first}$$

$$6 + (7 \times 9 + 1) \quad \text{Then Multiply the 7 and 9}$$

$$6 + (63 + 1) \quad \text{Then Add the 63 and 1}$$

$$6 + (64) \quad \text{Brackets are done so the last operation is to Add the 6 and 64}$$

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### Engineering Example:

The equation (Bernoulli's equation) below shows how fluid changes in a pipe. This equation can be used to find a change in pressure from one section of pipe to another. This can be achieved by rearranging this equation.

$$\frac{P_1}{\rho g} + \frac{u_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{u_2^2}{2g} + z_2$$

If  $P_1$  (initial pressure) is **1200 N/m<sup>2</sup>**,  $u_1$  (initial velocity) is **0.8 m/s**,  $\rho$  (density) is **1000 kg/m<sup>3</sup>**,  $g$  (gravity) is **9.81 m/s<sup>2</sup>**,  $z_1$  (initial height) is **0m**,  $u_2$  (secondary velocity) **1 m/s** and  $z_2$  is **0m**.

$$\frac{1200}{1000 \times 9.81} + \frac{0.8^2}{2 \times 9.81} + 0 = \frac{P_2}{1000 \times 9.81} + \frac{1^2}{2 \times 9.81} + 0$$

$$\frac{1200}{1000 \times 9.81} + \frac{0.64}{2 \times 9.81} = \frac{P_2}{1000 \times 9.81} + \frac{1}{2 \times 9.81}$$

First do the squared terms

$$\frac{1200}{1000 \times 9.81} + \frac{0.64}{2 \times 9.81} - \frac{1}{2 \times 9.81} = \frac{P_2}{1000 \times 9.81}$$

Rearrange the equation

$$\frac{1200 \times 1000 \times 9.81}{1000 \times 9.81} + \frac{0.64 \times 1000 \times 9.81}{2 \times 9.81} - \frac{1 \times 1000 \times 9.81}{2 \times 9.81} = P_2$$

Make  $P_2$  the subject

$$\frac{11772000}{9810} + \frac{6278.4}{19.62} - \frac{9810}{19.62} = P_2$$

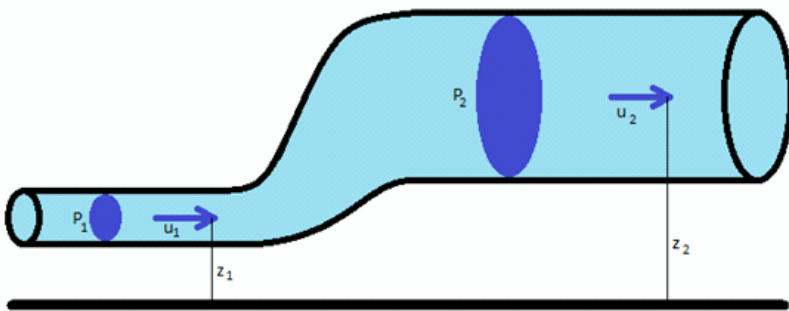
Multiply

$$1200 + 320 - 500 = P_2$$

Divide

$$1020 = P_2$$

Addition and Subtraction



## 7.2 Mathematical Notation, Symbols and Operators

In order to save ourselves time we often use symbols as shorthand. We look at three new symbols  $\Delta$ ,  $\sum$  and  $\pi$  which are used to describe operations, just like (+), (-), (x) and (-i-).

### The Delta $\Delta$ Operator

We use the Greek capital letter  $\Delta$  (called delta) to represent the difference (or change) between the start and end values of a quantity. For any quantity A:

$$\Delta A = A_{\text{final}} - A_{\text{initial}}$$

where  $A_{\text{final}}$  is the final amount of A and  $A_{\text{initial}}$  is the initial amount of A so  $\Delta A$  is equal to the difference in these two values.

Note: If  $\Delta A$  is positive then there is an increase in A and if  $\Delta A$  is negative then there is a decrease in A.

### The Sigma Operator

The Greek capital letter  $\sum$  (called sigma) is used to represent a sum, normally long sums which would take a while to write out completely. The general form for a sum using sigma is:

$$\sum_{i=1}^n X_i = X_1 + X_2 + X_3 + \dots + X_n$$

Below we have what each of the terms in the sum means:

- $x_i$  are the numbers that are being summed; it could be an expression involving i.
- i is called the summation index; it identifies each term in the sum.
- $i = 1$  tells us to start summing at  $i = 1$ .
- n tells us to stop summing when we have reached  $i = n$ .

For a numerical explanation, consider take the sum  $1 + 2 + 3 + 4 + 5$ . This can be simplified into the  $\sum_{i=1}^5 i$ . From the notation we can see that we are summing the number i from when i is equal to 1 to 5.



#### Example:

What is the value of:  $\sum_{i=4}^7 i^2$

Solution: To begin we 'read' what the operation is telling us to do. We know it is a sum from the E and the terms we are summing are the squares of the index. We are starting with 4 and ending at 7 (and assume that each number is an integer). This then will give us:

$$\sum_{i=4}^7 i^2 = 4^2 + 5^2 + 6^2 + 7^2 = 16 + 25 + 36 + 49 = \mathbf{126}$$



## 7.3 Fractions

Fractions are a way of expressing ratios and are synonymous to division. They are written in the form:

$$\frac{A}{B}$$

- The expression on the top of a fraction A is called the numerator.
- The expression on the bottom B is known as the denominator.

Note: Fractions are used as a nicer way of showing division ie:  $a \div b = \frac{a}{b}$

### Simplifying Fractions

Plenty of fractions can be reduced to simpler forms, for example  $\frac{2}{4}$  is the same as  $\frac{1}{2}$ . The best way of simplifying is to find a number that is a factor of the numerator and the denominator so that it cancels.



**Example:** Simplify  $\frac{350}{1000}$  into its simplest form

Solution: We first notice that a factor of 10 can be taken out of both the numerator and denominator, leaving us with:

$$\frac{10 \times 35}{10 \times 100} = \frac{\cancel{10} \times 35}{\cancel{10} \times 100} = \frac{35}{100}$$

The only remaining factors of 35 are 7 and 5. 5 is also a factor of 100 so we can take it out as a factor as well:

$$\frac{35}{100} = \frac{5 \times 7}{5 \times 20} = \frac{\cancel{5} \times 7}{\cancel{5} \times 20} = \frac{7}{20}$$

We can not simplify any further because the numerator and denominator share no more factors



**Example:** Simplify  $\frac{2x+6xy}{4x^2+10x^3}$

Solution: We note first all the terms are even this means we can take a factor of 2 out. Also all the terms have an x in them meaning we can take a factor of x out.

Since we now have a factor of 2x on both the numerator and denominator it will cancel, leaving us with a fraction that can't be simplified any more. The following steps are illustrated below.

$$\frac{2x + 6xy}{4x^2 + 10x^3} = \frac{2(x + 3xy)}{2(2x^2 + 5x^3)} = \frac{2x(1 + 3y)}{2x(2x + 5x^2)} = \frac{\cancel{2x}(1 + 3y)}{\cancel{2x}(2x + 5x^2)} = \frac{1 + 3y}{2x + 5x^2}$$



**Note:** Don't fall into the trap when simplifying shown in this incorrect example:

$$\frac{a+b}{b+c} = \frac{a+\cancel{b}}{\cancel{b}+c} = \frac{a}{c} \quad \text{WRONG!}$$

If you still are uncertain then use  $a = 1$ ,  $b = 2$ ,  $c = 3$  and then be shocked to find that  $\frac{1}{3} = \frac{3}{5}$ !

## Multiplying Fractions

Multiplying two fractions together is a simple process, the numerator is the product of the numerators and the denominator is the product of the denominators.

$$\frac{a}{b} \times \frac{x}{y} = \frac{a \times x}{b \times y}$$

**Note:** If we are multiplying a fraction by a number that isn't in fraction form, such as an integer or  $\pi$ , then it will only multiply the numerator. We do this by treating it as a fraction in the form  $\frac{\text{number}}{1}$  as shown in the next example.



**Example:** Find the Following

1.  $\frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} \times \frac{5}{7}$

2.  $2\sqrt{5} \times \frac{3}{10}$

Solution:

1.  $\frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} \times \frac{5}{7} = \frac{1 \times 2 \times 4 \times 5}{2 \times 3 \times 5 \times 7} = \frac{40}{210} = \frac{4}{21}$

2.  $2\sqrt{5} \times \frac{3}{10} = \frac{2\sqrt{5}}{1} \times \frac{3}{10} = \frac{2\sqrt{5} \times 3}{1 \times 10} = \frac{15\sqrt{5}}{10} = \frac{3\sqrt{5}}{2}$

## Dividing Fractions

The inverse of the fraction is where the numerator and denominator switch, e.g. fraction  $\frac{a}{b}$  has an inverse of  $\frac{b}{a}$ .

Division is the inverse of multiplication so when we divide by a fraction we are essentially multiplying by the inverse of that fraction. So a general example of division of fractions would look like:

$$\frac{\left(\frac{a}{b}\right)}{\left(\frac{c}{d}\right)} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$



**Example:** Find the following:  $\frac{9}{16} \div \frac{3}{4}$

Solution:

$$\frac{9}{16} \div \frac{3}{4} = \frac{9}{16} \times \frac{4}{3} = \frac{9 \times 4}{16 \times 3} = \frac{36}{48} = \frac{3}{4}$$



## Adding and Subtracting Fractions

To be able to add and subtract two fractions they both need to have the same denominator called a 'common denominator'. Take  $\frac{1}{4} + \frac{1}{5}$  for example. We need a new denominator for both in order to add them. The easiest denominator to pick is almost always the product of the two so in this case  $4 \times 5 = 20$ .

$$\frac{1}{4} + \frac{1}{5} = \left(\frac{1}{4} \times \frac{5}{5}\right) + \left(\frac{1}{5} \times \frac{4}{4}\right) = \frac{5}{20} + \frac{4}{20} = \frac{9}{20}$$

Subtracting fractions works exactly the same way, the denominators need to be the same but you then subtract at the end.



**Example:** Find  $\frac{1}{3} + \frac{1}{6}$  Express your answer as a simplified fraction.

Solution: The first thing is to check whether the denominators are the same and if not what can we do to fix that. Notice that in this case we don't need to change the second fraction because 3 is a factor of 6. What we would do instead is change the first fraction to have 6 as the denominator by multiplying by  $\frac{2}{2}$ .

$$\frac{1 \times 2}{3 \times 2} + \frac{1}{6} = \frac{2}{6} + \frac{1}{6} = \frac{3}{6}$$

From here we can see that the fraction can be simplified further. A factor of 3 can be taken out of the top and bottom leaving us with:

$$\frac{3 \times 1}{3 \times 2} = \frac{1}{2}$$



**Example:** Express  $\frac{1}{x+1} - \frac{1}{x-1}$  as a single fraction.

Solution: to do this we will multiply the denominators to get  $(x+1)(x-1)$  as the denominator.

$$\begin{aligned} & \left(\frac{1}{x+1} \times \frac{x-1}{x-1}\right) - \left(\frac{1}{x-1} \times \frac{x+1}{x+1}\right) \\ &= \frac{x-1}{(x+1)(x-1)} - \frac{x+1}{(x+1)(x-1)} = \frac{(x-1) - (x+1)}{(x+1)(x-1)} = \frac{-2}{(x+1)(x-1)} \end{aligned}$$

Being able to work with fractions is beneficial in that you don't always need a calculator to calculate them unlike if you exclusively used decimals for all your calculations.





## 7.4 Percentages

Percentages express how large one quantity is relative to another quantity; so for example percentage yields express how large the yield of a reaction is compared to the theoretical yield. A percentage of a quantity is a way of expressing a number as a fraction of 100. The symbol for percentage is %.

In general if we have quantities  $x$  and  $y$  and we want to express how large  $x$  is compared to  $y$  (i.e. what percentage is  $x$  of  $y$ ?) we can use the formula:

$$\text{Percentage} = \frac{x}{y} \times 100\%$$



**Example:** What percentage is 8 of 160?

Solution:

$$\frac{8}{160} \times 100\% = 5\%$$

We may also want to calculate the percentage of a quantity for example, what is 28% of 132? In general we can use the formula:

$$x = \frac{\text{Percentage}}{100\%} \times y$$



**Example:** What is 28% of 132?

Solution:

$$x = \frac{28\%}{100\%} \times 132 = 36.96$$



## 7.5 Rounding, Significant Figures and Decimal places

### Rounding

We round numbers as follows:

1. Decide to which 'nearest' number or decimal point to round to.
2. Leave it the same if the next digit is less than 5 (rounding down).
3. Increase it by 1 if the next digit is 5 or more (rounding up).

**Engineering Example:** When we look at Gravity (g) for Fluid mechanics this is rounded to 3 significant figures (s.f.) to 9.81 from 9.80991. Whereas for Geotechnics g is rounded to the nearest whole number as there is less accuracy needed for these calculations

### Significant Figures

Significant figures are used to give us an answer to the appropriate precision. We use s.f. as a short hand for significant figures. So we write a number to a given number of significant figures as follows:

- When asked to write a number to n significant figures we only display the first n non-zero digits.
- Therefore an answer may require rounding up or down. For example, a temperature of  $37.678^{\circ}\text{C}$  to 3 s.f. is  $37.7^{\circ}\text{C}$ .
- For positive decimals less than 1 we start counting the number of significant figures from the first non-zero number. For example, 0.009436 to 2 s.f. is 0.0094 and **not** 0.00.
- We use significant figures on negative numbers in the same way. For example,  $-137.54$  to 2 s.f. is  $-140$ .

**IMPORTANT:** When multiplying or dividing numbers expressed to different numbers of significant figures we leave our answer to smallest number of significant figures used in the numbers we have multiplied or divided.



**Example:** Round 0.3154 to 3 significant figures.

Solution: So the answer is 0.315 (we round down to 0.315 as the digit after the 5 is a 4).

### Decimal Places

The correct number of decimal places is used to indicate the correct precision when adding or subtracting. Consider the number 4.5312 which is given to 4 d.p. (decimal places). However we could also write this number to 1 d.p. which would be 4.5.



**Example:** An angle is taken between two points of  $134.952^{\circ}$ . Round the angle to 1 decimal place.

Solution: The answer is  $135.0^{\circ}$  (we round up to  $135.0^{\circ}$  as the digit after the 9 is a 5, this makes the next digit a 10 and so increases the 4 to a 5).



## 7.6 Physical quantities, Units and Conversions

Often we don't just have numbers on their own, they are connected to physical quantities that can be measured such as time, force, mass or velocity. When describing a quantity, units are needed to define what it means physically. There is a lot of difference between 200 degrees Celsius, and 200 degrees as an angle!

All quantities share the same format:

$$\text{Variable} = \text{Number} \times \text{Units}$$

E.g. suppose we have the mass  $m$  of the product is  $m = 1.2\text{kg}$ . Then the mass  $m$  is the variable, 1.2 is the number and kg kilograms are the units.

- Any terms in an equation that are added or subtracted need to have the same units in order to produce an answer that makes sense. For example the sum  $3\text{kg} + 12\text{s}$  doesn't make physical sense.
- However different units can be multiplied together and divided to make a new compound unit.



**Example:** Speed is calculated from the equation:

$$\text{Speed} = \frac{\text{Distance}}{\text{time}}$$

Distance is measured in meters ( $m$ ) and time in seconds ( $s$ ). What are the units of speed?

Solution: To do this we replace the terms in the equation with their units.

$$\text{Units of Speed} = \frac{m}{s}$$

This means that the speed is measured in meters per second  $\frac{m}{s}$ . This can also be denoted by **m/s** or **ms<sup>-1</sup>**.

**Note:** If a unit has a negative power then it is said to be per that unit. For example  $kNm^{-2}$  is pronounced as kilo-Newton per meter squared.

### Base Units

There are several fundamental quantities in science which all other quantities are built from. Some of these these with their SI units are:

- Length measured in meters  $m$ .
- Mass measured in kilograms  $kg$ .
- Time measured in seconds  $s$ .
- Force measured in newtons  $N$ .
- Temperature measured in kelvin  $K$ .

The main ones we are interested in for engineering are mass, weight, time, temperature and length. From these we can construct a whole host of other units such as newtons  $N$  and pascals  $Pa$ .



**Engineering Example:** We can see this we look at density of an of a fluid such as water

This is normally give as  $\text{kg/m}^3$  this means how many kilograms there are for each  $\text{m}^3$  of water there for

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

**Note:** Calculating units is a good way of checking an equation has been manipulated correctly since the units must be equal on both sides. For example if the left hand side of the equation has units of mass per temperature then the right hand side must do as well.

### Unit prefixes and Scientific Notation

The numbers used in engineering range dramatically from being very large like forces on super structures to incredibly small such as looking at steel at an atomic level so we use either prefixes or scientific notation to display numbers in a concise form. Take the kilogram for example, it means 1000 grams because it uses the prefix kilo k to mean  $\times 10^3$  (which is 1000).

The SI prefixes that you are expected to know are shown in this table below.

| Prefix | Symbol | Power of 10 |
|--------|--------|-------------|
| Pico   | p      | $10^{-12}$  |
| Nano   | n      | $10^{-9}$   |
| Micro  | $\mu$  | $10^{-6}$   |
| Milli  | m      | $10^{-3}$   |
| Centi  | c      | $10^{-2}$   |
| Deci   | d      | $10^{-1}$   |
| 1      | 1      | $10^0$      |
| Kilo   | k      | $10^3$      |
| Mega   | M      | $10^6$      |
| Giga   | G      | $10^9$      |
| Tera   | T      | $10^{12}$   |

**Example:** Express the world's population using an SI prefix to 1 significant figure.

Solution: At time of writing the world's population is 7.2 billion people to 1 significant figure or 7,000,000,000. This can be written as  $7 \times 1,000,000,000 = 7 \times 10^9$ .  $10^9$  has to prefix Giga, G, so the population would be 7 Giga-people.

Scientific notation (or standard form) is another way of writing incredibly large or small numbers. Instead of writing the number to the nearest SI prefix, we write the number as one (non-zero) digit followed by the decimal places all multiplied by the appropriate power of ten. For example 3112 is  $3.112 \times 10^3$  in standard form.



## Converting between Units

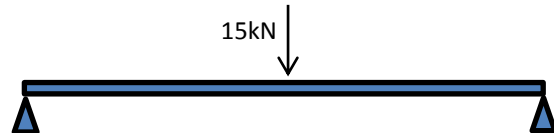
In engineering units are often of different size. In many examples you may be working with a force in kilonewtons (kN) and distances in meters (m) but the stresses may be given in an alternative unit, newtons per millimetre squared (N/mm<sup>2</sup>). When starting a problem it may be helpful to standardize the units to simplify the situation. This way the units are consistent throughout and you avoid having a tricky conversion at the end of solution. It is essential that your units are consistent, for example if you are adding 1km to 100m you must give your answer either as 1.1km or 1100m

### Engineering Example:

An equation you will see in your first year will be

$$\frac{\sigma}{y} = \frac{m}{I} = \frac{E}{R}$$

If we are looking at a force on a beam in kN

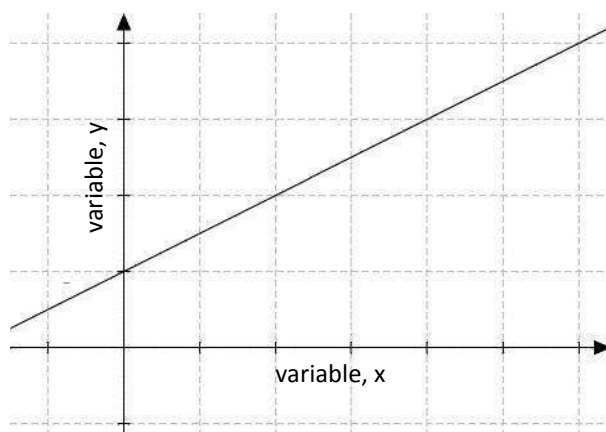


## 7.7 Graphs

Plotting data on a graph has the major advantage it is often clear to see trends and patterns that are present. In experiments we have:

- a control variable that we can change,
- an observed variable that we measure,
- and all other quantities we try and keep constant.

When plotting a graph of data we always plot the controlled variable along the horizontal (or  $x$ ) axis and the observed variable along the vertical (or  $y$ ) axis.



### Straight Line Graphs

Straight line graphs come in the form:

$$y = mx + c$$

where  $x$  is the controlled variable,  $y$  is the observed variable and  $m$  and  $c$  are constants.

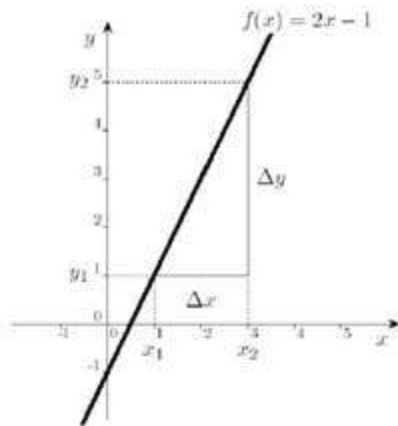
- $c$  is known as the intercept and is where the graph passes through the  $y$ -axis. Hence to find  $c$  we find the value of  $y$  when  $x = 0$ .
- $m$  is known as the gradient and describes how steep the line is. It can be found by drawing a 'triangle' on the straight line and using it to calculate the equation:

$$m = \frac{\Delta y}{\Delta x}$$

This can all be visualized on the graph below:



When  $x$  increases from 1 to 3 we have that  $y$  increases from 1 to 5. Hence



$$\Delta x = 3 - 1 = 2$$

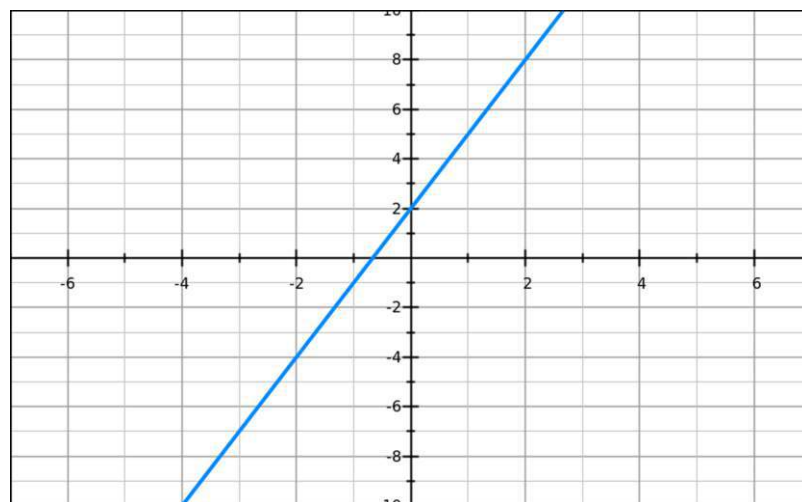
$$\Delta y = 5 - 1 = 4$$

So the gradient  $m = \frac{4}{2} = 2$

The intercept  $c$  is where the graph passes through the  $y$ -axis hence from the graph we can see  $c = -1$ .



**Example:** What is the equation of the straight line shown in the diagram below?



Solution: We know that since this a straight line it will be in the form  $y = mx + c$ . The value of  $c$  is the value of  $y$  when  $x = 0$ . At  $x = 0, y = 2$  from the graph. Therefore  $c = 2$ .

To find  $m$  we need to calculate  $\frac{\Delta y}{\Delta x}$ . Using the triangle method with the triangle in the graph above gives:

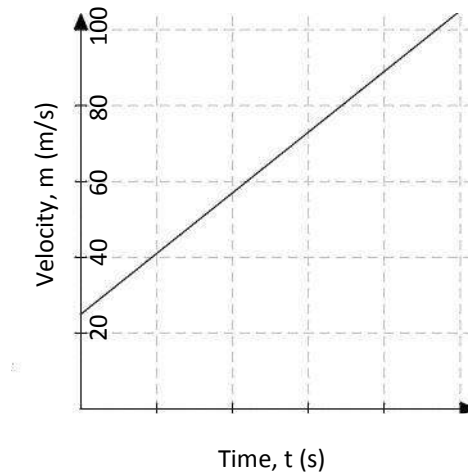
$$\text{Gradient} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{8 - (-8)}{2 - (-4)} = \frac{16}{6} = \frac{8}{3}$$

Since  $m = \frac{8}{3}$ , the equation of the line is  $y = \frac{8}{3}x + 2$



## Graphs with Units

Often graphs won't be those of just numbers and a line but instead will have some physical significance and hence include units. Consider the graph of a particle's velocity as a function of time:



The equation of the line is  $v = 8t + 25$ .  $v$  has units of meters per seconds and to make the equation work so everything on the right hand side must also have units of meters per second as well. That means that the  $c = 25$  has units of m/s. This 25 m/s is the starting velocity of the particle.

The units of  $m$  can be worked out from the equation:

$$\frac{\Delta y}{\Delta x} \text{ or in this case } \frac{\Delta v}{\Delta t}$$

$\Delta v$  has the same units as  $v$  (m/s) and similarly  $\Delta t$  has the units (s). This means  $m$ , the gradient, has units of  $\text{m/s} \div \text{s}$  or  $\text{m/s}^2$ . Since the gradient has units of distance per time squared, then it means that the gradient of the line shows acceleration. The particle will accelerate at  $8 \text{ m/s}^2$ .

## Graphing equations

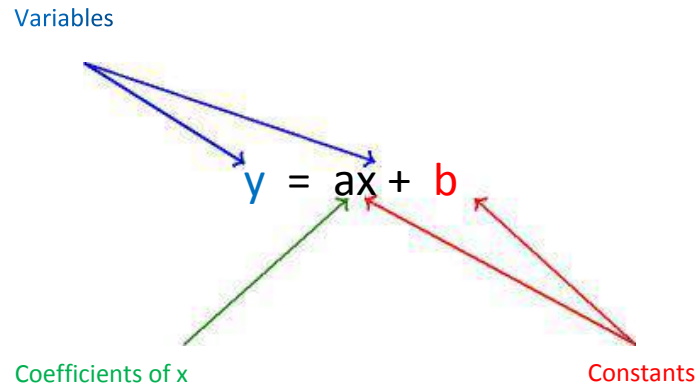
This is a very useful and powerful tool in engineering, as this allows us to plot and see relationships in complex problems beyond a simple straight line graph. This can be used to visualize such problems and find points of interest, these can be points where a value will equal 0 on either axis (roots) or low and high points in a curve such as (**stationary points**). We will look at different curves and lines that you may see in later sections.





## 7.8 Equations and Functions

Physical quantities are often linked in equations.



- Variables are quantities that can take different values; they can vary.
- Constants are fixed numbers so unlike variables cannot change.
- The coefficient of  $x$  is the constant before the  $x$ . In the same way for the equation  $y = 8t^2 + t$  the coefficient of  $t^2$  is 8 and the coefficient of  $t$  is 1.

### What is a Function?

Relationships between quantities are referred to as functions which we usually denote  $f(x)$ , pronounced 'f of x'.

In general we have  $y$  is a function of  $x$ :

$$y = f(x)$$

This means that  $y$  is equal to an expression 'made up' of  $x$ 's. For example  $y = x + 3$  or  $y = e^{3x}$ .

- So a function takes a quantity (a variable or number) and applies operations to it. This produces new quantity.
- Although we have used the notation  $f(x)$  there is no reason why we could not use  $g(x)$ .
- We can also have  $f(z)$  or  $g(\theta)$  if we have functions 'made up' of variables other than  $x$ . For example  $f(z) = z^2$  and  $g(\theta) = \cos(\theta)$ .

Note: We can often think of  $f(x)$  and  $y$  as interchangeable.





**Example:** The function  $f$  takes a variable, doubles it and then subtracts 2. The function  $g$  halves its input. What is:

1.  $f(3)$ ?
2.  $f(x)$ ?
3.  $g(8)$ ?

Solution:

1. The function  $f$  takes the input and doubles it then subtracts 2. With the input being 3, we can find  $f(3)$  by applying those steps to the number 3. When we double it we get 6. Subtract 2 then gives us 4. So our final answer is  $f(3) = 4$ .
2. We do these same steps but with the variable  $x$  instead of a number. First we double it to give  $2x$ . Then we subtract 2 to give  $2x - 2$ . We now have that  $f(x) = 2x - 2$ .
3. We apply the function  $g$  to 8. Halving 8 will give us 4, hence  $g(8) = 4$ .

## Functions with Multiple Variables

Functions can also be of multiple variables. The function:

$$y = f(x, z)$$

means that the function  $f$  takes in a value of  $x$  and also a value of  $z$  to give an output so  $x$  and  $z$  are both variables. For example:

$$y = f(x, z) = xz + 2x + 2$$

This function would input variables  $x$  and  $z$  and give their product plus  $2x$  plus 2 as an output. If  $x = 2$  and  $z = \frac{1}{2}$  then  $y = 7$ .

## 7.7 Powers

Powers are a way to write that we have multiplied a number by itself a given number of times. For example 4 squared is  $4^2 = 4 \times 4$  and 5 cubed would be  $5^3 = 5 \times 5 \times 5$ . So more generally, for any positive whole number  $n$ :

$$x^n = (x \times x \times \dots \times x)$$

**$n$  times**

- We call  $x$  the base.
- We call  $n$  the index of the base or its power.

We would either say  $x^n$  as “ $x$  raised to the power of  $n$ ”, “ $x$  to the power of  $n$ ” or “ $x$  to the  $n$ ”.



**Example:**

So we can write  $T \times T \times T \times T \times T$  as  $T^5$ . (Note that  $T$  is the base and the power is 5).



**Example:**

It is worth noting when a negative number is raised to an even power then the answer is positive

$$(-2)^2 = -2 \times -2 \times -2 \times -2 = \mathbf{16}$$

## Negative Powers

We can also have negative numbers as powers, these give fractions called reciprocals for example  $2^{-1} = \frac{1}{2}$ . So more generally:

$$x^{-n} = \frac{1}{x^n}$$

**Example:**

So we can write  $\frac{1}{vvvv}$  as  $\frac{1}{v^4}$  which is  $v^{-4}$

We can simplify multiple terms like in the examples below by collecting together the terms which are the same.

**Example:**

We can simplify  $cccdcd$  to  $c^4d^2$ . (Note that we have collected all the  $c$  and  $d$  together separately).

**Example:**

So we can simplify  $\frac{xxxyzzzz}{abbb}$  to  $\frac{x^2yz^4}{ab^3}$  which can then write as  $x^2yz^4a^{-1}b^{-3}$ .

## Special Cases

- Any non zero number raised to the power of 1 is itself, for example  $3^1 = 3$  and  $15^1 = 15$ . So we have:

$$x^1 = x$$

Note: Usually we do not write bases to the index of 1 we just write the base. So we would write  $x$  and not  $x^1$ .

- If a number is raised to the power of 0 then the answer is 1. For example  $4^0 = 1$  and  $(-457.391)^{-0} = 1$ . So we have:

$$x^0 = 1$$



## Roots

The opposite of squaring a number say  $4^2 = 16$  is finding the square root in this case  $\sqrt{16} = 4$ . Note the notation for square root is  $\sqrt{\quad}$ . The same is true for cubing and cube roots which are denoted  $\sqrt[3]{\quad}$ . For example  $3 \times 3 \times 3 = 27$  so  $\sqrt[3]{27} = 3$ . In general the opposite of raising a number  $x$  to the power of  $n$  is to find the  $n^{\text{th}}$  root which is denoted  $\sqrt[n]{\quad}$ .

We also use fractional powers as another, often more useful way, to write roots. So we have that  $\sqrt{x} = x^{\frac{1}{2}}$  and  $\sqrt[3]{x} = x^{\frac{1}{3}}$ . More generally we have.

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

**Note:** A root written in its fractional form is often must easier to manipulate with the algebraic rules for powers.



### Example:

So we can write  $\frac{\sqrt{x}}{x} = \frac{x^{\frac{1}{2}}}{x} = x^{\frac{1}{2}-1} = x^{-\frac{1}{2}}$ .

**Note:** A fractional power has the power as the numerator and the root as the denominator of the fraction.



### Example:

$\sqrt[3]{5^{12}}$  can be write as  $5^{\frac{12}{3}}$  which  $= 5^4 = 625$



**Example:** simplify the following:  $\frac{\sqrt[3]{ab^3(ac^4)^2}}{a^{\frac{4}{3}}b}$

The first thing that we can do is expand the brackets on the numerator to give:

$$\frac{\sqrt[3]{ab^3(ac^4)^2}}{a^{\frac{4}{3}}b} = \frac{\sqrt[3]{ab^3a^2c^8}}{a^{\frac{4}{3}}b}$$

The cube root can now be simplified

$$= \frac{a^{\frac{1}{3}}b^1a^{\frac{2}{3}}c^{\frac{8}{3}}}{a^{\frac{4}{3}}b}$$

All the powers can be collected

$$= a^{\frac{1}{3}-\frac{4}{3}+2}b^{1-1}c^{\frac{8}{3}}$$

This can be simplified

$$= a^{-1+2}b^0c^{\frac{8}{3}}$$

$$= ac^{\frac{8}{3}}$$

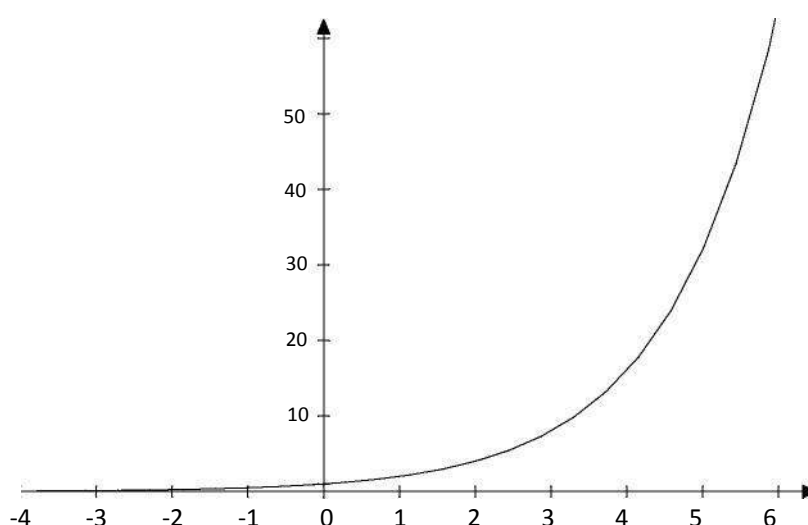


## 7.9 Exponentials

We may use the term 'exponential growth' as a way to describe the rate of a reaction as the temperature increases. In using this phrase we are describing a very fast rate of increase. We can represent this idea mathematically with one of the simplest exponential equations:

$$y = a^x$$

where  $a$  is a constant,  $x$  is the controlled variable (in the above case temperature) and  $y$  is the observed variable (in the above case rate of reaction).



Above we have the graph of  $y = 2^x$  and we can see the graph increases at very fast rate which is what we would expect with exponential growth. Other mathematical things to note are:

- The graph passes through the point  $(0, 1)$  as any variable raised to the power of 0 is equal to 1.
- The values of  $y$  are small and positive when  $x$  is negative.
- The values of  $y$  are large and positive when  $x$  is positive.

### The Exponential Function

So far we have used the word exponential to describe equations in the form  $y = a^x$  however the word is usually reserved to describe a special type of function. It is quite likely we will have seen the symbol  $\pi$  before and know  $\pi$  represents the infinite number 3.14159265... In this section we will be working with the number:

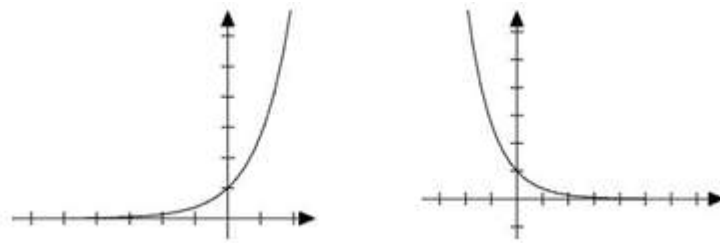
$$e = 2.7182818...$$

Like  $\pi$ ,  $e$  is another number that goes on forever! When we refer to the exponential function we mean  $y = e^x$ . This can be seen as a special case of the previous section when the constant is  $a = e$  in the equation  $y = a^x$ . When we refer to  $e^x$  we say 'e to the x'.

**Note:** Another notation for  $e^x$  is  $\exp(x)$ . They mean the same thing.



## Exponential Graphs



Exponential graphs follow two general shapes called growth and decay which are shown below.

- On the left exponential growth which is represented by the equation  $y = e^x$  note that the graph gets steeper from left to right.
- On the right exponential decay which is represented by the equation  $y = e^{-x}$  note that the graph gets shallower from left to right.

## Algebraic rule for Exponentials

We use very similar rules as the ones we had for powers. The only difference is that the base is now a constant and the power is a variable.

$$1. \quad a^x \times a^y = a^{x+y}$$

$$2. \quad \frac{a^x}{a^y} = a^{x-y}$$

$$3. \quad (a^x)^y = a^{x \times y}$$

**Note:** Above we have that  $x$  and  $y$  are variables and  $a$  is a constant for example  $a = e$  or  $a = 3$ . And in order to use the rules  $a$  must be the same for all terms. For example we can not simplify  $e^x \times 3^x \times 4^x$ .



## 7.10 Logarithms

Just as addition and multiplication have inverses (subtraction and division) so we have inverses for exponentials. These are called logarithms. A logarithm answers the question “how many of one number do we multiply to get another number?”



**Example:** How many 2's do we multiply to get 8

Solution:  $2 \times 2 \times 2 = 8$  so we need to multiply 3 of 2's to get 8. So the **logarithm is 3**.

We have a notation for logarithms for example the last example would be written as:

$$\log_2(8) = 3$$

where this tells us we need multiply 2 by itself 3 times to get 8.

So more generally we can write:

$$\log_a(x) = y$$

- This tells us we need multiply  $a$  by itself  $y$  times to get  $x$ . (In other words  $x = a^y$ ).
- We say that  $a$  is the base of the logarithm.

We can see clearly that as logarithms are the inverse of exponents there exists a relationship between the two given by:

$$y = a^x \Rightarrow x = \log_a(x) = y$$



**Example:** If  $10^x = 3$  then find  $x$ .

Solution: We need to multiply 10 by itself  $x$  times to get 3 so we have:

$$x = \log_{10}(3) = 0.477 \text{ to 3 s.f.}$$

Logarithms are useful for expressing quantities that span several orders of magnitude. For example when examining particle sizes in a soil sample we look at this on a logarithmic scale in order to clearly show the change in magnitude.



## Logarithms: The Inverses of Exponentials

As taking a logarithm is the inverse of an exponential to the same base. We can cancel them using the rules below.

$$\log_a(x) = y$$

### IMPORTANT:

1. When cancelling in the first case all the quantities in the logarithm must be contained in the “power” of the exponential. For example  $\log_a(a^x + 3) \neq x + 3$  as the 3 is not part of the power of the exponential.
2. When cancelling in the second case all of the power must be contained in the logarithm. For example  $a^{\log_a(x)+4} \neq x + 4$  as the 4 is not in the logarithm.



**Example:** Simplify the following:

1.  $\log_{10}(10^{5x^3+3x})$
2.  $3^{\log_3(x^7+1)}$
3.  $\log_3(10^{4x})$
4.  $a^{\log_a(4x^3)+2x}$
5.  $\log_8(8^{2x^4} + 7)$

Solution:

1.  $\log_{10}(10^{5x^3+3x}) = 5x^3 + 3x$
2.  $3^{\log_3(x^7+1)} = x^7 + 1$
3. We cannot simplify this as the logarithm has the base 3 and the exponential is to the base 10.
4. First note that  $a^{\log_a(4x^3)+2x} \neq 4x^3 + 2x$ . We do this as follows:

$$a^{\log_a(4x^3)+2x} = a^{\log_a(4x^3)} a^{2x} = 4x^3 + 2x$$

## Logarithms to the Base 10

Many equations in engineering include logarithms and we tend to use a few logarithms to different bases. The first one is to the base 10.

$$y = 10^x \Rightarrow x = \log(y)$$





## Logarithms to the Base e

Another common logarithm found in engineering is the logarithm to the base  $e$  known as the “natural logarithm”. We use the notation  $\ln(x)$  but this means the same as  $\log_e(x)$ . The natural logarithm is the inverse operation for the exponential  $y = e^x$  so we have the relationship as before:

$$y = e^x \Rightarrow x = \ln(y)$$



### Example:

If  $3 = e^x$  then  $x = \ln 3 \approx 1.09861$

As natural logarithms are the inverse operation to exponentials ( $y = e^x$ ) we have the following rules:

$$\ln(e^x) = x$$

$$e^{\ln(x)} = x$$

**Note:** These rules are the same as the more general ones earlier in the section however look different due to the  $\ln$  notation.



### Example:

We can simplify  $e^{\ln(x^4+2)} = x^4 + 2$ .



### Example:

We can simplify  $\ln(e^{(5x^4+2^x)}) = 5x^4 + 2^x$ .

## Laws of Logarithms

There are 3 laws of logarithms that we use to help algebraically manipulate logarithms:

1.  $\log_a(x \times y) = \log_a(x) + \log_a(y)$
2.  $\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y)$
3.  $\log_a(x^b) = b \times \log_a(x)$

**Note:** Remember we write  $\ln(x)$  for  $\log_e(x)$



**Example:**

We can simplify each of the following into a single logarithm:

1.  $\log 5 + \log 2$
2.  $\log_3 2 + \log_3 2 + \log_3 2$
3.  $3\log 3 - 2\log 2$
4.  $\ln 8 - \ln 4$

Solutions:

1. Using the first law we can see that:

$$\log 5 + \log 2 = \log 5 \times 2 = \log 10.$$

Since  $\log$  is used to denote the base 10 logarithm,  $\log 10 = 1$

2. This can be done using law 1 or by using law 3:

$$\log_3 2 + \log_3 2 + \log_3 2 = 3\log_3 2 = \log_3 8$$

3. Law 3 needs to be used to remove the co-efficient in the front of the logs:

$$3\log 3 - 2\log 2 = \log 3^3 - 2\log 2^2 = \log 27 - \log 4$$

4. Law 2 is used to give:

$$\ln 8 - \ln 4 = \ln \frac{8}{4} = \ln 2$$

## Converting between Logarithms to Different Bases

We may wish to convert between the different logarithms of different bases so for example we might want to convert  $\log(x)$  to  $\ln(x)$ . We do this using the following formula:

$$\log_a(x) = \frac{1}{\log_b(a)} \times \log_b(x)$$



**Example:** Convert  $\log(x)$  to be in the form  $\ln(x)$

Solution: Remember that  $\log(x) = \log_{10}(x)$  and  $\ln(x) = \log_e(x)$ . So using the formula above we have:

$$\log_{10}(x) = \frac{1}{\log_e(10)} \times \log_e(x)$$

$$\Rightarrow \log_{10}(x) = \frac{\ln(x)}{\ln(10)}$$



## 7.11 Rearranging Exponentials and Logarithms

When working with logarithms and exponentials the same rules for rearranging apply. We still use BODMAS in reverse order and the logarithms and exponentials are rearranged at the Orders stage. There are two rules that we utilise when rearranging these equations:

1.  $a^{\log_a x} = x$
2.  $\log_a(a^x) = x$

This also works if we are using  $\ln$  instead of  $\log$ ; it's just a different notation.

**Note:** When taking an exponential or taking a logarithm we must do the operation to the whole side of the equation.



**Example:** Make  $x$  the subject in the following equations:

1.  $y = 10^x - 1$
2.  $\ln(x + 2) = \ln(y) + \ln(2)$

Solution:

1. As we are doing BODMAS in reverse we start by adding 1 to both sides:

$$\begin{aligned} y + 1 &= 10^x && \text{Take a logarithm of base 10 of both sides} \\ \log_{10}(y + 1) &= \log_{10}(10^x) && \text{The right hand side simplifies to } x \\ x &= \log_{10}(y + 1) \end{aligned}$$

2. We can use the laws of logarithms to get the right hand side all as one natural logarithm:

$$\begin{aligned} \ln(x + 2) &= \ln(2y) && \text{Exponentiate both sides} \\ e^{\ln(x+2)} &= e^{\ln(2y)} && \text{Simplify this} \\ x + 2 &= 2y && \text{Subtract 2 from both sides} \\ x &= 2y - 2 \end{aligned}$$

### Engineering Example:

An equation that we see in geotechnics is  $\Delta H = \frac{H}{1+e} C_c \log\left(\frac{\sigma_{v1'}}{\sigma_{v0'}}\right)$  this equation shows us the change in soil movement when a load is applied. For example we may want to find the maximum load that we can apply to the soil so that we only have a certain amount of movement. This is important when working on tunnels for example as the amount of head room may change or in larger foundations such as those for roads as unexpected soil movement could damage the road.



## 8 Test Yourself Solutions



### Test Yourself Solutions – Re-arranging Equations

Re-arrange the following equations:

1. For the equation  $PV = nRT$  make T the subject of the equation  
We can simply divide each side by nR

$$T = \frac{PV}{nR} \quad (1 \text{ mark})$$

2. For the equation  $V = \pi r^2 \frac{h}{3}$  make h the subject of the equation.  
First we can multiply each side by 3 to clear the  $\frac{h}{3}$  to just be h

$$3V = \pi r^2 h$$

We can then divide by  $\pi r^2$  to get our h term on its own

$$h = \frac{V}{\pi r^2} \quad (1 \text{ mark})$$

3. For the equation  $K = \frac{1}{2}mv^2$  make m the subject of the equation

First we will multiply each side by 2 to clear the fraction

$$2K = \frac{2}{2}mv^2$$

$$2K = 1mv^2$$

$$2K = mv^2$$

Then we can divide each side by our  $v^2$  term

$$m = \frac{2K}{v^2} \quad (1 \text{ mark})$$

4. For the equation  $v^2 = u^2 + 2as$  make u the subject of the equation

First we will subtract  $+2as$

$$v^2 - 2as = u^2 \quad (1 \text{ mark})$$

We will now square root each side to clear our squared term for u

$$\sqrt{v^2 - 2as} = \sqrt{u^2}$$

$$u = \sqrt{v^2 - 2as} \quad (1 \text{ mark})$$





### Test Yourself Solutions – Re-arranging Equations

3. For the equation  $\frac{P_1}{\rho g} + \frac{u_1^2}{2g} + z_1 = \frac{P_2}{\rho g} + \frac{u_2^2}{2g} + z_2$  make  $P_2$  the subject of the equation

This may look a little daunting at first but follow the steps and it may surprise you.

To start with we can subtract  $z_2$  from each side

$$\frac{P_1}{\rho g} + \frac{u_1^2}{2g} + z_1 - z_2 = \frac{P_2}{\rho g} + \frac{u_2^2}{2g}$$

We can do the same for the whole  $\frac{u_2^2}{2g}$  this give us

$$\frac{P_1}{\rho g} + \frac{u_1^2}{2g} + z_1 - z_2 - \frac{u_2^2}{2g} = \frac{P_2}{\rho g} \quad (1 \text{ mark})$$

We now must time each term by  $\rho g$

**Note: Remember for to do the do each part of the equation**

$$\frac{P_1 \rho g}{\rho g} + \frac{u_1^2 \rho g}{2g} + z_1 \rho g - z_2 \rho g - \frac{u_2^2 \rho g}{2g} = P_2$$

This is give us the  $P_2$  on its own as we want but there is cancelling that we can do to tidy up the equation

$$P_1 + \frac{u_1^2 \rho}{2} + z_1 \rho g - z_2 \rho g - \frac{u_2^2 \rho}{2} = P_2 \quad (1 \text{ mark})$$

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### Test Yourself Solutions – Quadratic Equations:

1.  $x^2 + 6x + 9$

We can see from the equation that we will end up with two set of brackets each with an  $x$  in  
 $(x \quad)(x \quad)$

From here we need two numbers that add together to make 6 and multiply to make 9

Were a few different number that would fit one of our requirement but only one that fits both which is 3 and 3, so.

$$(x + 3)(x + 3)$$

From here we can see there is only one root for this equation

$$x + 3 = 0$$

There for (1 mark)

$$x = -3$$

2.  $x^2 - 2x - 15$

$$(x - 5)(x + 3)$$

$$x - 5 = 0 \text{ or } x + 3 = 0$$

$$x = 5 \text{ or } x = -3 \quad (1 \text{ mark})$$

3.  $x^2 - 10x + 21$

**Note:** As we have a positive term for  $c$  and a negative term for  $b$  we know that both terms inside the brackets must be negative, as two negative numbers multiplied together give a positive answer.

$$(x - 7)(x - 3)$$

$$x - 7 = 0 \text{ or } x - 3 = 0$$

$$x = 7 \text{ or } x = 3 \quad (1 \text{ mark})$$

4.  $2x^2 - 7x + 6$

$$? \times ? = 12$$

$$? + ? = -7$$

$$-3 \text{ and } -4$$

$$2x^2 - 4x - 3x + 6 \quad (1 \text{ mark})$$

$$2x(x - 2) - 3(x - 2)$$

$$(x - 2)(2x - 3)$$

$$x - 2 = 0 \text{ or } 2x - 3 = 0$$

$$x = 2 \text{ or } x = \frac{3}{2} \quad (1 \text{ mark})$$

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### Test Yourself Solutions – Quadratic Equations:

5.  $3x^2 + 5x - 2$

$$? \times ? = -6$$

$$? + ? = 5$$

$$3x^2 + 6x - x - 2$$

(1 mark)

$$3x(x + 2) - 1(x + 2)$$

$$(x + 2)(3x - 1)$$

$$x + 2 = 0 \text{ or } 3x - 1 = 0$$

$$x = -2 \text{ or } x = \frac{1}{3}$$

(1 mark)

6.  $x^2 + 4x - 3$

[Hint: This quadratic does not solve easily by factorizing]

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-4 \pm \sqrt{4^2 - 4 \times -3}}{2}$$

(1 mark)

$$x = -2 + \sqrt{7} \text{ or } x = -2 - \sqrt{7}$$

(1 mark)

7.  $x^3 + 6x^2 + 7x$

[Hint: Find a quadratic in this equation]

First we take one  $x$  out as a factor this leaves us with

$$x(x^2 + 6x + 7)$$

(1 mark)

This means that we have one of our roots this is  $x = 0$

From this point on we can treat this like we have before and ignore the  $x$  outside the brackets. We can do this with the quadratic formula or by completing the square.

$$x^2 + 6x + 7$$

$$(x + 3)^2 - 9 + 7 = 0$$

$$(x + 3)^2 = 2$$

(1 mark)

$$x + 3 = \pm\sqrt{2}$$

$$x = -3 \pm \sqrt{2}$$

$$\text{So the roots are } x = 0, x = -3 + \sqrt{2} \text{ and } x = -3 - \sqrt{2}$$

(1 mark)

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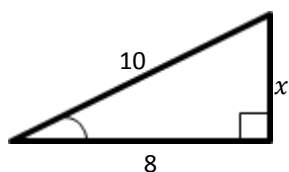




### Test Yourself Solutions– Trigonometry

1. Find the length of the side  $x$  :

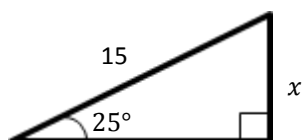
(1 mark)



$$\begin{aligned}a^2 + b^2 &= c^2 \\10^2 &= 8^2 + x^2 \\x &= \sqrt{10^2 - 8^2} \\x &= 6\end{aligned}$$

2. Find the length of the side  $x$ :

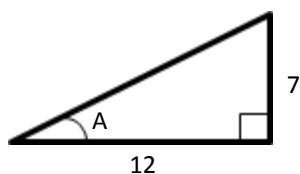
(1 mark)



$$\begin{aligned}\sin(\theta) &= \frac{O}{H} \\ \sin(25^\circ) &= \frac{O}{15} \\ 15 \sin(25^\circ) &= 6.339\end{aligned}$$

3. Find the angle A in radians and degrees:

(2 marks)



$$\begin{aligned}\tan(\theta) &= \frac{O}{A} \\ \tan(A) &= \frac{7}{12} \\ A &= \tan^{-1}\left(\frac{7}{12}\right) \\ &= 36.26^\circ \text{ or } 0.5281 \text{ rads}\end{aligned}$$



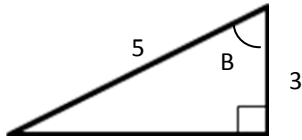




## Test Yourself Solutions– Trigonometry

4. Find the angle B in radians and degrees:

(2 marks)



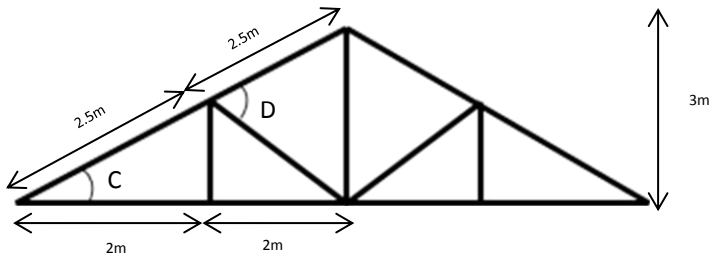
$$\cos(\theta) = \frac{A}{H}$$

$$\cos(B) = \frac{3}{5}$$

$$B = \cos^{-1}\left(\frac{3}{5}\right)$$

**53.13° or 0.9273 rads**

5. A truss has the following dimensions find angle C and D in degrees (4 marks)



To find C think of the truss as a set of right angled triangles. The truss to the centre is 4m and has a height of 3m therefore

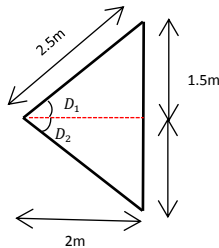
$$\tan(\theta) = \frac{O}{A}$$

$$\tan(C) = \frac{3}{4}$$

$$C = \tan^{-1}\left(\frac{3}{4}\right)$$

$$= 36.87^\circ$$

To find D we can divide the triangle up in to two right angle triangles to find  $D_1$  and  $D_2$ :



By similar triangles we can deduce that angle  $D_1$  is the same as angle C. (Think of it as they both share the side CA so therefore the angle to the horizontal, angle C and angle  $D_1$ , must be identical). So:

$$D_1 = 36.87^\circ$$

You could check this using trigonometry.

Also, we can deduce that  $D_1$  must equal  $D_2$  as they are identical triangles with sides 2m, 2.5m, 1.5m therefore:

$$D = D_1 + D_2 = 36.87 + 36.87 = 73.74^\circ$$

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**Test Yourself Solutions – Differentiation** (1): Differentiate the following functions:

1.  $y = 2x^2 + 3x - 10$

$$\frac{dy}{dx} = (2 \times 2x^{2-1} + 3 - 0) \quad (1 \text{ mark})$$

$$\frac{dy}{dx} = 4x + 3 \quad (1 \text{ mark})$$

2.  $y = 12x^3 - \frac{3}{x^2} + 6x$

$$y = 12x^3 - 3x^{-2} + 6x$$

$$\frac{dy}{dx} = 3 \times 12x^{3-1} - (-2) \times 3x^{-2-1} + 6 = \quad (2 \text{ marks})$$

$$\frac{dy}{dx} = 36x^2 + 6x^{-3} + 6 \quad (1 \text{ mark})$$

$$\frac{dy}{dx} = 36x^2 + \frac{6}{x^3} + 6$$

3.  $y = \frac{9}{x^4}$

$$y = 9x^{-4}$$

$$\frac{dy}{dx} = -4 \times 9 x^{-4-1} \quad (1 \text{ mark})$$

$$\frac{dy}{dx} = -36 x^{-5}$$

$$\frac{dy}{dx} = -\frac{36}{x^5} \quad (1 \text{ mark})$$

4.  $T = 2\ln(12k)$

$$\frac{dT}{dk} = 2 \times \frac{1}{k} \quad (1 \text{ mark})$$

$$\frac{dT}{dk} = \frac{2}{k} \quad (1 \text{ mark})$$

5.  $p(r) = \frac{4}{3}\pi r^3 - 2e^{2r}$

$$\frac{dp}{dr} = \frac{4}{3}\pi \times 3 \times r^{3-1} - 2 \times 2e^{2r} \quad (2 \text{ marks})$$

$$\frac{dp}{dr} = 4\pi r^2 - 4e^{2r} \quad (1 \text{ mark})$$





**Test Yourself Solutions – Differentiation (1):** Differentiate the following functions:

$$6. \quad y = 12x^5 + 6x^{-3} - \sin\left(-\frac{1}{3}x\right) + 8\ln(4x) - \frac{3}{e^{3x}}$$

$$\frac{dy}{dx} = 5 \times 12 x^{5-1} + (-3) \times 6x^{-3-1} - \left(-\frac{1}{3} \cos\left(-\frac{1}{3}x\right)\right) + \frac{8}{x} - (-3) \times 3e^{-3x} \quad (3 \text{ marks})$$

$$\frac{dy}{dx} = 60x^4 - 18x^{-4} + \frac{1}{3}\cos\left(-\frac{1}{3}x\right) + \frac{8}{x} + 9e^{-3x} \quad (1 \text{ mark})$$

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### Test Yourself Solutions – Differentiation (2) : Differentiate the following functions:

1.  $y = \cos(3x^3 + 2x^2)$

Using the **chain rule**:  $y = \cos(u)$  where  $u = 3x^3 + 2x^2$  (1 mark)

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -\sin(3x^3 + 2x^2) \times (9x^2 + 4x) \quad (1 \text{ mark})$$

$$\frac{dy}{dx} = (e^x - 4x - 9x^2)\sin(3x^3 + 2x^2)$$

1.  $y = \ln(x) \times 4x^2$

Using the **product rule** where if:  $y = uv \Rightarrow \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$  where  $u = \ln(x)$  and  $v = 4x^2$  (1 mark)

$$\frac{dv}{dx} = 8x \text{ and } \frac{du}{dx} = \frac{1}{x} \quad (1 \text{ mark})$$

$$\frac{dy}{dx} = 8x\ln(x) + 4x^2 \times \frac{1}{x}$$

$$\frac{dy}{dx} = 8x\ln(x) + 4x \quad (1 \text{ mark})$$

3.  $f(x) = \frac{4e^{4x} + 4x^3}{7x^2}$

Using the **quotient rule** where  $u(x) = 4e^{4x} + 4x^3$  and  $v(x) = 7x^2$  (1 mark)

$$f'(x) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2} = \frac{7x^2(16e^{4x} + 12x^2) - 14x(4e^{4x} + 4x^2)}{49x^4} \quad (2 \text{ marks})$$

$$f(x) = \frac{2(8xe^{4x} + 2x^3 - 4e^{4x} - 4x^2)}{7x^3} \quad (1 \text{ mark})$$

4.  $P(t) = \ln(t^3 + 3t + 1) \times \sin(t)$

Using the **product rule** where  $u = \ln(t^3 + 3t + 1)$  and  $v = \sin(t)$  (1 mark)

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$= \ln(t^3 + 3t + 1) \cos(t) + \sin(t) \frac{3t^2 + 3}{t^3 + 3t + 1} \quad (2 \text{ marks})$$





### Test Yourself Solutions – Differentiation (2) (continued)

5.  $y = \sin(x) (x^3 + 6)e^{-x}$

Using the **product rule** three times where  $u = \sin(h)$  and  $v = (h^3 + 6)$  and  $w = e^{-h}$  (1 mark)

$$\frac{dQ}{dh} = uv \frac{dw}{dh} + uw \frac{dv}{dh} + vw \frac{du}{dh} \quad (1 \text{ mark})$$

$$\frac{dQ}{dh} = \sin(x) (x^3 + 6)(-e^{-x}) + \sin(x) (3x^2)e^{-x} + \cos(x) (x^3 + 6)e^{-x} \quad (2 \text{ marks})$$

$$\frac{dQ}{dh} = e^{-x}(\cos(x) (x^3 + 6) - \sin(x) (x^3 - 3x^2 + 6)) \quad (1 \text{ mark})$$

**BONUS:**  $y = 7e^{-(3x^2+2x+1)} \sin(x^2 + 6)x^3$

Using the **product rule** where  $u = 7e^{-(3x^2+2x+1)}$  and  $v = \sin(x^2 + 6)$  and  $w = x^3$

$$\frac{dy}{dx} = uv \frac{dw}{dx} + uw \frac{dv}{dx} + vw \frac{du}{dx}$$

$$= 7e^{-(3x^2+2x+1)} \times \sin(x^2 + 6) \times 3x^2 + 7e^{-(3x^2+2x+1)} \times x^3 \times 2x \cos(x^2 + 6) + \sin(x^2 + 6) \times x^3 \times 7(-6x - 2)e^{-3x^2+2x+1}$$

$$\frac{dy}{dx} = 7x^2e^{-(3x^2+2x+1)}(\sin(x^2 + 6)(3 - x(6x + 2)) + 2x^2 \cos(x^2 + 6))$$



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## Test Yourself Solutions – Integration

**A**

$$1. \int 6x^2 + 3x \, dx = \int 6x^2 + \int 3x = \frac{6x^{2+1}}{2+1} + \frac{3x^{1+1}}{1+1} + C \quad (1 \text{ mark})$$

$$= 2x^3 + \frac{3x^2}{2} + C \quad (1 \text{ mark})$$

$$2. \int (x^7 - \cos(x)) \, dx = \int x^7 - \int \cos(x) = \frac{x^{7+1}}{7+1} - \sin(x) + C \quad (1 \text{ mark})$$

$$= \frac{x^8}{8} - \sin(x) + C \quad (1 \text{ mark})$$

$$3. \int_0^4 ((5x^2 + e^{2x})) \, dx = \left[ \frac{5x^{2+1}}{2+1} + \frac{1}{2}e^{2x} \right]_0^4 = \left[ \frac{5x^3}{3} + \frac{1}{2}e^{2x} \right]_0^4 \quad (1 \text{ mark})$$

$$= \left( \frac{5(4)^3}{3} + \frac{1}{2}e^8 \right) - \left( 0 + \frac{1}{2}e^0 \right) \quad (1 \text{ mark})$$

$$= 1596.65 \quad (1 \text{ mark})$$

$$4. \int_{\pi/2}^{\pi} \left( \frac{3}{x^2} + \sin(x) \right) \, dx = \left[ -\frac{3}{x} - \cos(x) \right]_{\pi/2}^{\pi} \quad (1 \text{ mark})$$

$$= \left( -\frac{3}{\pi} - \cos(\pi) \right) - \left( -\frac{3}{0.5\pi} - \cos\left(\frac{\pi}{2}\right) \right) = \left( -\frac{3}{\pi} - (-1) \right) + \frac{6}{\pi} + 0 \quad (1 \text{ mark})$$

$$= \frac{3}{\pi} + 1 \quad (1 \text{ mark})$$

$$5. \int \left( \frac{3x^3 + 4x^2 + 5x}{x^3} \right) \, dx = \int \frac{3x^3}{x^3} + \int \frac{4x^2}{x^3} + \int \frac{5x}{x^3} = \int 3 + \int \frac{4}{x} + \int \frac{5}{x^2} \quad (2 \text{ marks})$$

$$= 3x + 4\ln|x| + \frac{5x^{-2+1}}{-2+1} + C \quad (1 \text{ mark})$$

$$= 3x + 4\ln|x| - \frac{5}{x} + C$$





## Test Yourself Solutions – Integration

**B**

1

$$\int 2\sin(x) \cos(x)$$

$$u = \sin(x) \therefore \frac{du}{dx} = \cos(x) \Rightarrow dx = \frac{du}{\cos(x)} \quad (2 \text{ marks})$$

Substitute  $u$  and  $du$  in accordingly

$$\int 2u \times \cos(x) \times \frac{du}{\cos(x)}$$

The  $\cos(x)$  cancels

$$2 \int u \, du \quad (1 \text{ mark})$$

Now integrate:

$$2 \left( \frac{u^2}{2} \right) + C = u^2 + C \quad (1 \text{ mark})$$

Substitute  $u = \sin(x)$  back in

$$\sin^2(x) + C \quad (1 \text{ mark})$$

**Note:** Remember that integration is the opposite of differentiation. Here it might be worth considering '*what would I have to differentiate to get  $2\sin(x)\cos(x)$ ?*'. By inspection you can see that  $\sin^2(x)$  differentiates to  $2 \times \sin(x) \times \cos(x)$  therefore we can say that  $2\sin(x)\cos(x)$  must therefore integrate to  $\sin^2(x)$

2

$$\int x^5(1 + 8x^6)^{\frac{1}{3}} dx$$

$$u = 1 + 8x^6 \therefore \frac{du}{dx} = \frac{1}{48x^5} \Rightarrow dx = \frac{1}{48x^5} du \quad (2 \text{ marks})$$

Substitute  $u$  and  $du$  in appropriately

$$\int x^5 u^{\frac{1}{3}} \times \frac{du}{48x^5}$$

$x^5$  cancel

$$\frac{1}{48} \int u^{1/3} du \quad (1 \text{ mark})$$

Integrate:

$$\begin{aligned} \frac{1}{48} \left( \frac{u^{4/3}}{\frac{4}{3}} \right) &= \frac{1}{48} \left( \frac{3}{4} u^{4/3} \right) + C \\ &= \frac{1}{64} u^{4/3} + C \end{aligned} \quad (2 \text{ marks})$$

Substitute  $u = 1 + 8x^6$  back in:

$$\frac{1}{64} (1 + 8x^6)^{4/3} + C \quad (1 \text{ mark})$$





## Test Yourself Solutions – Integration

### C

$$1. \quad \frac{dy}{dx} = \frac{6x^3 + 3x^2 + 2x}{y}$$

Separate the variable

$$y \, dy = (6x^3 + 3x^2 + 2x)dx \quad (1 \text{ mark})$$

Integrate both sides

$$\int y \, dy = \int (6x^3 + 3x^2 + 2x)dx \quad (1 \text{ mark})$$

Multiple through by 2

$$\frac{1}{2}y^2 = \frac{6}{4}x^4 + \frac{3}{3}x^3 + \frac{2}{2}x^2 + C \quad (2 \text{ marks})$$

Take  $x$  out

$$y^2 = 3x^4 + 2x^3 + 2x^2 + C$$

$$y^2 = x^2(3x^2 + 2x + 2) + C$$

And root

$$y = x\sqrt{3x^2 + 2x + 2} + C \quad (2 \text{ marks})$$

### D

$$1. \quad \frac{dy}{dx} = 5x^5 + 2 \text{ and passes through the point } (3, 4)$$

$$y = \int (5x^5 + 2)dx$$

Substitute values in to find C

$$y = \frac{5}{6}x^6 + 2x + C \quad (1 \text{ mark})$$

$$4 = \frac{5}{6}(3)^6 + 2(3) + C \quad (1 \text{ mark})$$

$$C = -609.5 \quad (1 \text{ mark})$$

$$y = \frac{5}{6}x^6 + 2x - 609.5 \quad (1 \text{ mark})$$

$$2. \quad \frac{dy}{dx} = 3e^{3x} + \sin(x) \text{ and passes through the point } (0,1)$$

$$y = \int (3e^{3x} + \sin(x))dx$$

Substitute values in to find C

$$y = e^{3x} - \cos(x) + C \quad (1 \text{ mark})$$

$$1 = e^{3 \times 0} - \cos(0) + C \quad (1 \text{ mark})$$

$$C = 1 \quad (1 \text{ mark})$$

$$y = e^{3x} - \cos(x) + 1 \quad (1 \text{ mark})$$

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# Core Maths

Algebra

Trigonometry

Differentiation

Integration



# Algebra

**1.1**      **Rearranging Equations**

**1.2**      **Simultaneous Equations**

**1.3**      **Quadratics**

**Test Yourself**

# Trigonometry

2.1 Radians

2.2 Right-Angled Triangles

2.3 SOHCAHTOA

2.4 Inverse Function and Rearranging  
Trigonometric Functions

2.5 Trigonometric Identities

Test Yourself

# Differentiation

- |     |                                                       |                   |
|-----|-------------------------------------------------------|-------------------|
| 3.1 | Introduction to Differentiation                       |                   |
| 3.2 | Differentiating Polynomials                           |                   |
| 3.3 | Differentiating a Function with Multiple Terms        |                   |
| 3.4 | Differentiating Trigonometric Functions               |                   |
| 3.5 | Differentiating Exponential and Logarithmic Functions |                   |
| 3.6 | Chain Rule                                            |                   |
| 3.7 | Product Rule                                          | Test Yourself (1) |
| 3.8 | Quotient Rule                                         | Test Yourself (2) |
| 3.9 | Stationary Points                                     |                   |

# Integration

- 4.1 Introduction to Integration
- 4.2 Integrating Polynomials
- 4.3 Integrating  $x^{-1}$  and exponentials
- 4.4 Integrating Trigonometric Function
- 4.5 Finding the Constant of Integration
- 4.6 Integrals with Limits
- 4.7 Separating the Variable
- 4.8 Integration by Substitution

Test Yourself

# Further Material

5 Vectors

6 Matrices





# Refresher Topics

- 7.1 Addition, Subtraction, Multiplication and Division
- 7.2 Mathematical Notation, Symbols and Operators
- 7.3 Fractions
- 7.4 Percentages
- 7.5 Rounding, Significant Figures and Decimal Places
- 7.6 Physical Quantities, Units and Conversions
- 7.7 Graphs
- 7.8 Equations and Functions
- 7.9 Exponentials
- 7.10 Logarithms
- 7.11 Rearranging Exponentials and Logarithms



# Your Scores

This is a summary of your scores so far. Have a look to see which area you are strong at which areas might need going over.

**Note:** This work is **not assessed** - it is for your learning only.

**Rearranging Equations**

**Quadratics**

**Trigonometry**

**Differentiation (1)**

**Differentiation (2)**

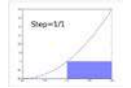
**Integration**

**TOTAL**

**/100**



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